

GOOD HEALTH AND HOW WE WON IT

**Upton Sinclair
and
Michael Williams**

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Transcriber's Notes

Obvious typographical errors have been silently corrected. Variations in hyphenation have been standardised but all other spelling and punctuation remains unchanged.

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GOOD HEALTH AND HOW WE WON IT

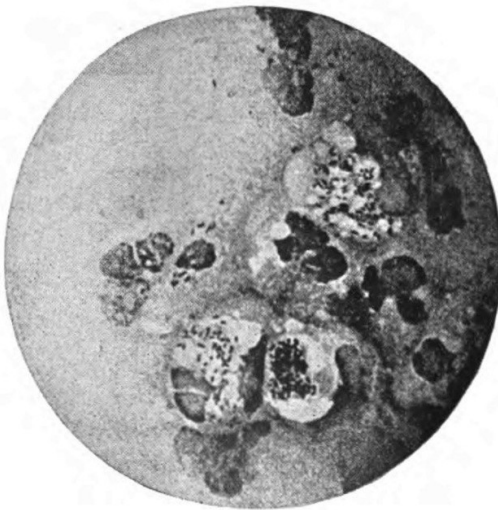


Fig. A.

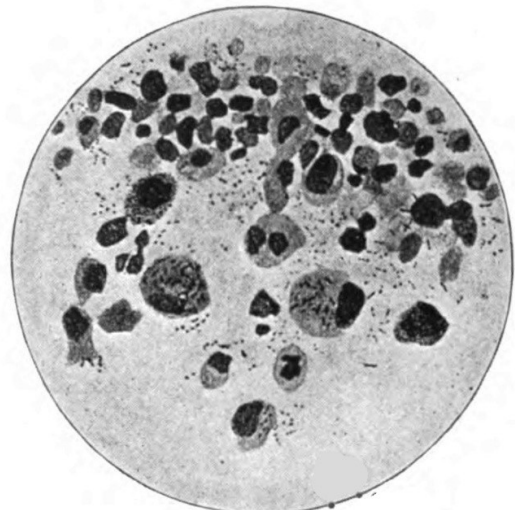


Fig. B.

"THE BATTLE OF THE BLOOD"

Micro-photograph of leucocytes (white and grayish bodies) in conflict with Germs (black dots and bodies). In Fig. A the germ is that of influenza, in Fig. B that of plague.

GOOD HEALTH
AND HOW WE WON IT

*WITH AN ACCOUNT OF THE NEW
HYGIENE*

BY
UPTON SINCLAIR
AND
MICHAEL WILLIAMS

*WITH SIXTEEN FULL-PAGE ILLUSTRATIONS
FROM PHOTOGRAPHS*

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[FRONTISPIECE](#)

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INTRODUCTION

BY UPTON SINCLAIR

Ten years ago, when I was a student at college, I fell a victim to a new and fashionable ailment called “la grippe.” I recollect the date very well, because it was the first time I had been sick in fourteen years—the last difficulty having been the whooping-cough.

I have many times had occasion to recall the interview with the last physician I went to see. I made a proposition, which might have changed the whole course of my future life, had he only been capable of understanding it. I said: “Doctor, it has occurred to me that I would like to have someone who knows about the body examine me thoroughly and tell me how to live.”

I can recollect his look of perplexity. “Was there anything the matter with you before this attack?” he asked.

“Nothing that I know of,” I answered; “but I have often reflected that the way I am living cannot be perfect; and I want to get as much out of my body and mind as I can. I should like to know, for instance, just what are proper things for me to eat——”

“Nonsense,” he interrupted. “You go right on and live as you have been living, and don’t get to thinking about your health.”

And so I went away and dismissed the idea. It was one that I had broached with a great deal of diffidence; so far as I knew, it was entirely original, and I was not sure how a doctor would receive it. All doctors that I had ever heard of were people who cured you when you were sick; to ask one to take you when you were well and help you to stay well, was to take an unfair advantage of the profession.

So I went on to “live as I had been living.” I ate my food in cheap restaurants and boarding-houses, or in hall bedrooms, as students will. I invariably took a book to the table, and ate very rapidly, even then;

frequently I forgot to eat at all in the ardor of my work. I was a worshiper of the ideal of health, and never used any sort of stimulant; but I made it a practice to work sixteen hours a day, and quite often I worked for long periods under very great nervous strain. And four years later I went back to my friend the physician.

“You have indigestion,” he said, when I had told him my troubles. “I will give you some medicine.”

So every day after meals I took a teaspoonful of some red liquor which magically relieved the distressing symptoms incidental to doing hard brain-work after eating. But only for a year or two more, for then I found that the artificially digested food was not being eliminated from my system as regularly as necessary, and I had to visit the doctor again. He gave my ailment another name, and gave me another kind of medicine; and I went on, working harder than ever—being just then at an important crisis in my life.

Gradually, however, to my great annoyance, I was forced to realize that I was losing that fine robustness which enabled me to say that I had not had a day’s sickness in fourteen years. I found that I caught cold very easily—though I always attributed it to some unwonted draught or exposure. I found that I was in for tonsilitis once or twice every winter. And now and then, after some particularly exhausting labor, I would find it hard to get to sleep. Also I had to visit the dentist more frequently, and I noticed, to my great perplexity, that my hair was falling out. So I went on, until at last I was on the verge of a nervous breakdown, and had to drop everything and go away and try to rest.

That was my situation when I stumbled upon an article in the *Contemporary Review*, telling about the experiments of a gentleman named Horace Fletcher. Mr. Fletcher’s idea was, in brief, that by thorough and careful chewing of the food, one extracted from it the maximum of nutriment, and could get along upon a much smaller quantity, thus saving a great strain upon the bodily processes.

This article came to me as one of the great discoveries of my life. Here was a man who was doing for himself exactly what I had asked my

physician to do for me so many years previously; who was working, not to cure disease, but to live so that disease would be powerless to attack him.

I went at the new problem in a fine glow of enthusiasm, but blindly, and without guidance. I lived upon a few handfuls of rice and fruit—with the result that I lost fourteen pounds in as many days. At the same time I met a young writer, Michael Williams, and passed the Fletcher books on to him—and with precisely the same results. He, like myself, came near killing himself with the new weapon of health.

But in spite of discouragements and failures, we went on with our experiments. We met Mr. Fletcher himself, and talked over our problems with him. We followed the course of the experiments at Yale, in which the soundness of his thorough mastication and “low proteid” arguments were definitely proven. We read the books of Metchnikoff, Chittenden, Haig and Kellogg, and followed the work of Pawlow of St. Petersburg, Masson of Geneva, Fisher of Yale, and others of the pioneers of the new hygiene. We went to Battle Creek, Michigan, where we found a million-dollar institution, equipped with every resource of modern science, and with more than a thousand nurses, physicians and helpers, all devoting their time to the teaching of the new art of keeping well. And thus, little by little, with backslidings, mistakes, and many disappointments, we worked out our problems, and found the road to permanent health. We do not say that we have entirely got over the ill effects of a lifetime of bad living; but we do say that we are getting rid of them very rapidly; we say that we have positive knowledge of the principles of right living, and of the causes of our former ailments, where before we had only ignorance.

In the beginning, all this was simply a matter of our own digestions, and of the weal and woe of our immediate families. But as time went on we began to realize the meaning of this new knowledge to all mankind. We had found in our own persons freedom from pain and worry; we had noticeably increased our powers of working, and our mastery over all the circumstances of our lives. It seemed to us that we had come upon the discovery of a new virtue—the virtue of good eating—fully as important as any which moralists and prophets have ever preached. And so our interest in these reforms became part of our dream of the new humanity. It was not enough for us to have found the way to health for ourselves and our

families; it seemed to us that we ought not to drop the subject until we had put into print the results of our experiments, so that others might avoid our mistakes and profit by our successes.

Historians agree that all known civilizations, empire after empire, republic after republic, from the dawn of recorded time down to the present age, have decayed and died, through causes generated by civilization itself. In each such case the current of human progress has been restored by a fresh influx of savage peoples from beyond the frontiers of civilization. So it was with Assyria, Egypt and Persia; so Greece became the wellspring of art and the graces of life, and then died out; so Rome conquered the world, built up a marvellous structure of law, and then died out. As Edward Carpenter and others have shown us, history can paint pictures of many races that have attained the luxuries and seeming securities of civilization, but history has yet to record for us the tale of a nation passing safely through civilization, of a nation which has not been eventually destroyed by the civilization it so arduously won.

And why? Because when ancient races emerged out of barbarism into civilization, they changed all the habits of living of the human race. They adopted new customs of eating; they clothed themselves; they lived under roofs; they came together in towns; they devised ways of avoiding exposure to the sun and wind and rain—but they never succeeded in devising ways of living that would keep them in health in their new environment.

The old struggle against the forces of nature once relaxed, men grew effeminate and women weak; diseases increased; physical fibre softened and atrophied and withered away; moral fibre went the same path to destruction; dry rot attacked the foundations of society, and eventually the whole fabric toppled over, or was swept aside, to be built up again by some conquering horde of barbarians, which in its turn grew civilized, and in its turn succumbed to the virulent poison that seemed inherent in the very nature of civilization, and for which there seemed to be no antidote.

So much for the past. As to the present, there do not lack learned and authoritative observers and thinkers who declare that our own civilization is also dying out. They point out that while in many directions we have bettered our physical condition, improved our surroundings, and stamped

out many virulent diseases (smallpox, the plague and yellow fever, for instance), and have reduced average mortality, nevertheless we have but exchanged one set of evils for another and perhaps more serious, because more debilitating and degenerating set: namely, those manifold and race-destroying evils known as nervous troubles, and those other evils resulting from malnutrition, which are lumped together vaguely under the name of dyspepsia, or indigestion—the peculiar curse of America, the land of the frying-pan.

It is also plain, say the critics of our civilization, that society to-day cannot be regenerated by barbarians. To-day the whole world is practically one great civilization, with a scattering of degraded and dying little tribes here and there. Modern civilization seems to have foreseen the danger of being overrun some day as the ancient civilizations were, and to have forestalled the danger by the inventions of gunpowder and rum, syphilis and tuberculosis.

Are these critics right? I believe that they are, as far as they go; I believe that to-day our civilization is rapidly degenerating; but also I believe that it contains within itself two forces of regeneration which were lacking in old societies, and which are destined ultimately to prevail in our own. The first of these forces is democracy, and the second is science.

To whatever department of human activity one turns at the present day, he finds men engaged in combating the age-long evils of human life with the new weapon of exact knowledge; and their discoveries no longer remain the secrets of a few—by the agencies of the public school and the press they are spreading throughout the whole world. Thus, a new science of economics having been worked out, and the causes of poverty and exploitation set forth, we see a world-wide and universal movement for the abolition of these evils. And hand in hand with this goes a movement of moral regeneration, manifesting itself in a thousand different forms, but all having for their aim the teaching of self-mastery—the replacing of the old natural process of the elimination of the unfit by a conscious effort on the part of each individual to eliminate his own unfitness. We see this movement in literature and art; we see it in the new religions which are springing up—in Christian Science, and the so-called “New Thought” movements; we see it in the great health movement which is the theme of

this book, and which claims for its leaders some of the finest spirits of our times.

In the state of nature man had to hunt his own food, so he was hungry when he sat down to eat. But having conquered nature, and accumulated goods, he is able to think of enjoyments, and invents cooks and the art of cookery—which is simply the tickling of his palate with all kinds of stomach-destroying concoctions. And now the time has come when he wishes to escape from the miseries thus brought upon him; and, as before, the weapon is that of exact science. He must ascertain what food elements his body needs, and in what form he may best take them; and in accordance with this new knowledge he must shape his habits of life. In the same way he has to examine and correct his habits of sleeping and dressing and bathing and exercising, in accordance with the real necessities of his body.

This is the work which the leaders of the new movement are engaged upon. To quote a single instance: while I was “living as I had been living” and eating the preparations of ignorant cooks in boarding-houses and restaurants, Dr. Kellogg of Battle Creek was bringing all the resources of modern chemistry and bacteriology to bear upon the problem of the nutrition of man; taking all the foods used by human beings, and analyzing them and testing them in elaborate experiments; determining the amount of their available nutriment and their actual effect upon the system in all stages of sickness and health; the various ways of preparing them and combining them, and the effect of these processes upon their palatability and ease of digestion. Every day for nine years, so Kellogg told me, he sat down to an experimental meal designed by himself and prepared by his wife; and the result is a new dietary—that in use at the Battle Creek Sanitarium—which awaits only the spread of knowledge to change the ways of eating of civilized man.

This new health knowledge has been amassed by many workers and, as in all cases of new knowledge, there is much chaff with the grain. There are faddists as well as scientists; there are traders as well as humanitarians. It seemed to us that there was urgently needed a book which should gather this new knowledge, and present it in a form in which it could be used by the average man. There have been many books written upon this; but they are either the work of propagandists with one idea—containing, as we have

proved to our cost, much dangerous error; or else the work of physicians and specialists, whose vocabulary is not easily to be comprehended by the average man or woman. What we have tried to write is a book which sets forth what has been proved by investigators in many and widely-scattered fields; which is simple, so that a person of ordinary intelligence can comprehend it; which is brief, so that a busy person may quickly get the gist of it; and which is practical, giving its information from the point of view of the man who wishes to apply these new ideas to his own case.

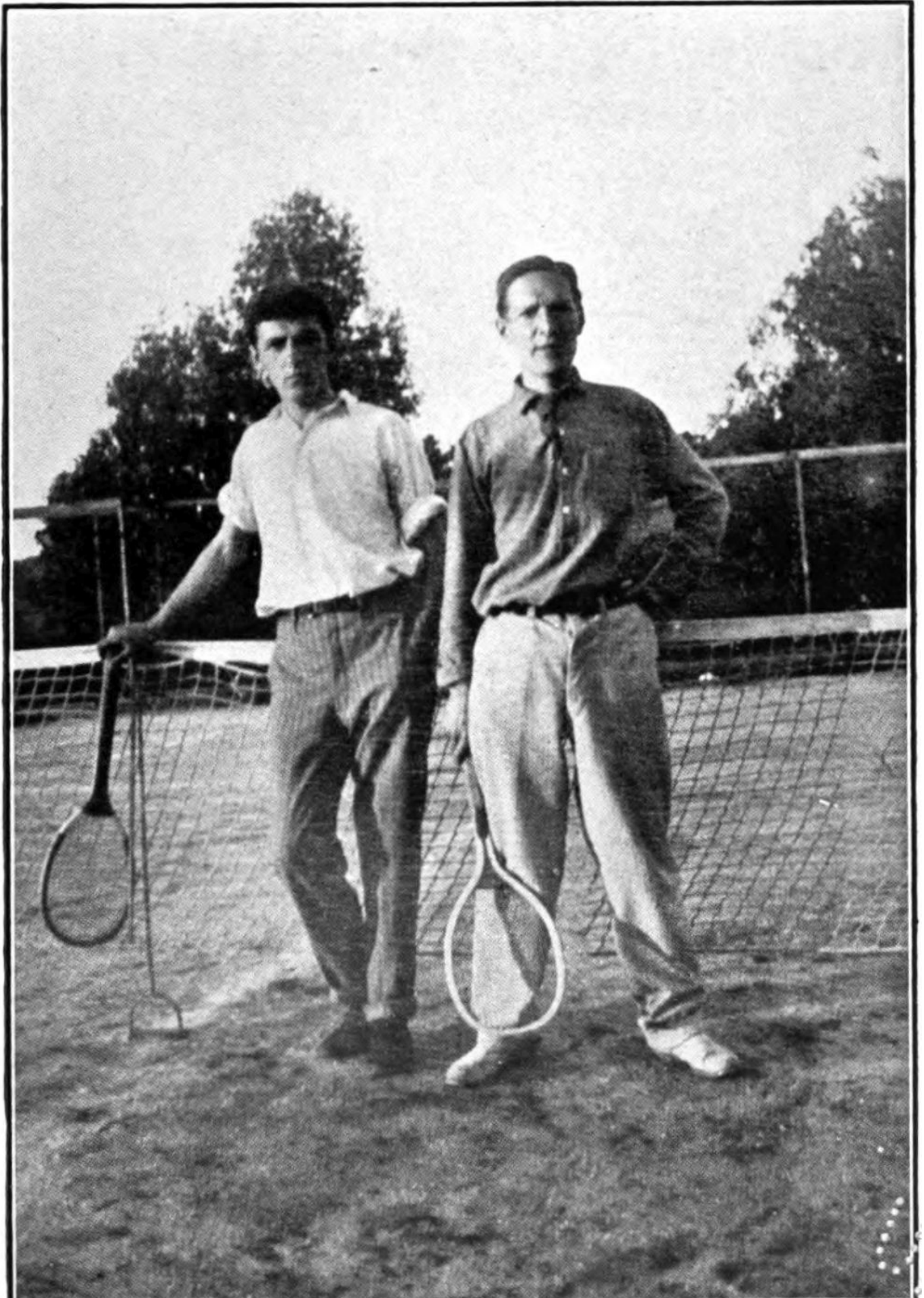
Michael Williams was recently persuaded to give a semi-public talk on the subject before an audience of several hundred professional and business people. He was compelled to spend the rest of the evening in answering the questions of his audience; and listening to these questions, I was made to realize the tremendous interest of the public in the practical demonstration which Mr. Horace Fletcher has given of the idea of Metchnikoff, that men and women to-day grow old before they ought to do so, and that the prime of life should be from the age of fifty to eighty. A broken-down invalid at forty-five, Mr. Fletcher was at fifty-four a marvel of strength—and at fifty-eight he showed an improvement of one hundred per cent. over his tests at the age of fifty-four; thus proving that progressive recuperation in the so-called “decline of life” might be effected by followers of the new art of health.

As a result of this address, Williams was invited by the president of one of the largest industrial concerns in the country to lecture to his many thousands of employees on the new hygiene; his idea being to place at their disposal the knowledge of this new method of increasing their physical and mental efficiency.

For business men and women, indeed, for workers of all kinds, good health is capital; and the story of the new hygiene is the story of the throwing open of hitherto unsuspected reserve-stores of energy and endurance for the use of all.

In writing upon this subject, the experiences most prominent in our minds have naturally been those of ourselves, of our wives and children, and of friends who have followed in our path. As the setting forth of an actual case is always more convincing than a general statement, we have

frequently referred to these experiences, and what they have taught us. We have done this frankly and simply, and we trust that the reader will not misinterpret the spirit in which we have done it. Mr. Horace Fletcher has set the noble example in this matter, and has been the means of helping tens of thousands of his fellow men and women.



MR. UPTON SINCLAIR AND MR. MICHAEL WILLIAMS
Resting from their favorite exercise.

I have sketched the path by which I was led into these studies; there remains to outline the story of my collaborator. Williams is the son of a line of sailors, and inherited a robust constitution; but as a boy and youth he was employed in warehouses and department stores, and when he was twenty he went to North Carolina as a tuberculosis patient. Returning after two years, much benefited by outdoor life, he entered newspaper work in Boston, New York, and elsewhere, and kept at it until four years ago, when again he fled South to do battle with tuberculosis, which had attacked a new place in his lungs. After a second partial recuperation, he went to San Francisco. At the time of the earthquake he held a responsible executive position, and his health suffered from the worry and the labors of that period. A year later there came the shock and exposure consequent upon the burning of Helicon Hall. Williams found himself hovering upon the brink of another breakdown, this time in nervous energy as well as in lung power. A trip to sea failed to bring much benefit; and matters were seeming pretty black to him, when it chanced that a leading magazine sent him to New Haven to study the diet experiments being conducted at Yale University by Professors Chittenden, Mendel and Fisher. He found that these experiments were based upon the case of Horace Fletcher, and had resulted in supporting his claims. This circumstance interested him, suggesting as it did that he himself might have been to blame for his failure with Mr. Fletcher's system. So he renewed the study of Fletcherism, and later on the same magazine sent him to Dr. Kellogg's institution at Battle Creek, with the result that he became a complete convert to the new ideas. Like a great many newspaper men, he had been a free user of coffee, and also of alcohol. As one of the results of his adoption of the "low proteid" diet, and of the open-air life, he was able to break off the use of all these things without grave difficulty. A bacteriological examination recently disclosed the fact that his lungs had entirely healed; while tests on the spirometer showed that his breathing capacity was far beyond that of the average man of his weight and size. In less than three months, while at the Battle Creek Sanitarium, tests showed a great gain in the cell count of his blood, and in its general quality. Also, his general physical strength was increased from 4635 units to 5025, which latter figure is well above the average for his height, 68.2 inches.

In conclusion, we wish jointly to express our obligation to Mr. Horace Fletcher, to Dr. J. H. Kellogg, to Professor Russell H. Chittenden, to Professor Lafayette B. Mendel, and to Professor Irving Fisher for advice, criticism and generous help afforded in the preparation of some of the chapters of this book. The authority of these scientists, physicians and investigators, and of others like Metchnikoff, Pawlow, Cannon, Curtis, Sager, Higgins and Gulick, whose works we have studied, is the foundation upon which we rest on all questions of fact or scientific statement. They are the pathbreakers and the roadbuilders,—we claim to be simply guides and companions along the journey to the fair land of health. The journey is not long, and the road is a highway open to all.

I

THE BATTLE OF THE BLOOD

The new ideas of living which are the subject of this book have proceeded from investigation of the human body with the high-power microscope. The discoveries made, which have to do, not so much with the body itself as with the countless billions of minute organisms which inhabit the body, may be best set forth by a description of the blood. "The blood is the life," says Exodus, and modern science has confirmed this statement. From the blood proceeds the life of all the body, and in its health is the body's health.

If you should prick your finger and extract a drop of your own blood, and examine it under a microscope, you would make the fascinating discovery that it is the home of living creatures, each having a separate and independent existence of its own. In a single ounce of blood there are more of these organisms than there are human beings upon the face of the globe. These organisms are of many kinds, but they divide themselves into two main groups, known as the red corpuscles and the white.

The red corpuscles are the smaller of the two. The body of an average man contains something like thirty million of millions of these corpuscles; a number exceeding the population of New York and London are born in the body every second. They are the oxygen conveyers of the body; the process of life is one of chemical combustion, and these corpuscles feed the fire. No remotest portion of the body escapes their visitation. They carry oxygen from the lungs and they bring back the carbon dioxide and other waste products of the body's activities. They have been compared to men who carry into a laundry buckets of pure water, and carry out the dirty water resulting from the washing process.

The other variety of organisms are the white cells or leucocytes, and it is concerning them that the most important discoveries of modern investigators have been made. The leucocytes vary in number according to the physical condition of the individual, and according to their locality in

the body. Their function is to defend the body against the encroachments of hostile organisms.

We shall take it for granted that the reader does not require to have proven to him the so-called “germ theory” of disease. The phrase, which was once accurate, is now misleading, for the germ “theory” is part of the definite achievement of science. Not only have we succeeded in isolating the specific germ whose introduction into the body is responsible for different diseases, but in many cases, by studying the history and behavior of the germ, we have been able to find methods of checking its inroads, and so have delivered men from scourges like yellow fever and the bubonic plague.

THE DEFENSES OF THE BODY

An experiment that is often tried in operating rooms furnishes a vivid illustration of the omnipresence of these invisible, yet potent, foes of life. In order to impress upon young surgeons the importance of maintaining antiseptic conditions, they are instructed to thoroughly wash their hands and arms in antiseptic soap and water; then they are told to leave their arms exposed for a few minutes, after which a microscopic examination of the bared skin will result in exposing the presence of myriads of germs. Many of these are, of course, harmless; some are even “friendly”—since they make war upon the dangerous kinds. But others are the deadly organisms which find lodgment in the lungs and cause pneumonia and tuberculosis; or the thirty odd varieties of bacilli which cause the various kinds of grippe and influenza and “colds,” which plague the civilized man; or others which, finding entrance into the digestive tract, are the cause of typhoid and other deadly fevers.

So it appears that we live within our bodies somewhat in the same fashion as isolated barons lived in their castles in the Dark Ages, beleaguered constantly by hordes of enemies that are bent upon our destruction—these being billions upon billions of disease germs. Every portion of the body has its defenses to protect it against these swarms. The skin is germ-tight in health; and each of the gateways to the interior of the body has its own peculiar guard—tears, wax, mucous membrane, etc. As Dr. Edward A. Ayers points out,—“Many of these entrances are lined with

out-sweeping brooms—fine hairs similar to the ‘nap’ or ‘pile’ of carpet or plush—which constantly sweep back and forth like wheat stalks waving in the breeze. You cannot see them with the low-powered eye, but neither can you see the germs. They sweep the mucous from lungs and throat, and try to keep the ventilators free from dust and germs. Behind the scurf wall and the broom brigade of the mucous membranes, the soldier corpuscles of the blood march around the entire fortress every twenty-eight seconds” (the time occupied by the blood in its circulation through the body).

HEALTHY BODIES ARE GERM-PROOF

And again (to quote another authority, Dr. Sadler), “All the fluids and secretions of the body are more or less germicidal. The saliva, being alkaline, discourages the growth of germs requiring an acid medium. The normal gastric juice of a healthy stomach is a sure germ-killer. In the early part of digestion, lactic acid is present, and there soon appears the powerful hydrochloric acid, which is a most efficient germicide....

“The living, healthy tissues of the body are all more or less germicidal; that is, they are endowed with certain protective properties against germs and disease. This is true of many of the other special secretions, like those found in the eye and elsewhere in the body, when they are normal. The blood and lymph, the two great circulating fluids of the body, are likewise germicidal. In some conditions of disease, there may be found various substances in the blood which can destroy germs.”

THE WHITE CELLS ON GUARD

And this definitely brings us to the other kind of inhabitants of the human blood, the leucocytes, or white blood corpuscles,—and so to the germ theory of health, which science is showing to be no less true than the germ theory of disease. In their natural state these cells are transparent, spherical forms of the consistency of jelly drops, which float in the bloodstreams or creep along the inner surface of the vessel. Their function was for a long time not understood; the discovery of the real facts, perhaps the most epoch-making discovery ever made concerning the human body, the world owes to the genius of Metchnikoff, the head of the Pasteur Institute of Paris.

These cells are the last reserves of the body in its defense against the assault of disease. Whenever, in spite of all opposition, the hostile germs find access either to the blood or to the tissue, the white cells rush to the spot, and fall upon them and devour them.

In their fight against the hordes of evil bacteria that invade the blood, where the battles are waged, the body's defenders have four main ways of battling. Again we quote from Dr. Ayers: "The blood covers some germs with a sticky paste, and makes them adhere to one another, thereby anchoring them so that they become as helpless as flies on fly paper. The paste comes from the liquid of the blood, the plasma. Another blood-weapon (the 'lysins') dissolves the germs as lye does. A third means of defense is the ability of the white blood corpuscles to envelop and digest the living germs. One white cell can digest dozens of germs, but it may mean death to the devouring cells.

The fourth and recently discovered weapon, or ammunition, of the blood is the opsonins. Wright and Douglas in London in 1903 coined the word, which comes from the Latin *opsono*: "I cook for the table," "I prepare pabulum for." This is precisely what the opsonins do in the blood. They manifest this beneficial activity when invading disease germs appear. They attract white blood cells to the germs and make the bacteria more eatable for the cells. They are appetizers for the white blood cells; or sauces, which help the white blood cells to eat more of the bacteria than they could do without this spur to their hunger. Wright and Douglas demonstrated beyond peradventure the ability of the white blood cells to eat a larger number of bacteria when the latter are soaked in opsonins. They also showed that this opsonic sauce, or appetizer, which stimulates the blessed hunger of the white blood cells for disease bacteria, could be artificially produced, and hypodermically introduced into a patient's blood, thus increasing that blood's power of defense by raising the quantity of opsonins. They also worked out a practical laboratory technique by means of which the opsonins can be measured, or counted, with a considerable degree of exactitude, thereby making it possible to estimate within limits of accuracy any one's ability to resist bacterial invasions. If the blood is rich in opsonins, its power to fight disease is strong. Opsonins are now inoculated into the blood at several institutions, notably McGill University in Montreal, and at the Battle Creek Sanitarium.

HOW THE WHITE CELLS DO THEIR WORK

The process by which the white cells fight for us may be watched in the transparent tissue of a frog's foot or the wing of a bat. If a few disease germs are introduced into this tissue, the white cells may be seen to accumulate on the wall of the blood vessel just opposite where the germs have entered. "Each cell begins to push out a minute thread of its tissue," writes Dr. Kellogg, in describing the process, "thrusting it through the wall of its own blood vessel. Little by little the farther end of this delicate filament which has been pushed through the wall grows larger and larger, while the portion of the cell within the vessel lessens, and after a little time each cell is found outside the vessel, and yet no openings are left behind. Just how they accomplish this without leaving a gap behind them is one of the mysteries for which Science has for many years in vain sought a solution. The vessel wall remains as perfect as it was before. Apparently, each cell has made a minute opening and has then tucked itself through, as one might tuck a pocket handkerchief through a ring, invisibly closing up behind itself the opening made. Once outside the vessel, these wonderful body-defenders, moving here and there, quickly discover the germs and proceed at once to swallow them. If the germs are few in number, they may be in this way destroyed, for the white cells not only swallow germs, but digest them. If the number is very great, however, the cells sacrifice themselves in the effort to destroy the germs, taking in a larger number than they are able to digest and destroy. When this occurs, the germs continue to grow; more white cells make their way out of the blood vessels, and a fierce and often long-continued battle is waged between the living blood cells and the invading germs."

Now, it must be understood that this description is not the product of any one's imagination, but is a definitely established fact which has been studied by scientists all over the world. Because of the importance of the discovery, and of the new views of health to which it leads, we have placed a picture of this "battle of the blood" at the front of this book. It shows the leucocytes of the human body in conflict with the germs of influenza: the black dots being the germs, and the larger grayish bodies the leucocytes. We have chosen a photograph rather than a drawing, so that the reader may realize that he is seeing something which actually has existence. We request

him to study the picture and fix it upon his mind, for it is not too much to say that from it is derived every principle of health which is set forth in the course of this book.

THE PROBLEM OF HEALTH

The human body is a complex and intricate organism, in some wonderful and entirely incomprehensible way integrating the activities of all these billions of other living organisms. Each and every one of these latter has its function to fulfill, and the life of the individual body is a life of health so long as the unity of all its organisms is maintained. Outside of the body are millions of hostile organisms assaulting it continuously; and the problem of health is the problem of enabling it to make headway against its enemies for as long a period as possible. Every act of a human being has its effect upon this battle; at every moment of your life you are either strengthening the power of your own organism or strengthening your enemies. Once the organism is unable to beat back its enemies, health begins to fail and death and complete disintegration is the ultimate result.

It must be understood that the peril of these hostile germs is not merely that they devour the substance upon which the body's own organisms have to be nourished. If that were all, they might remain in the body as parasites, and by taking additional nourishment a man might sustain life in spite of them. Nor is it even that they multiply with such enormous rapidity; the peril is that they throw off as the products of their own activity a number of poisons, which are as deadly to the human body as any known. These poisons are produced much more rapidly than they can be eliminated from the system, and so they fill the blood, and death ensues.

Thus the problem becomes clear. In the first place, what can we do to keep disease germs from securing entrance to the body; and second, what can we do to strengthen the body's army of defense so that the fate of any which do find entrance may be immediate destruction?

HEALTH, LIKE DISEASE, IS CATCHING

In actual practice it is found that the second problem is by far the more important one. Some germs we can avoid. If we boil all the water that we

drink we will not be very apt to have typhoid. If we exterminate rats and mosquitoes and flies and fleas, we will not have yellow fever, or malaria, or plague. But we cannot hope to do this at present in the case of such diseases as, for instance, consumption, grippe, and influenza. If we live in a city, we take into our lungs and throat millions of the germs of these diseases every day. Therefore the one hope that is left is to keep ourselves in such a condition of health that the army of our bodies shall be able to destroy these germs. When the blood is in a healthy condition, the white cells are numerous, powerful, and active, but when the blood flows stagnantly, or when it is impoverished, then the white cells are few and the forces of disease obtain a foothold.

Healthy men can go through many epidemics with impunity. Because the Japanese army was an army of healthy men, its death rate from those diseases which usually follow in the wake of all armies was lower than the world had ever known before. Robert Ingersoll once said that if he had been God and had made the world, he would have made health “catching,” and not disease. As a matter of fact, health is catching. It abounds in the very air we breathe, in the water we drink, in the movements of every muscle and the play of every fibre and nerve of the body; it comes from and is nourished by each and every one of the bodily actions and functions; while disease is only secured by persistent transgressions of the proper way of living, and by injurious habits and customs that result in lowering the “vital resistance.”

This vital resistance is the innate power of the body to keep itself strong; its very lifeforce. This is what we mean when we say that this or that person has “a good constitution,” or has “a weak constitution.” This is the capital in the bank of each individual life, placed there by Nature at the birth of that life, and increased or diminished by each and every action of our bodies, and also of our minds. As Rokitansky, the eminent German scientist, said, “Nature heals. This is the first and greatest law of therapeutics—one which we must never forget. Nature creates and maintains, therefore she must be able to heal.”

Many of the most notable discoveries and experiments of modern science concur in demonstrating that the natural and innate healing power of the body is man’s greatest resource in combating disease and maintaining

health. It is the body itself which cures the sick man; his own vitality, and not the drug or medicants which he may take. These may assist the healing process, but they do not set going the healing processes themselves. More often, indeed, they are distinct detriments. They stamp out or banish the distressing symptoms of ailments, and thus in effect they silence the signal bells of danger which the body rings at the approach of disease.

Modern science has turned its forces upon this question of maintaining at its highest potentiality the ability of the body to resist disease. All the habits of the human race have been investigated in the light of this idea, and some have been found to be wise and others to be unwise. These conclusions, with the evidence therefor, are the subjects of our book.

OUR FOOD IS THE CHIEF FACTOR

It has been found that the most important problems connected with health are those of nutrition—the questions of what and when and how and how much food we ought to eat.

Every language under the sun contains a prayer somewhat similar to that which we have in the Anglo-Saxon tongue, “Give us this day our daily bread.” If we stop to think for a moment, we realize that next to the air we breathe, and the water we drink, our food is the most important consideration in the maintenance of life. All this is the veriest commonplace; yet the fact remains that it is very rarely indeed that we do stop to think upon the subject of our food. It is something that we take for granted, like life itself. In the regular routine of our days our meals become fixed habits, and the taking of food an almost involuntary custom. It requires some extraordinary event to arouse us to a just appreciation of the importance of knowledge on this subject. Or else the coming of one of the myriad forms of digestive diseases will serve the purpose of introducing the subject to our notice.

Our blood is made directly from what we eat, and that old Saxon proverb is true which says that every man has lain in his own trencher. Man is his food. Each human body is made by chemical action from its food. All our actions and all our thoughts come from what we eat, even as the movements of machinery proceed from the coal fed into the boilers of the engine which

operate the machine. If we eat the right food, namely, the food which contains the elements our bodies require in the proper proportions, we repair all waste, replace broken down tissue and supply ourselves with physical and mental energy for our toils and joys in life; while if we eat the wrong foods we quickly injure our delicate though powerful physical and mental machinery.

All this would seem to be obvious; yet most people would grant that they have still much to learn concerning what really constitutes the best foods, and about the best ways of preparing, or making, or using those foods. Few of us possess anything more definite to guide us in our eating than the habits we acquired as children, or habits picked up in later life from following the example of our friends, or the food fashions of the day—for there are such things as fashions in foods and in the eating of foods, even as there are fashions in clothes and the making and wearing thereof. In this place it is proposed to study the subject of food from one standpoint, namely, its effect upon the Battle of the Blood; its relation to the vital resistance of the body whereby health is maintained.

II

HOW TO EAT: THE GOSPEL OF DIETETICS ACCORDING TO HORACE FLETCHER

We shall first of all see what modern science has to tell us concerning the question of *how* we ought to eat.

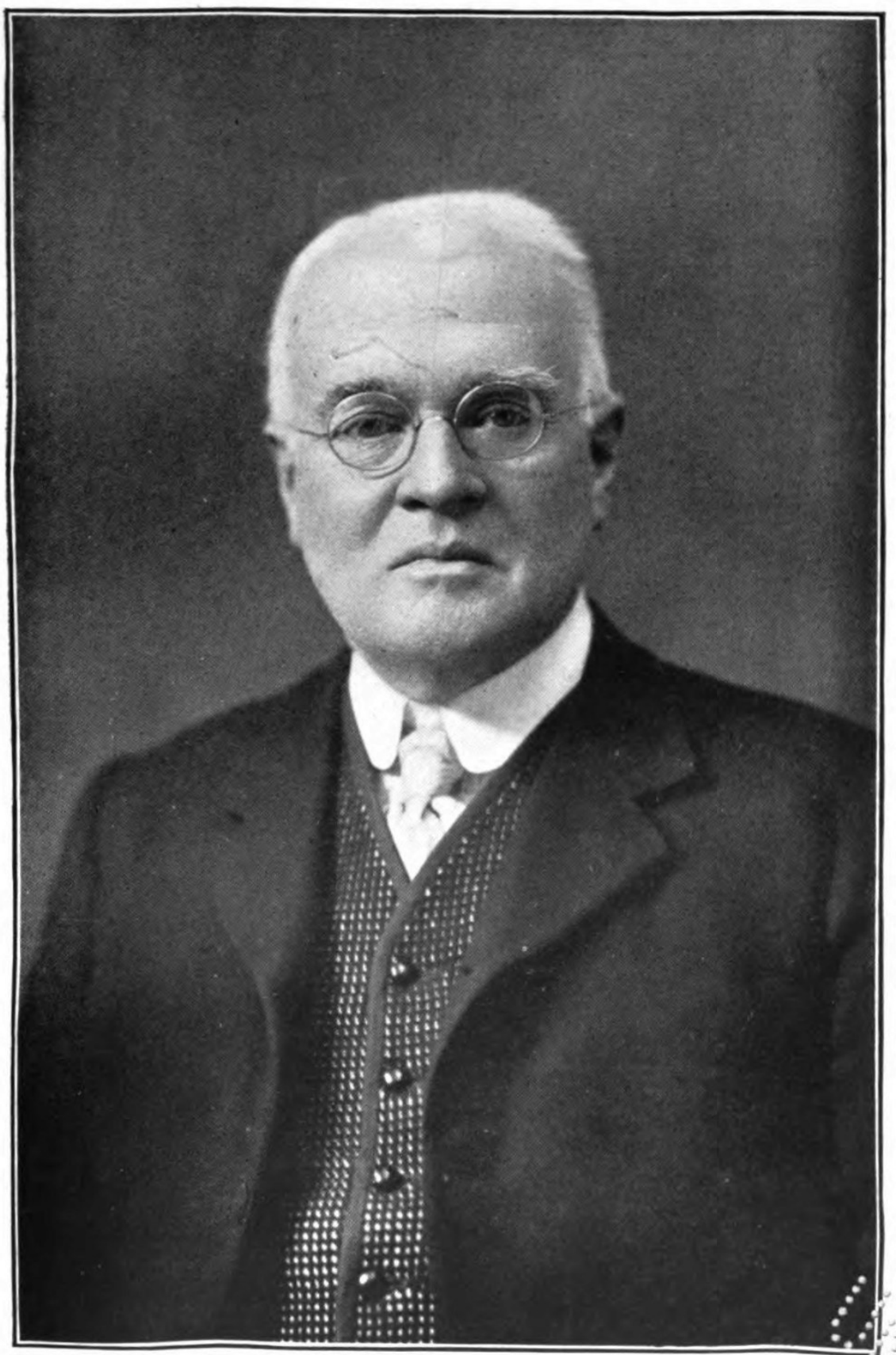
It may not seem possible that anything essential remains to be said at this late day on the subject of one of the commonest and decidedly most necessary of all human acts. That there should be knowledge of the utmost importance to learn regarding the actions and movements of the tongue, the teeth, and the jaws, may come with as much surprise to the majority of our readers as it did to us when we first hit upon this disturbing, but illuminating, fact.

The act of eating is the starting point of the long series of processes whereby our bodies are nourished. It is the only act of them all which lies within our control. We can directly supervise the work of our mouths; we can watch over the action of the teeth, and tongue, and palate; but we can not supervise the work of the stomach, or of the intestinal tube. Once we have swallowed our food, our mastery over it has ceased—except for some hit-or-miss participation in the further processes of its digestion by means of pills or potions. Realizing this, we come to recognize the basic importance of knowing the right way of eating.

THE STORY OF HORACE FLETCHER

This knowledge the world owes to Horace Fletcher, the American business man who has made many of the greatest physiologists of our times embark upon years-long series of experiments and inquiries into the problems of man's nutrition. As a result, the text-books of physiology are now being rewritten; and as a further result, tens of thousands of men and women, among them some of the best known authors, physicians,

clergymen, military men, and business men of both Europe and America, have been restored to health by the knowledge of how to eat their food.



MR. HORACE FLETCHER,
Whose books on dietetics and good health were the forerunners of the present movement.

This knowledge Mr. Fletcher gained at the very door of death, and in no more interesting and striking fashion could the importance of it be shown than by the relation of his remarkable case.

At the age of forty-five, after a varied and adventurous career, as miner, and explorer, and sailor, and hunter, Mr. Fletcher had won wealth, and retired from his business in order to devote himself to long-cherished interests in art and philosophy. He was still comparatively young, he was a member of many clubs, he had warm friends in all the capitals and countrysides of the world (Mr. Fletcher being one of the most untiring of globe-trotters), and in all ways except one he was equipped and ready for a long life of ease and enjoyment.

The one way in which he was not equipped was—in health.

HOW A STRONG MAN BROKE DOWN

Once he had been a man of robust physique, a champion gymnast and athlete; he had been president of the far-famed Olympic Club in San Francisco (which he founded, and where the pugilist Corbett was discovered), and had won plaudits even from famous professionals for his prowess with the gloves.

But he had overdrawn his account at the bank of life. He had expended more vital resistance than he had stored up; to such an extent, indeed, that when Mr. Fletcher went to the insurance companies at the time he retired from business he was rejected by them all; he was obese; he was suffering from three chronic diseases, and he was dying fast. Such was the verdict given by the skilled and experienced medical examiners of the life insurance companies. And instead of entering upon a long life of ease and enjoyment, he was thus condemned, seemingly, to a short life of invalidism and suffering.

FIGHTING FOR LIFE

But Mr. Fletcher declined to accept any such decision as that. He decided that he would regain his health—not that he would *try* to regain his health, but that he *would* regain his health.

He first turned to the physicians. Possessed of wealth, he was able to secure the services of many of the most able specialists of the world. He visited the most celebrated “cures” and “springs” and sanitariums of Europe and America. Nothing availed. He found passing relief now and then, but no permanent good. He gained no health, in other words, but obtained merely temporary abatement of this or of that disease.

Then he turned to himself. He began the study of his own case. As he attributed most of his bodily woes to faulty habits of eating, the subject of nutrition became uppermost in his studies. He was, coincidentally, deeply immersed and interested in the study of practical philosophy; and in a very remarkable fashion these two subjects, these two interests, nutrition and practical philosophy, became fused into one subject, supplementing and completing each other and jointly forming the burden of the message of Hope, of the tidings of great joy, which it became the mission of Horace Fletcher to deliver to mankind.

MR. FLETCHER’S DISCOVERY

He discovered, or rather rediscovered, and applied, two great and simple truths:

First, that the complete chewing of all food, both liquid and solid, whereby a process of involuntary swallowing is established, foods being selected in accordance with individual tastes, is by far the most important and most necessary part of human nutrition. It is the key that unlocks the door of health, and opens the way to the real hygienic life.

Second, that nothing poisons the body, and aids the forces of disease, more than worry—which Mr. Fletcher has named Fearthought. It is our nature to look forward, to anticipate. We can anticipate in two ways—anticipate evil, or anticipate good. The first way is to use fearthought; the second way is to use forethought. Forethought will produce cheerfulness and health, even as unspoiled rose seeds will produce roses. Fearthought

will produce disease and trouble, even as the germs of putrefaction will produce sickness and death.

So great an authority in philosophy and psychology as William James has given the sanction of his use to Mr. Fletcher's phrases; and has also named him as a shining example of those exceptional men who find in some mental idea a key to unlock reservoirs of hidden and unsuspected energy. While there is no doubting the fact that Horace Fletcher is decidedly an exceptional man, yet the records prove that his key is not merely for the use of exceptional people, but that it is one susceptible of being used by everybody possessing willpower enough to enable them to say "yes" when offered something good.

Like other great discoveries, Mr. Fletcher's discovery of the right way to eat came partly as an accident. Happening to be in Chicago at a time when his friends were all away, and being forced to stay in the city, he took to lingering over his meals in order to pass away the time. He began to taste every spoonful of soup, to sip every mouthful of anything liquid, with great deliberation, noting the different tastes and searching out new flavors.

He chewed each morsel of meat or bread or fruit or vegetable until, instead of being gulped down, it was drawn in easily by the throat. And in this manner did he stumble upon his pathway to deliverance. He had not been "toying" with his food—as he then considered he was doing—for more than a few weeks before he noticed that he was losing a great deal of superfluous fat, that he was eating less, but with far greater enjoyment, than ever before in his life, that his taste for simpler foods increased as his taste for highly seasoned and complex dishes decreased, and that he was feeling better both physically and mentally than he had felt in many years.

THE MAGIC OF MASTICATION

What did these things mean? Some hidden virtue in the food he was eating? Some hitherto quite unsuspected tonic in the smoke of Chicago? Or a lesson in health furnished by the "how" of his eating? At this point there flashed through Mr. Fletcher's memory the story of Gladstone's advice to his children to chew each morsel of food thirty-two times (once for each tooth in their heads) if they would preserve their health. In that moment,

Mr. Fletcher began his investigation of the many processes that go to make up the simple act of mastication, an investigation which has now been going on for more than ten years, and which has resulted in directing public attention to the supremely important subject of nutrition with more emphasis, and in the arousing of more general interest and the production of more telling effect than any other circumstance or event has done in the history of physiologic science. The word "Fletcherizing" was first applied by Dr. J. H. Kellogg, of Battle Creek, after the analogy of "pasteurizing," in describing the act of mastication as recommended by Mr. Fletcher. "Fletcherism," as Mr. Fletcher's system of mental science and of physical culture through mastication has come to be known, after first being for years a stock jest of the newspaper funnyman, has now been recognized, even by those scientists who detest all "isms," as a most valuable bridge from the land of bad food habits and disease to the land of good food habits and health.

The bridge certainly afforded its builder a passage from one region to the other. Following a constant improvement in his general condition, beginning almost simultaneously with the adoption of his new way of life, Mr. Fletcher is to-day one of the strongest and most enduring men alive. Tests of his strength and endurance made at the Yale gymnasium at different times prove beyond a doubt that this is so. The following is a quotation from the report of Dr. William G. Anderson, director of the Yale Gymnasium:

DR. ANDERSON'S REPORT

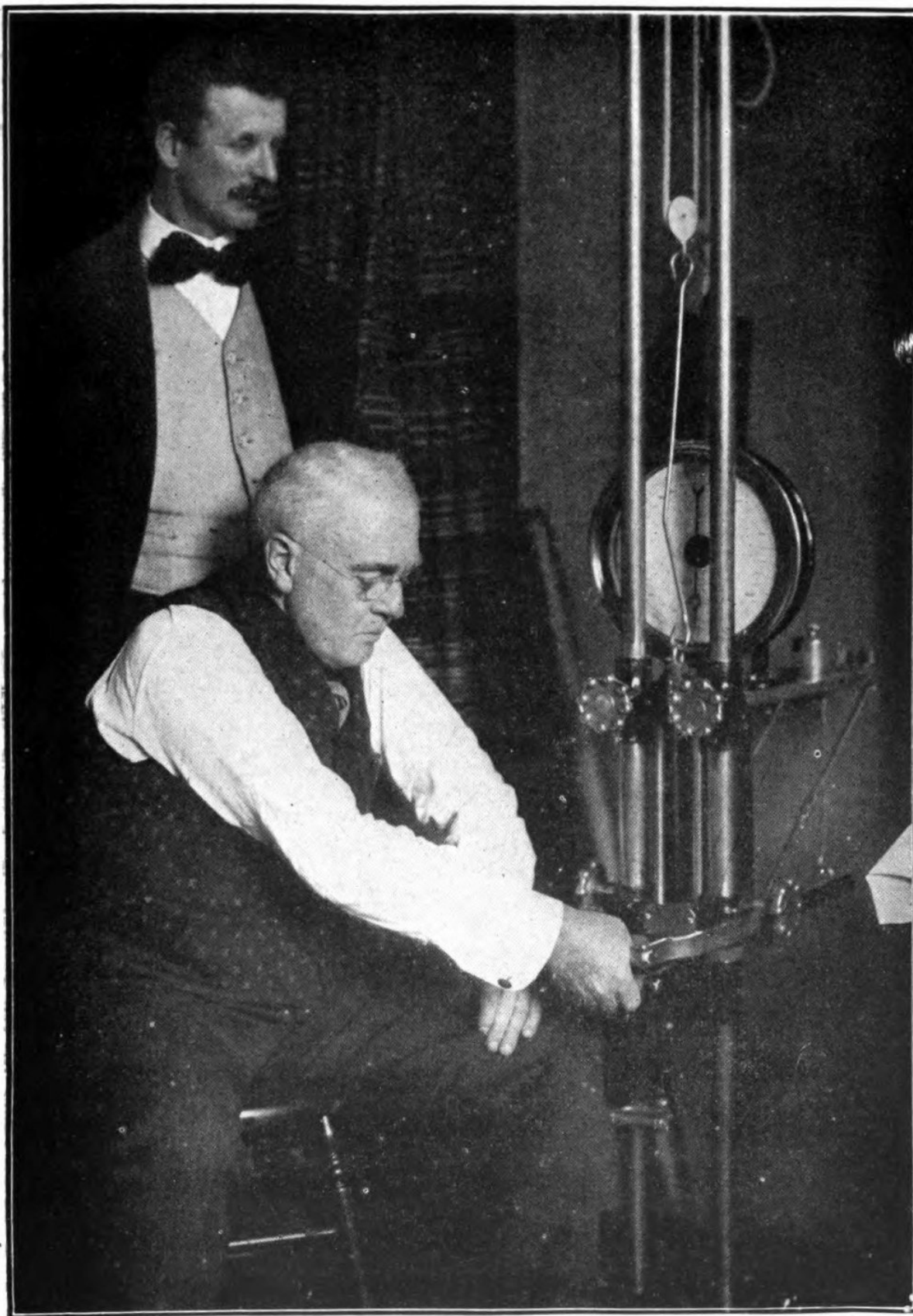
"In February, 1903, I gave to Mr. Horace Fletcher the exercises used by the 'Varsity' crew. He went through these movements with ease and showed no ill effects afterwards. At that time Mr. Fletcher weighed 157½ pounds, and was in his fifty-fifth year. On June 11, 1907, Mr. Fletcher again visited the Yale Gymnasium and underwent a test on Professor Fisher's dynamometer. This device is made to test the endurance of the calf muscles.

"The subject makes a dead lift of a prescribed weight as many times as possible. In order to select a definite weight, the subject first ascertains his strength on the Kellogg mercurial dynamometer by one strong, steady contraction of the muscles named—and then he finds his endurance by

lifting three-fourths of this weight on the Fisher dynamometer as many times as possible at two or three second intervals. One leg only is used in the lift, and as indicated, the right is usually chosen.

“Mr. Fletcher’s actual strength as indicated on the Kellogg machine was not quite four hundred pounds, ascertained by three trials. In his endurance test on the Fisher machine he raised three hundred pounds three hundred and fifty times and then did not reach the limit of his power.

“Previous to this time, Dr. Frank Born, the medical assistant at the Gymnasium, had collected data from eighteen Yale students, most of whom were trained athletes or gymnasts. The average record of these men was 87.4 lifts, the extremes being 33 and 175 lifts.



MR. HORACE FLETCHER

Making a World's Record on the Dynamometer without previous training. Dr. William G. Anderson, Director of the Yale Gymnasium, in the Background.

“You will notice that Mr. Fletcher *doubled* the best record made previous to his feat, and numerous subsequent tests failed to increase the average of Mr. Fletcher's competitors. Mr. Fletcher informs me that he had done no training nor had he taken any strenuous exercise since February, 1907. On two occasions only during the past year he reports having done hard work in emergencies; once while following Major-General Wood in the Philippines in climbing a volcanic mountain through a tropical jungle on an island near Mindanao for nine hours; and once wading through deep snow in the Himalayan Mountains, some three miles one day and seven miles the next day, in about as many hours. This last emergency experience came through being caught in a blizzard near Murree, in Northern India, at 8500 feet elevation, on the way to the vale of Kashmir. These two trials represented climatic extremes, and Mr. Fletcher states that neither the heat nor the cold gave him discomfort, a significant fact in estimating physical condition.

“Before the trial on the Fisher machine, the subject's pulse was normal (about 72); afterwards it ran 120 beats to the minute. Five minutes later it had fallen to 112. No later reading was taken that day.

“The hands did not tremble more than usual under resting conditions, as Mr. Fletcher was able to hold in either hand immediately after the test a glass brimming with water without spilling a drop. The face was flushed, perspiration moderate, heart action regular and control of the right foot and leg used in the test normal immediately following the feat. I consider this a remarkable showing for a man in his fifty-ninth year; 5 feet, 6½ inches in height, weighing 177½ pounds and not in training.”

In order to make a more thorough test of Mr. Fletcher's power of endurance under varying degrees of physical strain, he underwent on the 17th, 18th, 19th, 21st and 22nd of June, 1907, a number of other exceedingly severe tests, of which Dr. Anderson says: “After each test the respiration and heart action, while active, were healthy, and, under such conditions, normal.

“There was not the slightest evidence of soreness, stiffness or muscular fatigue either during or after the six days of the trials. Mr. Fletcher made no apparent effort to conceal any evidence of strain or overwork and did not show any. He informs me that he felt no distress whatever at any time. Should any one wish to become more familiar with the strenuousness of the movements selected, let him try them. The effort will be more convincing than any report.

“During the thirty-five years of my own experience in physical training and teaching, I have never tested a man who equalled Mr. Fletcher’s record.

“The later tests, given in June, 1907, were more taxing than those given in 1903, but Mr. Fletcher underwent the trials with more apparent ease than he did four years ago.

“What seems to me to be the most remarkable feature of Mr. Fletcher’s test is that a man nearing sixty years of age should show progressive improvement of muscular quality merely as the result of dietetic care and with no systematic physical training. The method of dietetic care, too, as given by Mr. Fletcher, is so unusual that the results seem all the more extraordinary. He tells me that during the four and a half years intervening between the first and the recent examinations he has been guided in his choice of foods and in the quality also, entirely by his appetite, avoiding as much as possible any preconceived ideas as to the values of different foods or the proportions of the chemical constituents of the nourishment taken.

“During this four year period he has more than ever catered to his body nourishment in subservience to instinctive demand. He has especially avoided eating until appetite has strongly demanded food, and has abstained from eating whenever he could not do so in comfort and enjoyment. Mastication of solid food and sipping of liquids having taste to the point of involuntary swallowing, according to his well-known theory of thoroughness in this regard, has also been faithfully followed.

“There is a pretty good evidence that taking food as Mr. Fletcher practices and recommends limits the amount ingested to the bodily need of the moment and of the day, leaving little or no excess material to be disposed of by bacterial agency. This might account for the absence of toxic products in the circulation to depress the tissue.

“The possible immunity from lasting fatigue and from any muscular soreness, resulting from the unaccustomed use, and even the severe use, of untrained muscles is of utmost importance to physical efficiency.

“My own personal observance and trial of Mr. Fletcher’s method of attaining his surprising efficiency, strengthened by my observation of the test-subjects of Professors Chittenden and Fisher who have come under my care meantime, lead me to endorse the method as not only practical but agreeable. As Mr. Fletcher states, both the mental and mechanical factors in selecting and ingesting food are important, the natural result of the care being a wealth of energy for expression in physical exercise.”

FLETCHERISM

So much for Horace Fletcher’s own case.

Yet when he first announced his discovery, his own family laughed at him, and the medical world called him crank. But by quiet, sane, persistent work—by applying to the propaganda of his idea the same methods that had brought him success in business, he succeeded in impressing the scientific world with the value of his method.

An extensive literature has grown up around Mr. Fletcher’s own books. The most important medical bodies in Europe and America have invited him to lecture before them. Hospitals in larger cities have printed his own code of the rules of mastication for distribution. And no large sheet of paper was required, for the whole system could be printed on a postal card, and room would be left for a picture of its author.

Why is complete mastication the best way of eating? Why does its practice lead to recovery of lost health, or increase of health; to increase of strength, to increase of endurance. Is it not a very tedious method, and thus of more trouble than its promised benefits are worth? Does it not waste time? Does it not lead to loss of enjoyment of food?

These are a few of the questions which a discussion of Fletcherism invariably arouses. We speak with a deep conviction of truth when we say that Fletcherism leads to saving of time, instead of loss of time; that it brings increase of sensuous enjoyment of food instead of decrease of it; and

that if it is tedious or a bore, then it is not Fletcherizing. The very essence of Fletcherism is the dropping of worry, the elimination of stress and strain. If you do as Fletcher says, instead of doing as somebody says that Fletcher says, you will chew for taste, and not for time; you will take a crust of bread, or a morsel of potato, for instance, into your mouth and roll it with your tongue, and press it against the roof of your mouth, and pass it to and fro, and crunch it, and crush it; and all the while you will not be counting the chews, nor even thinking about chewing, but on the contrary you will be thinking of the taste of the morsel, and seeking that taste—and finding it.

Yes, finding it, even in a crust of bread or in a morsel of potato, in those humble foods which the most of us seem to take more as matters of habit; for by giving the saliva in the mouth a chance to fulfill the work for which it is put in our mouths by nature, we find that the starch in the bread and in the potato is turned into a sweet, toothsome and partly digested morsel of sugar.

Here is a point that answers another of the questions which arose a paragraph or so back. This turning of the starch in bread into sugar by the action of saliva is only one of the numerous acts of digestion which is accomplished in the mouth by the teeth, the tongue, the palate, and the various kinds of juices, or saliva, which are in the mouth. Horace Fletcher pointed out, and medical science now confirms his assertions, that many of the most important parts of the digestive process are meant by nature to be carried out in the first three inches of the alimentary canal. And this is the only place in all the thirty feet or so of the alimentary canal where digestion is in our own control. If we bolt or insufficiently masticate our food, these mouth processes of digestion are simply not accomplished; and for this the whole system suffers sooner or later. The stomach and the intestines are called on to do a great deal of extra work, and much of this extra work is of a kind which they are unable to do. Consequently, what food can not be digested must decompose in the intestines, with the consequent production of poisonous fluids and gases which permeate the body. The whole machinery of digestion is thrown out of gear. All the various germs of disease race to be first to enter the disarranged mechanism, as criminals rush to a city that is in disorder. The blood not being as well nourished as it should be, the white army of the soldiers of the body begin to weaken and to die, and the forces of disease penetrate through their warding lines and

attack the fort of life from many sides, or else concentrate their strength in the form of some virulent sickness.

Thorough mastication, on the other hand, means the reverse of these conditions. Almost incredible seem the hundreds of stories which we personally know to be true of men and women who have used Mr. Fletcher's method as a means to enter the land of good health. In the opinion of Dr. Kellogg of Battle Creek, "There is no doubt that thorough mastication of food solves more therapeutic problems than any other thing that can be mentioned. It solves the whole question of the right combination of foods; solves the question of the quantity of foods, and the quality of foods, after one has got his appetite trained, his natural instinct trained; and when it comes to certain diseases like acidity of the stomach, hyper-acidity or hypo-acidity, dilation of the stomach or cirrhosis of the liver, or any other trouble with the digestive organs, if it does not effect a radical cure it makes it possible to tolerate a condition which otherwise would be deadly in a short time. It makes it possible for a patient to live a long time, enjoying comfortable health, where otherwise he would be crippled so that he could not live long at all."

Although we insist upon the fact that Fletcherism is simple, and easy, too, once you have really begun its proper use, yet we also know that there are many difficulties which the average man or woman has to face at the outset. Professor Fisher encountered these difficulties when experimenting with his students at Yale, and we are indebted to him for enumerating some of them. And these difficulties, like the habit of hasty eating itself, are products of our civilization.

We mean such difficulties as, first, *conventionality*, or the desire to eat what others eat, and the unwillingness to appear different; *politeness*, the desire to please one's host, or hostess, and eat "what's set before you," or to eat something which you know you don't want or which you know is bad for you, because you fear to offend somebody or other who has cooked it, or bought it for you; *food notions*, or the opinion that certain foods are "wholesome," and that certain foods should be avoided as injurious even if delicious to the taste; *narrowness of choice*, as at a boarding house table (and a great number of home tables!) which often supplies what is not wanted and withholds what is; and, lastly, habit, by which the particular

kinds and amounts of food which have become customary through the action and interaction of the causes previously named, are repeated day after day, without thought.

“Habit hunger” is another of our handicaps. Habit hunger is said by Mr. Fletcher to be responsible for a vast deal of overeating. He refers to the fact that when we are children we eat at least one-third more proteid or tissue-building foods, in proportion to our size, than we require as adults, for the reason that our growing frames must then be nourished and upbuilt; but when we reach the adult stage we are apt to maintain this excessive consumption of proteid food—and proteid, as we shall see later on, is the chief source of dietary ills.

These are some of the difficulties to be encountered by the person who sets out upon the road to health. But they are very slight barriers, indeed, to the person possessed of willpower, and when the benefits and pleasures to be gained are so enormously in excess of the few initiatory troubles, it is not to be wondered at that more than a million persons in England and America are already following Horace Fletcher’s system in whole or in part.

HOW CHEWING STIMULATES DIGESTION

Certain remarkable experiments conducted by Rogers, Metchnikoff, and Pawlow in Europe, and by Cannon and Kellogg in America, have thrown a new and interesting light upon the ideas of Fletcher; proving that the act of chewing the food gives to the nerves that control the digestive fluids an opportunity to assay the food, to test it and select for it the particular kind of digestive fluid which that particular kind of food requires. It appears that there are many different kinds of saliva, and each one of these kinds has a particular kind of work to do, which no other kind is able to do. Metchnikoff has shown that if one takes cane sugar into the mouth with or without other food, there is manufactured by the salivary glands a certain peculiar fluid which digests cane sugar. If the cane sugar is not taken into the mouth, then that substance is not made. The saliva that flows into the mouth when there is food there but no cane sugar with the food, will not digest cane sugar. So it readily can be seen that if cane sugar should be hastily swallowed, it is much less likely to be properly digested. And this holds good with nearly all other kinds of food.

THE “FOOD FILTER”

“But how is a person to know when he has chewed a mouthful long enough?” the reader asks. Mr. Fletcher answers that nature has provided us with a food filter—an automatic safety device. Professor Hubert Higgins, formerly demonstrator of anatomy at Cambridge University in England, and Professor Hasheby of Brussels, Belgium, have lately conducted a series of experiments which throw light on this question on its scientific side. At the back of the tongue there are a number of little knobs, which are really taste buds, or apparatus for the tasting of food. During the time that mastication is going on, the mouth is closed and is completely air tight, and germproof. This fact one can readily demonstrate by filling out the lips with air. The mouth is full of air, yet one can breathe behind this curtain of air, showing that the mouth is thoroughly cut off. This is what happens during mastication, for of course one should masticate with the lips closed. Now, when the food has become sufficiently ensalivated, or mixed up, the circumvallate papillæ at the back of the throat, where the taste buds are, relax, and behind that the soft palate forms a negative pressure. This soft palate is muscled just as it is in the horse—which is an animal that masticates, but is not found in the dog, which is an animal that bolts its food. Whenever the food is ready for the body, the soft palate relaxes, and is sucked back, and the swallowing of a mouthful of the prepared food takes place involuntarily.

The body is thus supplied with as perfect a protection as could be devised, and perfectly automatic; all that is necessary being that one should masticate the food until it naturally disappears. One must not attempt to keep the food too long in the mouth, but let it have its own course. There are some sorts of food which, when one has chewed them three or four times, are sucked up, showing that they have received all the mouth treatment that nature requires they should. With other foods one can masticate up to one hundred and fifty times, and still they are not sucked up.

This food filter is a perfectly instinctive apparatus; but as people have acquired the habit of flavoring foods with artificial sauces and relishes, most of them have spoiled this protective device. In the words of Mr. Fletcher himself: “This is a gift of Nature to man which we have been

neglecting. It is not a gift which has been given to me and a few others alone. I think everybody could acquire the use of it if they would give Nature a chance by eating slowly, by eating with a sense of enjoyment, and by never eating save when they are really hungry and in a mood to enjoy the food.”

III

THE YALE EXPERIMENTS

At Yale University, Professor Russell H. Chittenden, Director of the Sheffield Scientific School, Lafayette B. Mendel, Professor of Physiological Chemistry, and Irving Fisher, Professor of Political Economy, have carried on a long series of experiments, begun six years ago as a test of the claims made by Fletcher. The net results of these experiments up to date (for they are still in progress) may be put into a nutshell. The following statement was drawn up by one of the writers of this book and submitted to Professors Chittenden and Fisher, who have accepted it as a summary of their present views:

“The commonly accepted standards which claim to tell the quantity of food needed each day by the average man are based upon many careful observations of what men actually do eat.

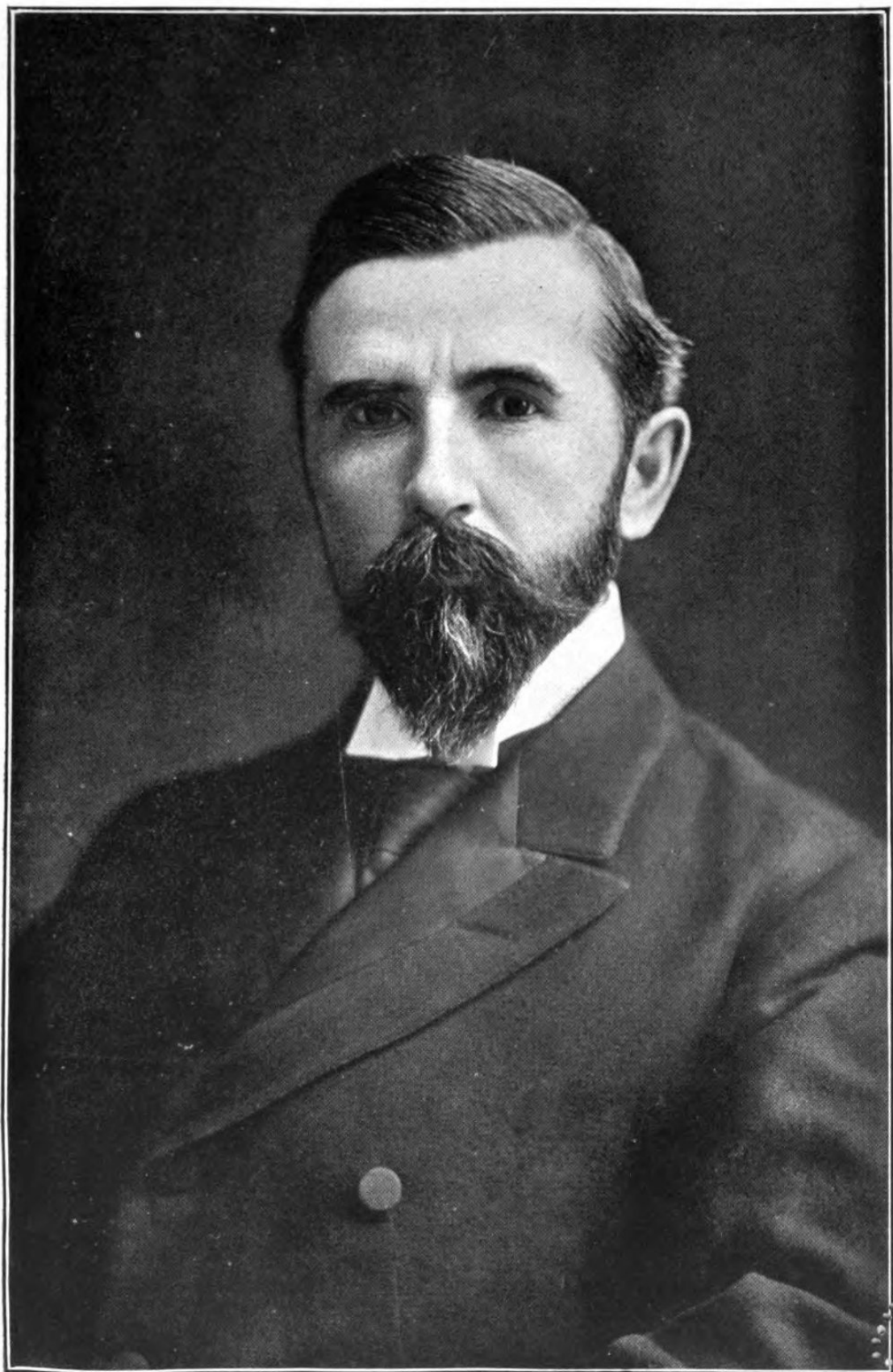
“We challenge these standards, however, as the exact science of to-day cannot accept as authority common customs and habits in any attempt to ascertain the right principles of man’s nutrition, since experiments have demonstrated how readily one set of habits may be substituted for another and how easily wrong habits become hardened into laws. The evidence presented by observers of common customs, while they must be duly considered, cannot, therefore, be taken as proof that these habits and customs are in accord with the true physiological needs of the body.

“We believe that the following propositions have been demonstrated as truths by the experiments we have made at Yale.

“People in general eat and drink too much.

“Especially do they eat too much meat, fish and eggs.

“This is so because meat, fish and eggs are the principal proteid-containing foodstuffs.



PROF. RUSSELL H. CHITTENDEN, PH.D., LL.D., SC.D.,
Director Sheffield Scientific School, Yale University. He has conducted many dietary
experiments from the physiologist's point of view.

“Proteid is an essential food element, absolutely necessary for the upbuilding of tissue, for the maintenance of life. It is one of three main elements into which all foodstuffs may be divided—the others being Carbohydrates (the sugars and starches) and Fat. While it is indispensable, it is also the element which the body machinery finds most difficult to dispose of. Proteid is ‘nitrogenous.’ Nitrogen is never wholly consumed in the body furnace as fats, sugars and starches are. There is always solid matter left unconsumed, like clinkers in a furnace; which clinkers the kidneys and liver have to labor to dispose of. If the clinkers are produced in excess of the ability of these organs to handle them without undue wear and tear, damage of a serious, and sometimes permanent, nature follows. The ideal amount of proteid is the amount which will give the body all of that substance which it needs without entailing excessive work upon the body machinery.

“Excessive consumption of proteid foodstuffs—like meat, fish and eggs—is the greatest evil affecting man’s nutrition. The excess of proteid not only remains unburned in the bodily furnace, but this waste matter very often decays in the body, forming a culture bed for germs which effect the whole system, a condition scientifically known as autointoxication, or self-poisoning of the body through the action of the germs of putrefaction, and of other germs, which are bred in the colon, or large intestine. The researches of Metchnikoff, Bouchard, Tissier, Combe, and other eminent scientists, have shown that autointoxication is the source of a great number of the most serious chronic diseases which afflict mankind.

“We say, then, that the existing dietary standards place in all cases the minimum of proteid necessary for the average man’s daily consumption at far too high a figure. It may be safely said that it is placed twice as high as careful and repeated experiments show to be really necessary.

“There can be little doubt that the habit of excessive eating and drinking, combined with the habit of too hasty eating and drinking, especially of meat, fish and eggs, are probably the most prolific sources of many bodily disabilities affecting men and women, and are consequently the greatest

deterrents to the attaining by men and women of a high grade of efficiency in work, of better health, of greater happiness, and of longer life.

“We believe that it has been demonstrated as a fact that health can be bettered, endurance increased, and life lengthened, by cutting down the commonly accepted standards of how much meat, eggs, fish and other proteid food we should eat and drink by about one-half.”

After Horace Fletcher had attracted the notice of the scientific world in 1902, Professor Chittenden invited him to become the subject of a series of experiments at Yale, where the Sheffield Scientific School possessed an equipment suitable for an elaborate inquiry of this kind much superior to any to be found in Europe.

FLETCHER’S CLAIMS SUPPORTED

Professor Chittenden first made certain, by experiments which precluded any chance of error, that Horace Fletcher’s claims were justified so far as Horace Fletcher himself was concerned. But this, of course, by no means solved the problem. Mr. Fletcher might simply be a physiological curiosity—a digestive freak—of whom there are many known cases. He lived and thrived on an amount of proteid food startlingly less than was deemed necessary by all existing standards, but this could not be taken as proof that people in general could do likewise. Only an exhaustive series of tests on a large number of people of varying ages and conditions of life could prove this. Professor Chittenden resolved to make these tests.

At the very outset, however, he faced this difficulty. If Mr. Fletcher’s was merely a freak case, there would be a grave danger in putting other men upon his dietary. Mr. Fletcher was flourishing on a daily consumption of proteid foodstuffs amounting to an average of only 45 grams, and the fat, sugar and starch consumed by him were in quantities only sufficient to bring the total food value of the daily food up to a little more than 1600 “calories,” or units of fuel energy. The Voit standard—which is the typical one, the one most commonly accepted, and which is based upon thousands

of studies of what men and women actually eat—demands that the average man shall eat at least 118 grams of proteid, with a total fuel value of 3000 large “calories” for the daily ration.

To make clear to the non-scientific reader just what quantity of foodstuffs is represented by 50 grams of proteid, which is 5 grams more than that consumed daily by Mr. Fletcher in his tests, and is approximately the amount consumed daily by other men in the Yale experiments, it may be said that 50 grams is about equal to 772 grains, which are equal to about $1\frac{3}{4}$ ounces. This quantity would be represented by the proteid contents of $9\frac{1}{2}$ ounces of lean meat, or 7 eggs, or 27 ounces of white bread. Nine and one-half ounces of meat (using comparisons furnished by Dr. Edward Curtis) is about the weight of a slice measuring 7 by 3 inches and cut $\frac{1}{4}$ of an inch thick. Twenty-seven ounces of bread represent somewhat less than two loaves, the standard loaf weighing one pound (16 ounces). Of course, few people ever eat 7 eggs, or 2 loaves of bread in a day; but the vast majority of people in America do eat a great deal more proteid than would be represented by 7 eggs, or 2 loaves of bread or a slice of meat of the size named, since proteid is found in a great number of other foodstuffs besides those mentioned.

CHITTENDEN'S EXPERIMENTS ON HIMSELF

Professor Chittenden realized that to ask a number of men to subsist on a ration similar to that which nourished Mr. Fletcher might possibly result in seriously weakening their constitutions. This is the problem which has often confronted other scientists, and Professor Chittenden solved it in a way characteristic of the true scientist—the devoted warrior in humanity's cause who wages warfare against the forces of evil. He began his experiments upon himself.

The result rewarded his self-sacrificing spirit; for within a few months a severe case of muscular rheumatism (which had plagued him for years, refusing to yield to treatment) disappeared; and with it went a recurrent bilious headache. And it may be stated that these have never returned. Professor Chittenden has adopted as a habit of life the dietary which he began as an experiment five years ago. At that time he was a hearty eater of three meals a day, meals rich in meat and other proteid foodstuffs.

THE OTHER CHITTENDEN TESTS

Professor Chittenden then began experiments with a group of university professors and instructors, with a group of thirteen enlisted men of the army, and a group of eight college athletes in training. All three of these groups of men were subjected to careful laboratory observations for continuous periods of many months, during which the proteid ration was reduced from one-half to one-third what had been customary. The professors and athletes followed their customary vocations during the period of observations, while to the ordinary drills of the soldiers were added severe gymnasium work under the supervision of Dr. Anderson.

Results were as follows: The subjects usually lost some weight, especially such as were fat. But it was found that having got down to a new standard, they held this steadily. They all maintained muscular and nervous vigor. Careful tests determined that the soldiers and athletes positively gained in muscular strength. All kept in good health; and many got rid of illnesses with which they had been suffering in the beginning. Appetite was thoroughly satisfied; and quite a number of the subjects permanently adopted the new method of living. Nine of the soldiers went in a body to a new station, and from thence they afterwards wrote, through one of their number, to Professor Chittenden, saying: "The men are in first-class condition as regards their physical condition, and all of them feeling well. We eat little meat now as a rule and would willingly go on another test."

At the beginning of the experiments these soldiers were subsisting on a daily ration which allows one and one-quarter pounds of meat per day apiece; and toward the end of the experiments they were subsisting and increasing their strength on a daily ration of meat equivalent to about one small chop or less!

These experiments constituted the first series made by Professor Chittenden. He later carried through a series with dogs: prior experiments having supported the view that the dog, a typical high proteid-consuming animal, declined or died when forced to subsist on quantities of proteid less than the amount ordinarily consumed. Professor Chittenden, however, challenged here the methods, as well as the results, of previous investigators. In previous experiments with dogs the animals had been

invariably handicapped by being confined in dark and dismal quarters, too cramped to permit of exercise, and at times unsanitary in condition. He reversed these conditions—and reversed the results. His dogs lived and thrived on a diet far less rich in proteid than former investigators deemed necessary.

PROFESSOR CHITTENDEN'S CONCLUSIONS

Summing up the conclusions reached by him after arduous years of experiment and study, Professor Chittenden declares that 60 grams of proteid (about the quantity which a single small chop would supply) are all that are required by the average man of 150 pounds body weight. This is one-half the Voit standard, and far below the common practices of the majority of mankind in Europe and America.

“But there should be no practical use of the terms ‘standard diets’ and ‘normal diets’ by people in general,” says Professor Chittenden. “What is needed to-day is not so much an acceptance of the view that man needs so many grams of proteid per kilogram of body weight, as a full appreciation of the general principle that the requirements of the body for proteid food are far less than the common customs of mankind, and that there are both economy and gain in following this principle in practice.”

HOW TO INCREASE ENDURANCE

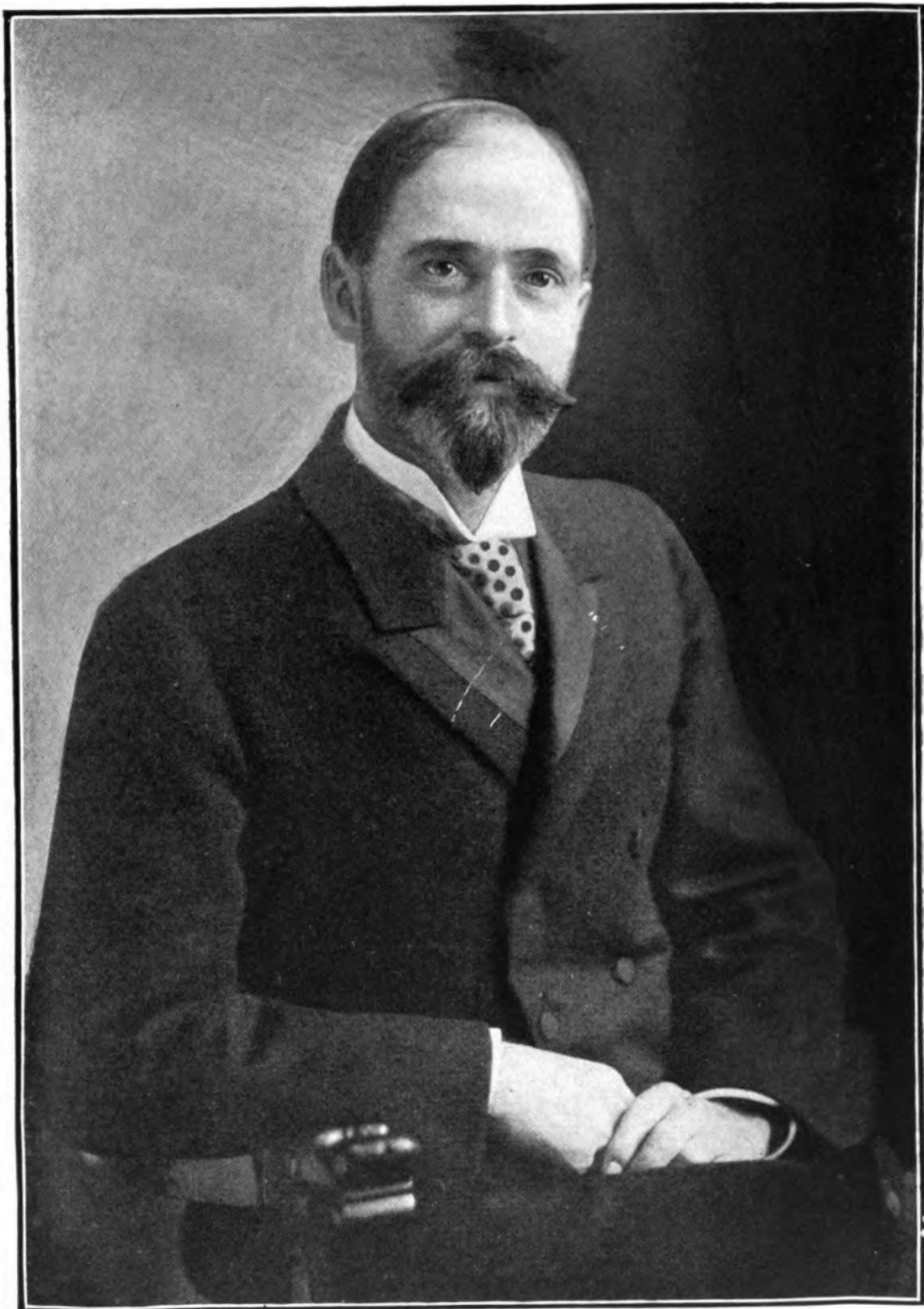
The most broadly interesting of these Yale food experiments are those having to do with the question of endurance. The vast majority of people are not ambitious to excel as athletes; they find better and more enjoyable forms of work in life than putting up big dumb-bells, or breaking records on the athletic field. Of course, everybody wants to be strong, and to have well-trained and active muscles; but on the whole, what the majority of people need is physical and mental stick-to-itiveness—the ability to work without deterioration, without running down like worn-out machinery. Professional men, day laborers, students and athletes, all need this invaluable quality of endurance—this quality that is the true capital in the bank of life to be at their command day in and day out, with a reserve ready

to be drawn upon whenever an emergency arises. And it is precisely here that the new art of health bestows its benefits upon those who follow it.

It was to ascertain the relation between diet and endurance in the light of the new knowledge shed upon the subject by Professor Chittenden's experiments, that Professor Irving Fisher inaugurated his own experiments at Yale University. He conducted two series of tests, as follows:

First, to ascertain the effect of thorough mastication on endurance, following the rules laid down by Horace Fletcher, with the help of nine healthy students.

Second, to ascertain the influence of flesh eating on endurance as compared with the effect of abstinence from flesh, with a group of forty-nine persons, splitting the group as follows,—first, athletes accustomed to a flesh, or high proteid dietary; second, athletes accustomed to a low proteid, or non-flesh dietary; third, sedentary persons accustomed to a low proteid, or non-flesh dietary.



PROF. IRVING FISHER, PH.D.,
Professor of Economics at Yale University. His investigations have had to do largely with the
cost of necessary food.

The flesh-eaters were Yale men, including some of the best known athletes of the university. The abstainers were nurses and physicians attached to the Battle Creek Sanitarium.

Professor Fisher's interest in the subject was that of a political economist. Meats, as a general rule, are the most expensive part of the national diet, and it is apparent that if a fleshless, or low proteid, diet will increase endurance, it will also increase the national earning capacity, and thus add to the national wealth. When Professor Fisher began his experiments he encountered a singular fact, which was that the science of physiology had given very little attention to the study of endurance. "That strength and endurance are not identical, is only partly recognized," he writes. "The strength of the muscle is measured by the utmost force that it can exert once; its endurance, by the number of times it can repeat any exertion within its strength. The repetition of such exertion, if not stopped by the refusal of the will, is finally stopped by the reduction of the strength of the muscle till it is unable to perform further. Thus endurance may be expressed in terms of loss of strength. It is related to fatigue, and it is only through the study of fatigue and fatigue poisons, made by Mosso and others, that light has been thrown on the nature of endurance."

When these tests were held Professor Fisher had not then invented the machine for registering endurance which is now in use in the Yale gymnasium; therefore, three simple tests were employed: first, holding the arms horizontal as long as possible; second, deep knee bending; third, leg raising with the subject lying on his back.

VICTORY FOR THE LOW PROTEID DIET

The results of the competitive tests were all in favor of the flesh-abstaining athletes. In the first test, which was holding the arms horizontal, only two of the fifteen flesh-eaters succeeded in holding their arms out over a quarter of an hour; whereas twenty-two of the thirty-two abstainers surpassed that limit. None of the flesh-eaters reached half an hour, but fifteen of the thirty-two abstainers exceeded that limit. Of these, nine

exceeded one hour, four exceeded two hours, and one exceeded three hours, the last going exactly two hundred minutes, or three hours and twenty minutes.

In the leg raising test the record showed little difference. None of the abstainers reached their absolute limits. The highest record for the abstainers was one thousand times. A flesh-eater reached one thousand, three hundred and two, but did so after the one-thousand mark had already been set for him by an abstainer, and he went into the test with the expressed intention of defeating his rival. Professor Fisher states that it was evident from his fatigue at the end of the test that he could not have repeated the performance on the next day, as did his flesh-abstaining rival.

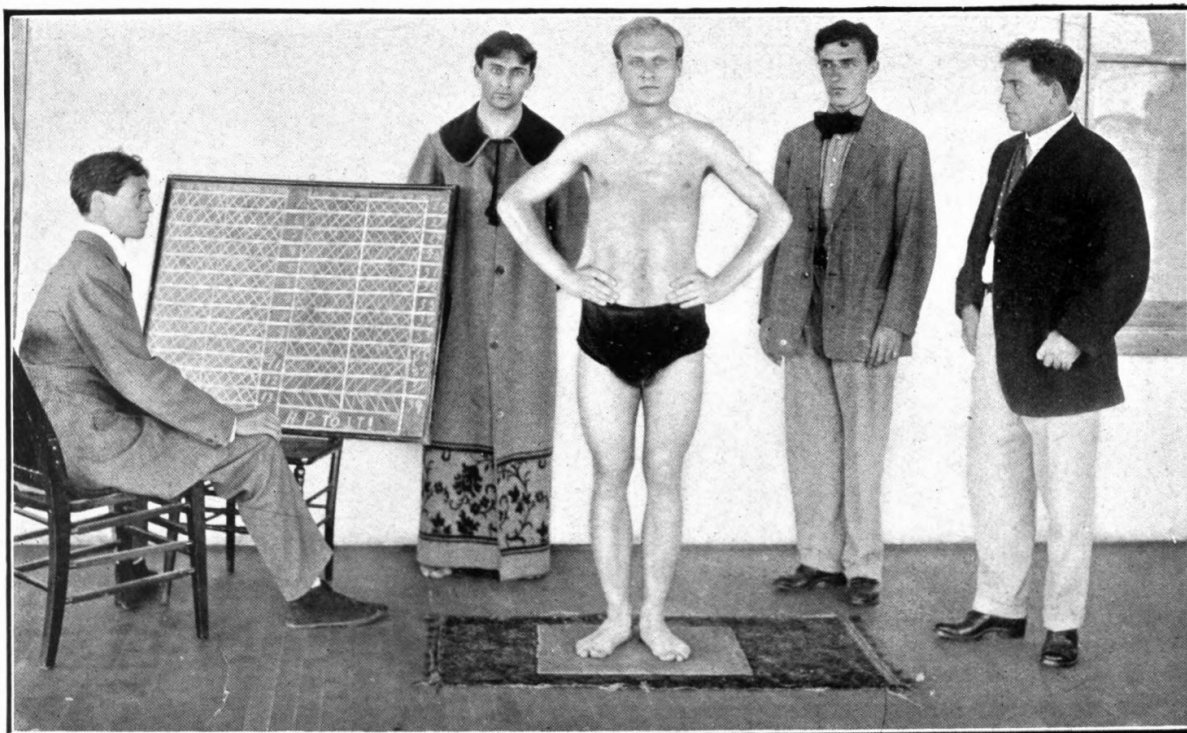
In respect to deep-knee bending, Professor Fisher pointed out that of the nine flesh-eaters who went into this contest, only three went above three hundred and twenty-five times, while of the abstainers, seventeen surpassed this figure. Only nine of the flesh-eaters reached one thousand, as against six of the twenty-two abstainers. None of the flesh-eaters surpassed two thousand, while two of the abstainers did. One abstainer, an athlete, S. A. Oberg, did two thousand and four hundred dips or deep knee bends, almost doubling the highest figure set by the flesh-eating athlete, which was one thousand, two hundred and ninety-two. Most of the Yale flesh-eating athletes were so severely crippled by their efforts in this particular set of movements that Professor Fisher resolved not to employ them again, and went to work on his device for mechanically registering endurance. One of the Yale athletes, who in the deep-knee bending test had reached five hundred times, fainted. Several had to be carried down the gymnasium stairs, and others were made so stiff and sore that for days they could not walk up and down stairs with comfort, while in the case of the abstainers from flesh foods there were comparatively little painful after-effects. Two of the abstainers, one a Yale athlete, were almost free from physical after-effects. The Yale man ran on the track of the gymnasium after his performance, and took a long walk afterward; while the other athlete, Oberg, a Sanitarium nurse, who made the highest record of all, two thousand four hundred times, continued his duties and found little annoyance from stiffness or soreness. (Another flesh-abstaining athlete, John E. Granger, of Battle Creek Sanitarium, has since made a new record of five thousand and two dips in two hours and nineteen minutes.)

Professor Fisher tried many means to stimulate the flesh-eating athletes to do their very best. He called upon their “Yale spirit” to rally to their aid, and he states that the advantage of rivalry as between the flesh-eaters and the abstainers was decidedly upon the side of the flesh-eaters, for their tests, with two exceptions, came after all the records of the abstainers had been completed. The Yale men felt that their tests would go on record as tests of Yale athletes, and Professor Fisher states that the “Yale spirit” which aided them appeared to be as great a stimulus as any “vegetarian” spirit could possibly be.

THE RESULT OF THE MASTICATION TEST

As to the experiment with the nine healthy students, Professor Fisher says:

“The results of the experiment demonstrated so great an increase of endurance as to seem at first incredible. It certainly was a surprise, both to the men and to me. But statistics which I have been collecting during the last two years have prepared me to find great differences and changes in endurance. The special result of the present experiment is to show that diet is an important factor in producing such alterations. The fact that endurance, even among persons free from disease, is one of the most variable of human faculties—far more variable than strength, for instance—is evident to any one who has made even a superficial examination. Some persons are tired by climbing a flight of stairs, whereas the Swiss guides, throughout the summer season, day after day spend their entire time in climbing the Matterhorn and other peaks; some persons are “winded” by running a block for a street car, whereas a Chinese coolie will run for hours on end; in mental work, some persons are unable to apply themselves more than an hour at a time, whereas others, like Humboldt, can work almost continuously through eighteen hours of the day.



MR. JOHN E. GRANGER BREAKING THE WORLD'S RECORD FOR DEEP KNEE BENDING.
The spectator at the extreme right is Mr. Alonzo A. Stagg, coach of the Chicago University football team. Mr. Michael Williams is between the two.

"It is, to say the least, remarkable that hitherto so little effort has been directed toward discovering the factors which explain such differences in endurance. That exercise is one of the most and perhaps the most important factor has long been recognized. A correspondent assures me that by means of moderate *regular* exercise he succeeded in increasing his endurance between 100 and 200% in three weeks as measured by leg-raising and "dipping." The influence of diet has always been regarded as small or negligible, and the opinion has almost been universal, until recently, that a diet rich in proteid promotes endurance. Even among those whose researches have led them to the opposite conclusion, there is very little conception of the extent to which diet is correlated with endurance. Such a person, a medical friend of the writer, stated, when the present experiment was planned, that he did not think the dietetic factor strong enough compared with others to produce any marked effect. We have all heard, of course, of the enthusiastic reports of vegetarians as to their increased endurance, but these we have discounted as exaggerations. The result of the present experiment, however, would seem to indicate that one's improvement in endurance is usually not less, but greater, than he himself is

aware of. Probably it is also true that we may lose a large fraction of our working power before we are distinctly conscious of the fact.

“While the results of the present experiment lean toward ‘vegetarianism,’ they are only incidentally related to that propaganda. Meat was by no means excluded; on the contrary, the subjects were urged to eat it if their appetite distinctly preferred it to other foods.

“The sudden and complete exclusion of meat is not always desirable, unless more skill and knowledge in food matters are employed than most persons possess. On the contrary, disaster has repeatedly overtaken many who have made this attempt. Pawlow has shown that meat is one of the most, and perhaps the most, ‘peptogenic’ of foods. Whether the stimulus it gives to the stomach is natural, or in the nature of an improper goad or whip, certain it is that stomachs which are accustomed to this daily whip have failed, for a time at least, to act when it was withdrawn.

“Nor is it necessary that meat should be permanently abjured, even when it ceases to become a daily necessity. The safer course, at least, is to indulge the craving whenever one is ‘meat hungry,’ even if, as in many cases, this be not oftener than once in several months. The rule of selection employed in the experiment was merely to *give the benefit of the doubt* to the non-flesh food; but even a *slight* preference for flesh foods was to be followed.

“Under flesh foods are included all meat and ‘stock’ soups. It has been shown that although these extracts of meat contain a large amount of nitrogen, it is not in the form of proteid which can be utilized, but only of waste nitrogen which must be excreted. Apparently the sole virtue of such soups is that they supply the ‘peptogenic’ stimulus above referred to.

ANYBODY CAN APPLY THE NEW KNOWLEDGE

“The practical value of the experiment consists in the fact that any layman can apply it, with or without a knowledge of food values, though with more advantage if he possess than if he lack such knowledge. If the dietetic rules of the present experiment are followed, no self-denial as to foods is required. It is, however, absolutely necessary that there should be

self-control enough to break up the habit of hurried eating to which modern civilization has brought us—habituating us, as it were, to eat against time.

“Experience indicates that appetite does not lead to a diet fixed in amount or constituents, but moves in undulating waves or cycles. The men who took part in the experiment were encouraged, after any of the symptoms which seemed to be associated with high proteid (such as heaviness, sleepiness, stiffness, or soreness after exercise, or catching cold), to cut down on their proteid and substitute fat to restrain the gastric juice. This advice was intended to make application of the theories of Folin that we usually carry a reservoir of proteid, enough to supply our needs for body-building for a fortnight. If this reservoir is exhausted, proteid starvation occurs and the body feeds on itself; if it is filled too far it overflows and causes the evils of excessive proteid. If this theory is correct, the art of eating may consist largely in maintaining a golden mean, such that the proteid reservoir is neither empty nor overflowing much. Many persons fear to reduce their proteid to the Chittenden minimum for fear of proteid starvation; but the experience of those who have tried it would seem to show that this fear is groundless, *provided* no violence is done to natural appetite. This may be trusted, so it would appear, to raise a warning in the form of ‘nitrogen hunger,’ before the danger point is reached.” In other words, the body will ask in the language of hunger for proteid food, if you are not eating as much as you should. Professor Fisher considers that an amount of meat equivalent to about one small chop will supply all the proteid necessary in the daily ration, since proteid is also consumed in bread, potatoes and nearly all other foods.

It might be added that one of the writers has found the remedy for continual bilious headaches in the rigid exclusion from his diet of all foods that are rich in proteids, including meat, fish, eggs, milk, cheese, peas and beans; and maintains weight and working efficiency upon such amount of proteid as he derives from ordinary breadstuffs. He has found that the craving for high proteid foods soon disappears if it is not gratified; and that the quantity of bread, potatoes, etc., which the average person would eat at dinner and supper supplies all the nitrogen which his system needs, without leaving any to cause autointoxication.

IV

HOW DIGESTION IS ACCOMPLISHED

In order not to interrupt the narrative of the Yale experiments, we have foregone defining certain of the technical terms which it was necessary to use. It will be well, before going further, to give a simple description of the manner in which the food we eat is transformed in the body into tissue building material and energy: a process the many parts of which are grouped by physiologists under the name of Metabolism.

When you take a mouthful of food it enters on a journey through the body in which it traverses more than thirty feet of the intestinal tube before that part of it which the body cannot use is ejected; the process of metabolism begins the moment the lips touch it. The six salivary glands which are located in the mouth manufacture saliva, which flows out through numerous openings, and mixes with food as it is chewed. The saliva not only moistens the food, thus allowing it to be more easily swallowed, but it also has a most important chemical office, converting all starchy food matter into sugar, and thus performing the first and one of the most essential steps in the process of digestion.

After the food has been masticated and saturated with saliva, it passes down the throat through the gullet, which performs a peculiar muscular contraction, thrusting downward the particles of food. The conversion of the starch in food into sugar, or glucose, which is begun by the saliva in the mouth, is continued as the food passes into and down the gullet, but stops almost completely when the food once reaches the stomach.

THE WORK OF THE STOMACH

It is in the stomach, on the other hand, that most of the work of digesting the albuminoids, or proteids, of food is performed by the gastric juice. The stomach is a pear-shaped bag, that holds about three pints of material, or three-quarters of an ounce for every inch of the individual's height. Food

enters it through the gullet on the upper left hand side, just below the heart. Myriads of glands in the walls of the stomach are active in the formation of either pepsin, or an acid fluid which, when combined with pepsin, forms the gastric juice.

At the back of the stomach, partly overlapping it, lies the liver, which discharges a liquid called the bile into the alimentary canal just below the stomach. Behind the stomach, lies a large gland called the pancreas, which discharges a remarkable fluid, named pancreatic juice, into the intestine through the same opening which the liver uses for its bile. Connected with the stomach is the small intestine, which is the narrow portion of the alimentary canal, and the largest and most important of all the digestive organs. It is some twenty-five feet in length, and its walls are everywhere covered with glands which secrete and exude mucous and other fluids.

At the lower end of the intestine is the colon or large intestine which is not a digestive organ in itself, but is a reservoir in which the food is stored up for a short time, to allow opportunity for complete absorption of the digested portions.

THE ELEMENTS OF FOOD

Although there may be thousands of different dishes, and combinations of foodstuffs, fundamentally they are all closely akin, and can be all resolved into a few quite simple elements: Proteid, Carbohydrate, Fat or Mineral Salts, or combinations of these; the Proteid class having many subdivisions, and the Carbohydrates being made up of the various sugars and starches.

We also know definitely just what use the body makes of these various substances. The Proteid is the up-builder of tissue, the essential foodstuff without which life cannot exist. If we compare the human body to an engine, as nearly all physiologists seem bound to do, we may say that Proteid is like the brass, or other metal, of which the structure is composed. The various Carbohydrates and the Fats are the fuels from which are derived the energy which animates and operates the mechanism. The Mineral Salts are used to supply various important bodily needs, such as

elements required by the bones, or the delicate tissue in the eyes, the enamel for the teeth, and so forth.

THE WORK OF THE DIGESTIVE JUICES

As there are five main food elements, namely, proteid, starch, sugars, fats, and salts, so also there are five main digestive fluids, the saliva, the gastric juice, the bile, the pancreatic juice, and the intestinal juice.

The saliva is an alkaline fluid that digests starch. Its work is checked by the presence of acid substances; which explains why the digestive action of saliva ceases soon after it enters the stomach. Hence the importance of giving the saliva ample opportunity to perform its function, by complete mastication, is obvious.

The gastric juice, of which about seventy ounces is formed by the stomach daily, contains in addition to hydrochloric acid, a quantity of pepsin, which with the acid dissolves all sorts of proteids or albuminous substances, like meat and eggs; and it also contains rennet, which coagulates milk. The gastric juice digests proteids by converting them into pepsin, an exceedingly soluble substance which passes readily into the blood.

The bile manufactured by the liver has the function of digesting fats. Fats are not changed chemically, as are starches and proteids. They are only broken up into particles so small that the cells of the mucous membrane can take them up and effect their removal into the blood stream.

The pancreatic juice is able to perform the work of all the three digestive fluids which we have already named. In fact, it is even more powerful than saliva in the digestion of starch, since it is able to digest raw as well as cooked starch, which the saliva cannot do. It is also able to convert proteid into peptone, as does the gastric juice; and it emulsifies fats, as does the bile.

The intestinal juice digests cane sugar, and is supposed to have a digestive influence upon all the other food elements.

The mineral salts which are taken into the body are dissolved by all the digestive fluids which we have named, some by the saliva and the juices of the intestinal tube, and others, which require acids for their solution, by the gastric juice.

Nearly all these digestive fluids are also powerful antiseptics and are able to destroy germs when the health of the body as a whole is good. The gastric juice, for instance, acts as an antiseptic, preserving the stomach contents from putrefaction during the digestive process. It is a remarkable fact that the gastric juice, although it is so essential to life, is a deadly poison, which, when introduced into the blood produces insensibility and death.

These digestive juices and organs are able completely to dispose of all the food elements which are introduced into the body, save proteid alone. The sugars and starches are either completely absorbed and oxidized, or stored up in the form of surplus fat. The oxidation or burning up of proteid, however, is never complete. There is always a certain amount of unburnable substance left behind from the processes of metabolism, which the liver and kidneys of the body have to dispose of. If only as much proteid as is needed by the body for the upbuilding of its tissue, and the repair of waste, is taken, the body can very readily handle it; but an excess of proteid is highly disadvantageous. Professor Chittenden, in his great work, "The Nutrition of Man," has set forth in elaborate detail the process of the assimilation of proteid. It appears that there are many kinds of proteid; the proteid of eggs is different from that of meat, and that again from the proteid of beans, and so on; and human proteid is different from all. Consequently, the body is obliged to transform every kind of proteid which is brought to it. This proteid is then absorbed by the blood, and carried to the tissues, which are kept perpetually bathed in a supply of nutritive material. The taking of more proteid than is needed would not be so dangerous if it were simply passed on without being digested; nor even if it were digested and transformed, and then promptly eliminated. But what actually happens is that the new proteid taken in is passed through all the stages of assimilation, and drives out in front of it, so to speak, the proteid which has already been prepared, but has not yet been used. And the result is, of course, to throw a double strain upon the liver and the kidneys, the organs of elimination.

Professor Chittenden also points out the common blunder which is made in assuming that persons who are doing hard work need an additional amount of proteid substance. One commonly hears the phrase that laborers and athletes can eat meat in large quantities, and “work it off.” As we have seen, one can “work off” sugars and starches and fats completely; but one cannot “work off” proteid completely. Professor Chittenden is now recognized as the leading authority of the world upon this particular question; and he sets forth clearly in his book the fact that the quantity of proteid needed is not increased by muscular activity. One may work as hard as he pleases, but his body will use no more nitrogen, save only in the case where a sufficiency of other food elements is not supplied. Only as a last resort will the system undertake the labor of burning up proteid to make energy.

HOW MUCH SHOULD WE EAT

When foodstuffs are taken into the body, digested, assimilated, and used up, they produce the same amount of heat and other forms of energy as if burned outside of the body; and hence it follows that the number of calories, or units of heat, represented in a given foodstuff, is taken by scientists as a common measure of its food value.

A calory is a heat unit, which has been adopted as a means of estimating the nutritive value of foodstuffs. It represents the amount of energy required to raise the temperature of four pounds of water one degree Fahrenheit. The number of calories contained in food is obtained by burning the food and measuring the heat produced by means of a calorimeter.

It has been calculated that the normal, average person needs from one hundred and sixty to two hundred and forty calories of proteids every day, in order to build blood and tissues. He needs daily from five hundred to nine hundred calories of fats, which supply heat.

He needs of carbohydrates, which are the starches and sugars, and which the body uses to produce energy for work and heat, from one thousand to one thousand four hundred calories daily. It is declared by Chittenden and Kellogg, whose work has overset the old notions, that the total number of calories, or food units, should rarely exceed two thousand.

Two thousand calories are furnished respectively by twenty-eight ounces of bread, or ninety-six ounces of milk, or sixty-two ounces of potatoes, or nine ounces of butter. One quarter of each of these, or any other fractions which together equal unity, will make up a ration containing two thousand calories.

It is quite impossible, however, to make a hard and fast rule in this matter. Every individual differs from others in his requirements. Moreover, the weather, the season of the year, the amount and kind of work done, are all factors in the situation. Hard physical work and exposure to cold demands the largest food supply. A person who naturally perspires freely needs more food than a person who does not, because of the large amount of heat carried off from his body by the evaporation of sweat from the skin. Adults require food chiefly to repair waste and losses. Growing children require in addition to food to repair waste and losses, material for tissue building. According to the best authorities upon the diet of children, a growing infant utilizes fully one-third of its total intake of food in tissue building. When an adult becomes emaciated he requires more tissue building material than the normal adult, his need for it being practically the same as that of a growing child.

We give below a table showing the average number of food units or calories required daily by people of various heights and weights. This table is one drawn up by Dr. J. H. Kellogg, Superintendent of the Battle Creek Sanitarium. In calculating the number of calories required in a given case, the estimate should be based not upon the actual weight of the individual, but upon the weight of the average person of his height.

“Persons who are in good health,” says Dr. Kellogg, “and find their weight somewhat greater than the figures given in the table, should not necessarily consider themselves obese. While above the average in weight, their condition is probably natural, and no attempt should be made to reduce the weight to any considerable amount, as injury may result. The average for adults applies especially to healthy adults between twenty and thirty years of age. Most people who are above forty years of age have a natural tendency to increase of flesh, which requires no attention unless it becomes excessive. Any reduction in foods made by an obese person should be in

carbohydrates rather than in proteids or fats, unless these latter have been taken in excess.”

TABLE NO. 1

Showing for different ages the average height, weight, and the number of food units or calories required daily.

Boys

Age	Height in Inches	Weight in Pounds	Calories or Food Units
5	41.57	41.09	816.2
7	45.74	49.07	912.4
9	49.69	59.23	1,043.7
11	53.33	70.18	1,178.2
13	57.21	84.85	1,352.6

Girls

Age	Height in Inches	Weight in Pounds	Calories or Food Units
5	41.29	39.66	784.5
7	45.52	47.46	881.7
9	49.37	57.07	1,018.5
11	53.42	68.84	1,148.5

Men

Height in Inches	Weight in Pounds	Calories or Food Units			Total
		Proteids	Fats	Carbo- hydrates	
62	110.0	165	495	890	1650
64	121.0	181	543	1086	1810
66	132.0	198	594	1188	1980
68	143.0	215	645	1290	2150
70	154.0	231	693	1386	2310
72	165.0	247	741	1482	2470
74	176.0	264	792	1584	2640

Women

Height in Inches	Weight in Pounds	Calories or Food Units			Total
		Proteids	Fats	Carbo- hydrates	
57	78.4	118	344	688	1180

59	88.8	132	396	792	1320
61	99.2	149	447	894	1490
63	109.3	163	489	978	1630
65	120.2	180	540	1080	1800
67	130.7	195	585	1170	1950
69	143.0	215	645	1290	2150
71	155.0	232	696	1392	2320

PRACTICAL APPLICATION OF DIETARY RULES

While dieticians have ascertained the number of food units daily required by the average person, yet on no point do they reach more thorough agreement than in saying that the average person should not establish any hard and fast rules as to the quantity and kinds of food he consumes. It is really only an invalid, one who is in a physician's care, who needs to have his food regulated in this precise fashion. The average person should be careful to practice thorough mastication, and should see to it that the proteid part of his meals is not excessive, but he should avoid worrying about his food habits. Any person who fusses and fumes about the kind of foodstuffs and the number of calories they contain, will be apt to cause himself harm; for science has proved by laboratory experiments, which we shall describe later on, that worry, in fact any of the unpleasant emotions, exercises a prohibitive effect upon the flow of digestive juices.

The really important thing to do is to follow a simple dietary, which at the same time is well balanced in its food elements, well cooked, and tastefully served. The housewife will see to it that the foodstuffs she chooses represent more of carbohydrates and fats than of proteids; her guiding rule in this matter being that *the proportion of proteids to the other food elements be ten per cent*. The United States Department of Agriculture has prepared a list of foodstuffs, comprising all those in common use, which shows the proportion of their constituents, and their total energy value, in calories, per pound of material.

This is "Bulletin No. 28, Revised Edition," the work of two of the leading physiological chemists of America, W. O. Atwater and A. P. Bryant; and may be had on sending five cents to the Department. We have inserted in the Appendix a selected list of foodstuffs taken from this publication; and

we give here a rough classification of foods, from which one can see at a glance their leading elements:

FOODSTUFFS WHICH ARE RICH IN PROTEIDS

Eggs
White of Egg
Skimmed Milk
Buttermilk
Yogurt
Cottage Cheese
Nut Products

FOODSTUFFS WHICH ARE RICH IN FATS

Butter
Nut Oils
Olive Oil
Cream
Olives
Nuts (except chestnuts)
Egg Yolks

FOODSTUFFS WHICH ARE RICH IN CARBOHYDRATES

Potato
Rice
Breads
Cereal Preparations

PURE CARBOHYDRATES

Fruits (raw and cooked)
Fruit Juices
Fruit Jellies
Honey
Malt Honey
Marmalades

FOODSTUFFS WHICH ARE RICH IN PROTEIDS AND FATS

Nuts
Nut Butters
Eggs
Cheese
Nut Products

FOODSTUFFS WHICH ARE RICH IN PROTEIDS AND CARBOHYDRATES

Peas
Beans
Lentils
Chestnuts
Skimmed Milk
Gluten Preparations.

FOODSTUFFS WHICH CONTAIN ALL THE FOOD ELEMENTS IN FAIRLY GOOD
PROPORTION

Crackers
Batter Breads
Pastry
Malted Nuts
Custards
Puddings
Salads
Sandwiches
Soups (other than meat or fish soups).

V

HOW FOODS POISON THE BODY

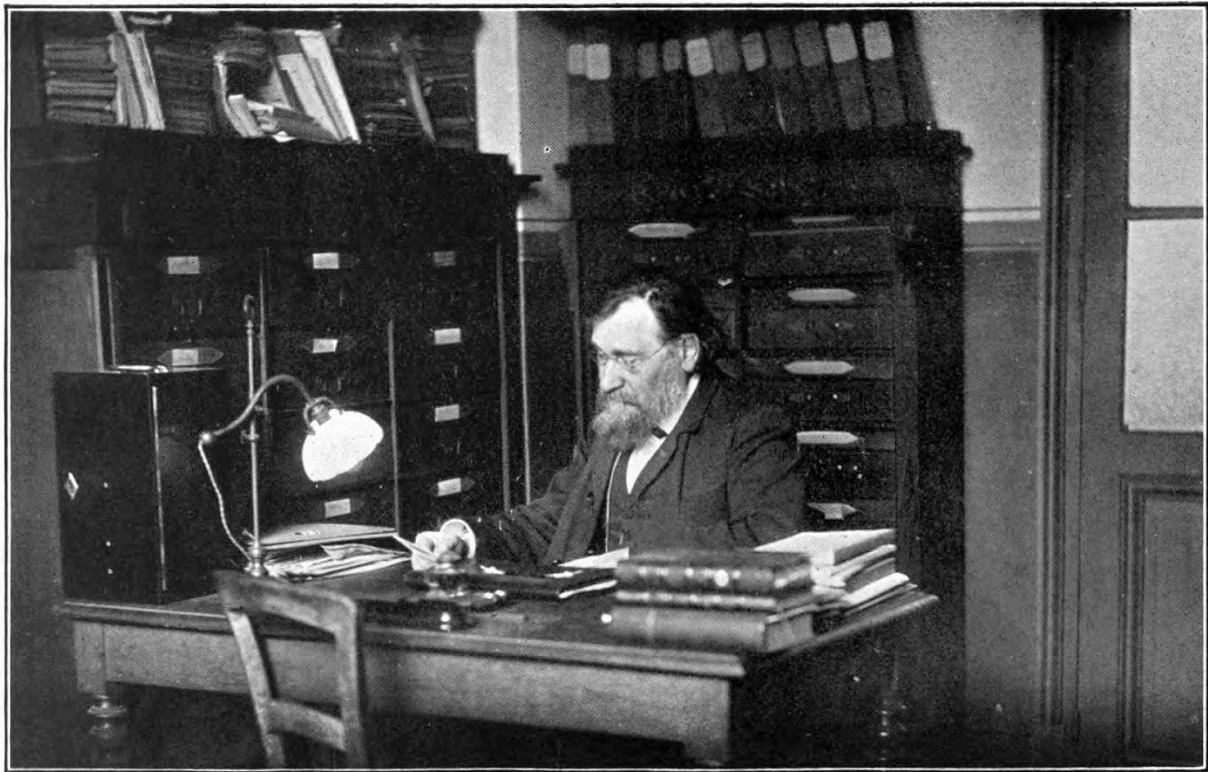
In our survey of the processes and organs of digestion, we saw that after food has traversed the stomach and small intestine it passes into the colon, where it must remain for some considerable time, while the absorption of its digested elements is completed. And this brings us to the most important of the discoveries of the new hygiene. It has been found that some of the foods which human beings eat are loaded with injurious bacteria, and with the poisons which these bacteria produce. And others of them are indigestible, and when they reach the colon, become a source of incubation for countless billions of other bacteria. It was demonstrated by Metchnikoff that these poisons are absorbed into the system, and are the cause of manifold evils. This is the process which is called “autointoxication.”

It would not be regarded as an exaggeration by the leading physiologists of the world to-day to speak of autointoxication as the primary source of nine-tenths of the afflictions from which humanity suffers. Any one would be prepared to admit that the banquet he had attended on the previous night was responsible for the headache which he has on the present morning; but the investigations of bacteriologists have revealed that the food habits of which banquets are typical are responsible for a chronic ailment, of which such diseases as gout, rheumatism, Bright’s disease, consumption, and pneumonia are merely symptoms.

THE INVESTIGATIONS OF METCHNIKOFF

Elie Metchnikoff, sub-director of the Pasteur Institute of Paris, is a philosopher, as well as a physiologist; a philosopher who brings to the support of his speculations the exact methods of the laboratory. He, with the other great leaders of the new art of health, is at last removing from science the reproach leveled at it by Metchnikoff’s great fellow-country-man and friend, Tolstoi, who said that science was useless to man, since it did not direct its attention to the problems which mean most to humanity, such as

the great questions of life and death, but confined its efforts to investigating useless birds and butterflies.



M. ELIE METCHNIKOFF, OF THE PASTEUR INSTITUTE OF PARIS. His researches have thrown great light on autointoxication. He believes that the normal life should be over 125 years long.

The books in which Metchnikoff has recorded the results of the investigations which for many years he has been making into the problems of old age and death, have caused a profound sensation in the scientific world. In these books, the great Russian emphatically and definitely ranks himself with the optimists. He states that scientific study of the constitution of man, and of the workings of man's nature, and of his environment in the world, do not support the view that man is born unto sorrow as the sparks fly upward—to quote the words of the Psalmist—but can really be fitted to live a useful and happy life, ending in a calm and peaceful old age—if man will but turn his attention to the knowledge by which he can really live in harmony with his environment. Metchnikoff has arrived at the conclusions that man and woman would live to be at least one hundred years old, if they could enable their bodies to eliminate those deadly toxins which are the product of the activities of the bacteria which inhabit the human body, as well as of the body's own organic processes.

Age is not always to be computed in years. As a common saying puts the case, "A man is as old as he feels, a woman as old as she looks." A famous French physiologist has altered this to read, "A man is as old as his arteries." The primary change produced by the coming of old age is the hardening and withering of the arteries. As the result of this withering process, a large number of the smaller arteries disappear, so that the blood supply of the muscles, brain, heart, and other important organs, is cut off. This is the change that is technically known as "arterio-sclerosis." It is quite often found in persons of less than fifty years of age. On the other hand, Harvey, the famous discoverer of the circulation of the blood, declared that in the post-mortem examination made of Old Parr, the celebrated Englishman who died at the age of one hundred and fifty-two years and nine months, he found not a trace of this degenerative change.

In the United States the average length of life is about forty-two years; but a large and growing school of modern scientists (comparative anatomists) declare that the natural age of the human family cannot be much less than from one hundred, to one hundred and twenty-five years. Any death that comes at least before one hundred years, is not a natural death but accidental or violent. From the point of view of science, death through disease is just as accidental and violent as the extinguishment of life in a railway wreck or by drowning in the sea; and the fact that the average life of man is to-day only about one-third of that which nature designed for him is due to the operation of autointoxication more than to any other cause.

Natural death in man is therefore more a possibility than an actual occurrence. Nevertheless, instances have been recorded of the actual appearance of the instinct in aged people, where the wished-for death came not because life was burdensome, not because of poverty, disease, or loneliness, but seemed to arrive as naturally as sleep to a younger person, or the wish for more extended life which all of us possess. Metchnikoff states that instances of veritable cases showing an instinct of death are extremely rare, yet this instinct really does seem to lie deep in the constitution of man. And if the cycle of human life followed an ideal course, he concluded men and women after living a healthy and useful life extending over at least a century, with their usefulness and satisfaction in life at maximum during the latter portion of that period, would then give themselves up calmly and

gracefully to the arms of death, as to the arms of a friend laying them down to earned and wished-for rest. Old age would have no terrors, and death no victory.

It has been, perhaps, Metchnikoff's crowning discovery, that the immediate cause of old age is not merely the accumulation of poisonous wastes, but is due to a destruction of the tiny cells which make up the tissues by certain cells of the body, which he describes as macrophages. These are of an especial kind, which wander through the body and devote their energies to the destruction of waste particles and organic débris—particles of material which are not used in the building up of tissue, just as particles of brick and wood might be left on the ground after the erection of a house. These macrophages enact the part of scavengers, very much like the turkey buzzards, which in southern cities eat up the refuse from the back alleys. Just so long as these wandering cells confine themselves to this useful and necessary work, all goes well; but when the vigor of the body cells has been lowered by the accumulation of tissue poisons, these scavenger cells turn traitor to the cause of the body and attack the very cells which they formerly guarded. They have been photographed in the very act of devouring nerve cells in the brains of old people.

HOW TO PREVENT DEGENERATION OF TISSUE

It can readily be seen that if the pernicious activity of these macrophages can be prevented, the coming on of degenerative changes in the body tissue will be much delayed. The practical question, which Metchnikoff therefore asked himself was, How may this revolt of the macrophages, this rebellion of the body's army, be prevented?

It is not possible to attack the macrophages themselves without at the same time doing damage to the body. For these wandering cells are more hardy and vigorous than the higher cells by which the bodily functions are performed, and which they attack, so whatever might be done to weaken the attack of the wandering cells would to a greater degree damage the higher cells of the body. The conclusion that Metchnikoff reached was that the only direction in which we can hope for success in the attempt to prolong human life, lies in giving attention to the predisposing causes which weaken the vitality of the higher body cells and thus expose them to the successful

attacks of the scavenger cells. In other words, if we are to prolong human life, we must make the conditions of life such that the premature accumulation of body wastes or poisons shall be prevented.

One of the first steps to take to affect that end is, obviously, the avoidance of the introduction of poisons, and poison-forming foods, into the body. Out of all proportion to all other causes which lead to the formation of body poisons, is the production of toxins in the colon or large intestine. Metchnikoff's studies show beyond a doubt that there is a close connection between the size of the colon and the duration of life in various birds and animals. Where the colon is used, and has attained large proportions, as in man, in the horse, and many other animals, life is comparatively short, and death is premature. Where the colon is rudimentary, or where only such foods are eaten as do not decay or ferment in the colon, then life is long.

Thus the most important problem, according to Metchnikoff, is how to prevent the development of poisons in the colon. He believes that the colon, indeed, is quite superfluous, and that man would be better off without it. He quotes several curious cases in which the colon has been removed from the body, and the subjects of the operations have recovered impaired health and lived for long periods afterwards. Since the colon cannot be generally removed from the body, however, the practical problem comes down to this: How may we avoid the evils which result from the fermentative and putrefactive processes which go on in this organ?

If the large intestine is kept clean, if only those foods which are antitoxic are eaten, then there will be very few poisons generated in the colon, and the health of the body will be maintained in a higher degree and for a much longer period than can be possible when toxic foods are freely partaken of. It is here that the great argument for vegetarianism on its scientific side arises. All meats and fish are not only "toxic" foods in themselves, but they are quite likely to contain parasites of various kinds.

Ordinary bread has been shown to contain a sufficient amount of proteid to supply all the body needs, as do also rice and other cereals and potatoes. Nuts and dried peas and beans are exceedingly rich in proteid, like meat, and therefore should be eaten sparingly. The best foods in the order of

excellence are given by Dr. Kellogg, as follows—the antitoxic foods being in italics: *fresh ripe fruits, cooked fresh fruits, cooked dried fruits, nuts, cooked cereals, rice, zweibach, toasted corn flakes, potato, cauliflower, and other fresh vegetables, honey, malted nuts, yogurt, or buttermilk, sterilized milk, and cream, peas, beans, lentils, raised bread, and sterilized butter.*

HOW TO ENLIST THE SERVICES OF FRIENDLY GERMS

Since the poisons which are produced in the colon are due to the growth and cultivation of germs, the remedy which naturally suggested itself to a bacteriological specialist like Metchnikoff was to find some harmless or comparatively harmless germ with which the poison-forming germs might be fought—or, in other words, to introduce into the body an extra battalion of soldiers to assist the warrior cells in the battle of the blood.

After years of study and research, Metchnikoff found this beneficent germ in various lactic acid forming microbes, particularly an especial microbe known by the name of Bulgarian bacillus, or Yogurt. This bacillus grows in milk, and in growing it produces large quantities of pure lactic acid. It does not decompose fats, nor does it produce alcohol, as do other lactic forming germs, such as those found in kumyss, matzoon, and kephir.

Milk is first sterilized by boiling for a few minutes, then allowed to cool and a quantity of the ferment is added. In a few hours a sour taste which is pleasant to all whose palates relish mild buttermilk, is developed. Metchnikoff advises that a pint or a pint and a half of this sour milk be taken daily. By this means large quantities of the acid forming and beneficent germs are taken into the intestine, and by degrees the poison producing germs are killed or driven out. Thus the work required of the kidneys, the liver, the skin, and other excretory organs is lessened, and the vigor of the living cells is maintained so that the macrophages do not attack and destroy them.

In Bulgaria where Yogurt is a staple article of food, there are more centenarians, and more vigorous old people to be found than anywhere else on earth. Not only are the Bulgarians and the Hungarians the longest lived races in Europe, but they show a remarkable freedom from appendicitis, colitis, and other diseases due to intestinal infections, circumstances which

called the attention of European physicians to a study of the milk ferment which produced Yogurt, and led to the scientific investigations, first by Masson of Geneva and later and more completely by Metchnikoff and Kellogg, which have placed its use both as a curative and a preventive agent upon a thoroughly scientific basis.

Its use is bound to supersede that of kumyss, kephir, matzoon, and other lactic acid ferments on account of the fact that these ferments are able to live only in the small intestine, while Yogurt bacillus thrives in the colon, where it may be found weeks after the administration of Yogurt has ceased. The importance of this fact will be seen at once when it is recalled that the colon is the chief seat of the anaerobic infection and poison production which are the causes of intestinal autointoxication. Thus the last word of modern science on this subject would seem after all to be but the confirmation of a means for reaching natural old age which has been known for hundreds of years. But to-day we are learning to use means for the prolongation of life by the light of knowledge; no longer blunderingly, handicapped by evil habits which nullify the value of the small fraction of hygienic truth which we possess. To-day, Hygeia, while it holds out to our lips an elixir of life, insists that if it is to have its maximum power, we must also breathe rightly, sleep rightly, and eat and drink rightly.

VI

SOME IMPORTANT FOOD FACTS

The importance of avoiding constipation will be obvious to those who have followed this account of the process of autointoxication; one should see that his daily bill of fare contains a generous supply of laxative foodstuffs, such as sweet fruits, ripe figs and prunes, acid fruits and fruits juices, fresh vegetables, fats and all grain preparations. It is of the utmost importance that the bowels should move regularly once a day. There is another reason for eating food in the shape of fruits or salads, which is that the body may have a sufficient supply of mineral salts.

Nuts and fruits are a splendid combination, since the fat of the nuts and the sugar of the fruits supply the energy and heat producing substances. Fruit sugar indeed is merely a digested form of starch—the digestive process having been accomplished by the heat of the sun in the ripening of the fruit. Fruits contain no fat and practically no starch, and with the exception of the fig, the banana, and a few others, they contain so small an amount of proteid that that element may be considered practically missing. Fruits are used for the sugar, the acids, and the water they contain. Nuts and fruits may be eaten and digested raw by persons who have sound teeth, and who will thoroughly masticate these foods.

Bananas should never be eaten until they are completely ripe, this condition being shown by the appearance of black or dark brown spots on the skin. When in this condition they are usually thrown into the garbage can by the fruit dealer.

Before eating them, one should scrape off the outside fluff, which is next to the skin, as experiments have shown this to be highly indigestible. Eaten when ripe no fruit is more nutritious or delightful than the banana. The only way in which unripe bananas should be used is baked, the same as apples, when they make a succulent and nutritious dish.

Sweet apples will digest more quickly than any other raw fruit substance; but if eaten raw, apples should be thoroughly ripened, and most thoroughly masticated, else hard pieces of apple will enter the stomach and give rise to fermentation. A mealy apple is considered by physiologists as a food substance almost completely predigested, and ready for absorption. If such an apple is reduced to a smooth pulp by chewing, it will pass out of the stomach within an hour. Baked, sweet apples are digested by persons whose stomachs will not tolerate any other fruit.

The acid of sour apples is an excellent corrective for foul conditions of the stomach, such as exist in biliousness. The germs of typhoid, of cholera, and others likely to produce acute disease, are quickly killed by solutions of citric and malic acids, the acids of the lemon or the apple. The juice of a lemon added to an ounce of water will render that water sterile within half an hour, even though it may contain the germs of typhoid fever and cholera. The antiseptic properties of fruit juice render it exceedingly valuable as a means of killing the germs in the stomach and the alimentary canal; a fact which explains the benefits derived from various “fruit cures,” which have been for many years practiced in Europe, and more recently have been employed in various parts of the United States.

The indigestion which many people complain of as arising from the use of fruit comes not from fruit in itself, but from its improper use in combination with other foods with which it does not agree. It is sometimes supposed, for instance, that fruits conduce to bowel disorders; but the truth is that an exclusive diet of fruit is one of the best known remedies for chronic bowel disorders. Care should be taken, however, to avoid fruit juices which contain a large amount of cane sugar; only the juices of sweet fruits should be employed, or else a mixture of sour and sweet fruit juices without sugar. Raisins, figs, prunes, sweet apples and sweet pears may be mixed with sour fruits. Fruit that is sweetened with sugar to a large extent is indigestible, since cane sugar often proves an irritant: while the combination of cream and sugar which is so often used with many fruits is a very bad one. Fruits should be eaten with vegetables only if both are thoroughly masticated, for the reason that the cellulose in vegetables takes a long time to digest, while fruit takes a very short time, and is held in the stomach and ferments. Fruit combines well with cereal foods, breads, and the like, and with nuts.

WHAT COOKING DOES FOR GRAINS

Cooking does for grains what the sun does for fruit; it performs a preliminary digestion. In undergoing digestion the starch in food passes through five stages: first, it is converted into amyloextrin, or soluble starch; second, erythroextrin; third, achroöextrin; fourth, maltose; and fifth, levulose, or fruit sugar. Cooking can carry the starch through the first three of these processes, rendering it ready for almost instant conversion into maltose, on coming into contact with the saliva in mouth and stomach. In the intestine maltose is converted into levulose or fruit sugar and the process of digestion is completed. Modern science has shown by experiments that the preliminary digestive work done by cooking varies greatly with the method of cooking adopted. There are practically three methods used in the cooking of cereals, kettle cooking (that is, boiling and steaming), over cooking, or roasting, and toasting, or dry cooking. Kettle cooking changes the raw starch into soluble starch; in other words, it carries the starch through the first step of the digestive process. Baking, or very prolonged kettle cooking, will convert the starch into erythroextrin, the second stage of starch digestion. Toasting, or dry cooking, in which the starch is exposed to a temperature of about 300 Fahrenheit, advances the starch one step farther, yet.

ABOLISH THE FRYING PAN!

One important thing to remember in connection with cooking is that fried foods, the use of which is so prevalent in America is an unmitigated evil. "Of all dietic abominations for which bad cooking is responsible, fried dishes are the most pernicious," says Dr. Kellogg. "Meat fried, fricasseed, or otherwise cooked in fat, fried bread, fried vegetables, doughnuts, griddle cakes, and all similar combinations of melted fat or other elements of food are most difficult articles of digestion. None but the most stalwart stomach can master such indigestibles. The gastric juice has little more action upon fats than water. Hence a portion of meat or other food saturated with fat is as completely protected from the action of gastric juice as is a foot within a well-oiled boot from the snow and water outside."

This same reason explains why rich cake, shortened pie crust and pastry generally, as well as warm bread and butter disagree with sick stomachs and are the cause of many diseases. Not only does the interfering with the digestion of the food by its covering of fat set up fermentation, but the chemical changes occasioned in the fat itself develop exceedingly injurious acids which irritate the mucous membrane of the stomach, causing congestion and sometimes even inflammation. The frying-pan is an implement that should be banished from every kitchen in the land.

For many years past America has been deluged with various breakfast foods, the virtues of which have been loudly trumpeted. Yet in the ordinary process of cooking these breakfast foods, oatmeal, cracked wheat, etc., it is seldom that more than half the starch completes even the first stage of conversion. Hence it cannot be acted upon at all by the saliva, which does not begin the process of digestion with raw starch. The use of imperfectly cooked cereals is without doubt responsible for a great share of the dyspepsia prevailing among Americans. Oatmeal porridge, and similar preparations, unless most thoroughly cooked, are not wholesome foods, and when cream and sugar are added, there is a combination calculated to create a marked form of dyspepsia. Cereals must be cooked dry in order to be thoroughly cooked, and when prepared by dry cooking or toasting, they are well adapted to the human stomach, are easily digested and in combination with fruits and nuts, constitute a good dietary. Cereals must not only be cooked dry in order to be promptly digested, but they should also be eaten dry. Experiments show that an ounce of dry, well cooked cereal food when well masticated will produce two ounces of saliva; whereas mush, gruel, and other moist cereal foods cause the secretion of only a very small quantity of saliva, less than one quarter of the amount produced by the same food in a dry state.

In connection with the cooking of cereals, it is well to remember that the chief vegetable proteid, gluten, is also rendered very much more easily digested by thorough cooking. On the other hand, the digestibility of animal proteids, in the form of both meat and eggs, is greatly diminished by cooking.

The potato is another important foodstuff; when it is well cooked it is one of the most nutritious and wholesome of all foods. The starch of the potato

is more easily digested than that of cereals, as has been shown by numerous experiments conducted of late in Germany and in America. A good way of preparing potatoes so as to increase their digestibility is to cut them into slices after cooking and then place in an oven until slightly browned; but the admixture of fat of any sort should be avoided.

On the other hand, cabbage is one of those vegetables which is less likely to create stomach trouble when eaten raw than if cooked. The food value of cabbage, however, is so small that it is hardly worth eating, save as a relish. The same remark may be made as to such other foods as celery, spinach, and greens of all sorts. They are only valuable for the sake of the small quantity of mineral salts they contain, and for the sake of adding another taste to the bill of fare. Onions have a higher nutritive value, but this is offset by their containing an irritating volatile oil, which when onions are used too freely may harm the mucous membrane. The onion plays its best part in cookery when used as a flavoring substance.

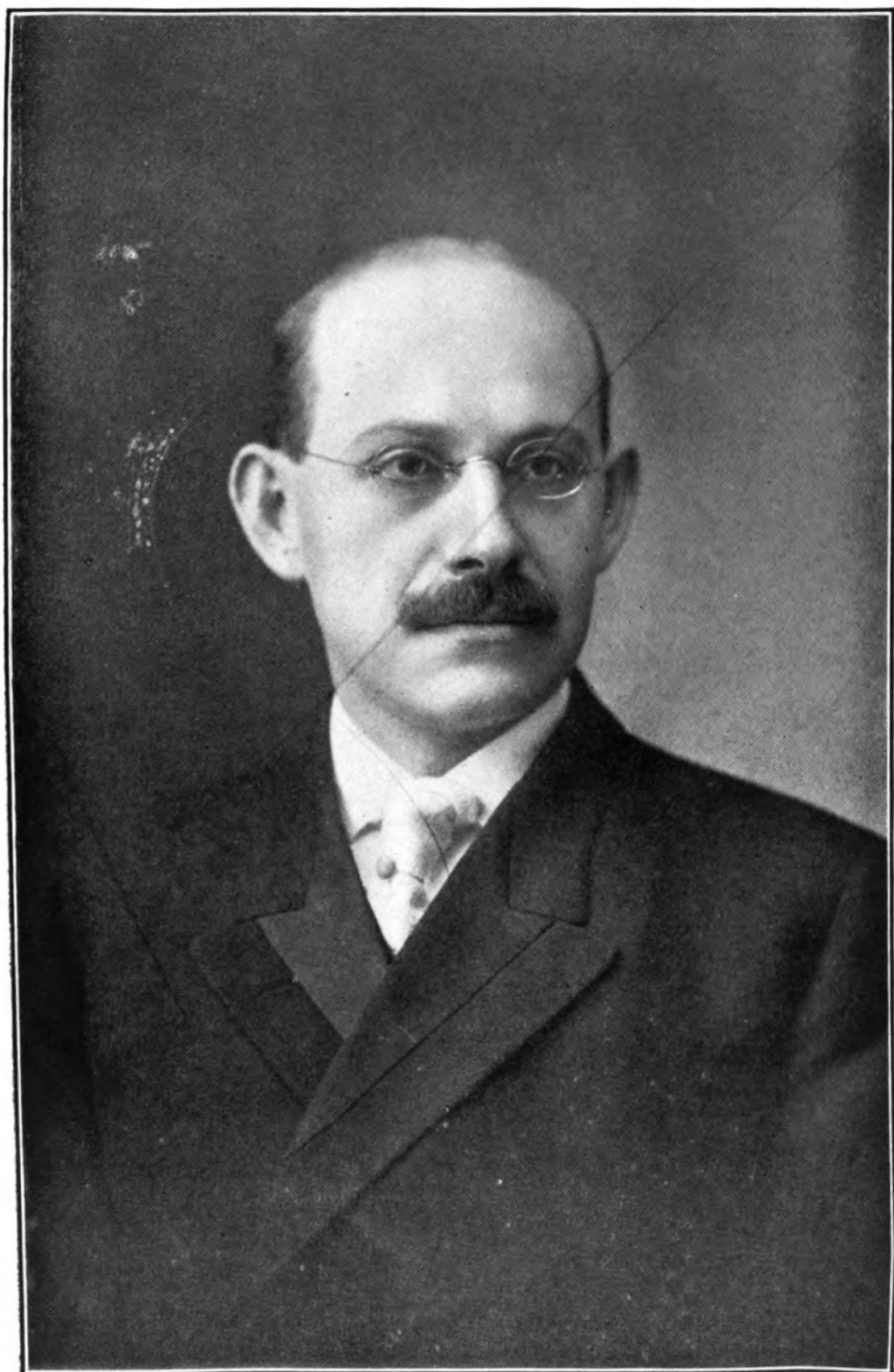
The mushroom is another article of food, popular among those who can afford it, which modern science shows to be practically unfit for human use. Paradoxically enough, although chemical analysis of mushrooms show them to be so rich in proteids as to earn for them the name of vegetable beefsteak, yet researches have shown that these proteids are not available by the body, and hence that mushrooms have no nutritive value whatsoever.

DAIRY PRODUCTS NEED ATTENTION

Milk is commonly considered a wholesome and easily digested food, but this is true only in a modified sense. Thousands of infants die annually because of indigestion set up by the use of cows' milk, and hundreds of adults are more or less injured by the too free use of unsterilized cows' milk, which produces biliousness, sick headache, inactive bowels and a variety of other disturbances. These are not alone due to the toughness of the curds which are formed by milk, and which set up fermentative and putrefactive processes in the stomach unless the milk is thoroughly cooked beforehand.

Federal departments at Washington were, not long ago, almost crippled by the prevalence of typhoid fever among the employees; and the public

health service under Surgeon-General Walter Wyman traced more than ten per cent. of the cases to the milk supply. Professor Lafayette B. Mendel of the Sheffield Scientific School of Yale University, told one of the writers of this book that he went to a certain city that had suffered an epidemic of typhoid, and made a map showing each house that had contained a case of typhoid fever. He made a similar map showing the houses where certain milkmen stopped—and the two maps were almost completely identical. It has also been established beyond a doubt that tuberculosis is communicated from the cow to the human being, and in certain sections of the world it is believed that milk from tubercular cows is the chief channel of infection. It has been shown that even if the udder of a cow be healthy, a tubercular cow may give infected milk, and that the presence of a single tubercular cow in a herd may be responsible for the infection of the milk of healthy animals. Several international medical congresses have lately declared that all milk should be boiled in order to kill the germs.



**PROF. LAFAYETTE B. MENDEL, PH.D., YALE UNIVERSITY,
Who has carried on researches in conjunction with Prof. Chittenden.**

The United States Department of Agriculture issued in Circular No. 111 of the Bureau of Animal Industry, and in Circular No. 114, the recommendations made by a conference of some twenty of the foremost scientists of the United States, and few more important documents concerning the public health have ever been issued by a government. In brief, these recommendations may be thus stated: Raw milk is highly dangerous. Boiling or pasteurizing kills the disease germs and makes the milk safe without seriously impairing the taste or digestibility. Milk produced under the most ideal conditions, such as “certified” milk, is only relatively safe. Pasteurization, when properly done, makes the milk absolutely safe.

Butter, of course, is subject to all the arguments that can be advanced against milk, with the additional one that it is even more subject to infection with germs than milk itself, since the time that elapses between its manufacture and its consumption is usually far longer than the time that elapses between the drawing of milk from the cow and its use. Only butter that is made from sterilized cream should be used.

Cheese, of course, is open to all the objections urged against unsterilized milk and butter, and in addition has a disagreeable quality all its own. The cheese eater may at any time swallow a serious or even a fatal dose of “cheese poisons,” which are substances produced in cheese by the action of germs. These are not ordinarily present in sufficient quantity to render their presence apparent; nevertheless, a great number of cases of cheese poisoning are annually reported by various boards of health all over the country. Cheese made from sterilized milk is less open to these objections. A delicious cottage cheese may be made from Yogurt milk.

The too free use of sugar at the table and in cooking, not only in its pure form, but in the shape of preserves, syrups and sweet beverages, has been shown to be a most prolific source of injury to the stomach. Sorghum, maple sugar, and maple syrup are essentially the same as cane sugar and molasses. It has been shown that if we eat freely of fruits we will obtain all the sugar our system requires in a form that is easily digestible.

The constitution needs quite a good deal of fat; wholesome fats are contained in nuts, and in cereals, and are also provided liberally by ripe olives and olive oil. Emulsified fats are those in which the minute particles are broken up; and these are far more readily absorbed by the tissues of the body. The fat in ripe olives is emulsified fat—as likewise is olive oil when used in mayonnaise dressing. It should not be mixed with vinegar, however, as vinegar is an irritating substance that works harm, when used freely, to the mucous membrane of the stomach. Lemon juice is not only much safer, but makes a much more delicious dressing.

The objection which applies to vinegar, applies also to pepper, mustard, and other condiments and spices.

The too free use of salt, of which nearly everyone is guilty, is another habit upon which modern physiologists frown. While salt is essential, it is contained as an element in many foods, and there is no more reason why it should be sprinkled upon each and every article of food that is taken than we should have castors containing all the other kinds of inorganic salts, that the system needs, and which are supplied to it in fresh foods. Salt using is merely a habit, and a disastrous one, since it has been shown to be one of the factors in the causation of kidney troubles, such as Bright's disease.

The large use of glucose in the form of candy, syrups, adulterated honey, and various sweets which are in common use, is said by physiologists to be responsible for a large number of cases of diabetes, a disease which is rapidly increasing in America. There is now produced a malt sugar, called malt honey or "meltose," which can be used freely for all the purposes that cane sugar is used.

The case of food reform against fish would merely lead to the relating of the arguments against meat. Fish contains nearly seven per cent. of uric acid. It is exposed like meat to the presence of tape worms and other parasites. Even when fresh out of the water its flesh is filled with fatigue poisons, the result of its struggles to escape from the net or the hook; and Mosso of Turin and other authorities have shown that these fatigue toxins have a bad effect upon the body. No food will so quickly decompose and putrefy as fish, and unless perfectly fresh it will always be found full of the

putrefactive bacteria which are the active agents in causing autointoxication.

It may be stated, however, that the person who follows that careful and helpful mode of eating recommended and practiced with such marked benefits by Horace Fletcher and his converts, will assuredly minimize the dangers that lurk unsuspected by the uninformed in many of our commonly used foods, and will derive a greater benefit from all food than it is possible for those to gain who eat in the hasty and careless fashion characteristic of most Americans.

VII

HOW OFTEN SHOULD WE EAT?

WE have discussed the question how to eat and what to eat; there remains the question of when to eat. English people, as a rule, eat four meals a day. The French are practically a two meal a day nation, eating a very light breakfast.

Of late years there has been a strong tendency on the part of American dieticians to advocate a reduction in the number of daily meals, the ideal aimed at being the establishing of the custom of two meals a day, with at least six hours intervening between them.

It may be asked whether appetite is not a safe guide to follow, and whether it is not the part of wisdom to follow personal inclination in the choice and quantity and number of meals. Does not a study of dietetic customs and habits definitely decide the essential rules of dietetics? While it is true that habits and customs are very strong factors in everybody's life, yet it is also true that they are very unreliable guides. We are constantly acquiring new habits, and sloughing off old ones; and even the most deeply impressed of habits may be changed for others. And while the common customs of mankind would seem to indicate that three or four meals a day is the rule, at least among civilized nations, yet the facts are that the most primitive people take one meal a day, and the great majority of people in the world, as a rule, eat certainly less than three.

TWO MEALS A DAY THE BEST

Physiological facts argue for the two meal plan, or else for very light and easily digested food, if an extra meal be taken.

Healthy digestion requires at least five hours for its completion, and one hour for rest before another meal is taken. This makes six hours necessary for the disposal of each meal. If food is taken at shorter intervals than this, when ordinary food is eaten, the stomach will be allowed no time for rest.

Again, if a meal is taken before the preceding meal has been digested and has left the stomach, a portion remaining, one is likely to undergo fermentation, in spite of the preserving influence of the gastric juice; thus the whole mass of food will be rendered less fit for the nutrition of the body, and the stomach itself will be likely to suffer injury from the acids developed.



MR. UPTON SINCLAIR'S CHILDREN, Well nourished on two meals a day.

These facts make it plain why eating between meals is a gross breach of the requirements of good digestion. The habit of nibbling at confectionery, fruit, nuts, and other things between meals, is a positive cause of dyspepsia. No stomach can long endure such usage. There is a continual irritation of the mucous membrane of the stomach, and a continual excitation of the glands, which, in the long run, work great harm.

The same reasons which are advanced against the habit of eating between meals fit the case of irregularity of meals. Those who have regular duties, regular hours of work, should have regular meal hours. The human system is continually forming habits, and seems in a great degree dependent upon

the performance of its functions in accordance with the habits that are formed. This fact is especially observed in respect to digestion. When meals are taken at regular times the stomach becomes accustomed to receiving food at those times, and is prepared for it. If meals are taken irregularly, the stomach is taken by surprise, so to speak, and is never in that state of rest in which it should be for the prompt and perfect performance of its functions. The habit which many business and professional men form, in the stress of their occupations, of allowing their meal hours to be intruded upon, at times depriving themselves of a meal, will undermine the best digestion in the long run. There is no physiologist who would not endorse the following words of Kellogg: "Every individual ought to consider the hour for meals a sacred one, not to be intruded upon under any ordinary circumstances. Eating is a matter of too momentous importance to be interrupted or delayed by ordinary matters of business or convenience. The habit of regularity in eating should be cultivated."

DON'T EAT BEFORE SLEEPING

The meal which most people would find it advantageous either to drop altogether, or to reduce in quantity, is supper. The physiological law which is now come to be recognized is, that the brain must be active to insure good digestion; and that the stomach must be empty to insure good sleep. That sense of drowsiness which so often follows a hearty meal is not a physiological condition; it is not evidence of a naturally sedative effect in eating; but is really an evidence of indigestion. Those who practice eating before retiring often sleep soundly until an hour or two after midnight, then awake, and find difficulty in getting to sleep again. This is due to irritation of the solar plexus set up by the labor of digesting under unfavorable conditions. The lack of appetite for breakfast after a late supper is evidence of the exhausted state of the stomach. Fruits and cereals are the ideal supper rather than the ideal breakfast—though good at any time!

DRINKING AT MEALS

It is nearly always the case that a hasty or over-hearty eater is also in the habit of drinking copiously of water or other fluids at his meals. He "washes his food down" instead of legitimately drinking. The body, of

course, needs liquid, but, as a rule, meal times are not the times for the taking of this liquid supply; except for what is contained in the food itself. The hasty eater thus associates two great evils.

Liquid of any kind in large quantity is inimical to digestion, because it delays the action of the gastric juice, and weakens its digestive qualities, and also checks the secretion of saliva. In case the fluid taken is very hot, as tea, coffee, cocoa, or a considerable quantity of soup—it relaxes and weakens the stomach. On the other hand if it is very cold, it checks digestion by cooling the contents of the stomach, and reducing its temperature to a degree at which digestion cannot proceed. Even a small quantity of cold water, ice cream, or other very cold substance will create a serious disturbance if taken into a stomach where food is undergoing digestion. The process of digestion cannot be carried on at a temperature that is less than the body, which is about one hundred.

The old notions about the processes of digestion were chiefly drawn from the experiments of Dr. Beaumont made nearly a hundred years ago up in Northern Michigan, around Mackinac; with a Canadian hunter, Alexis St. Martin, as the subject. Most people have probably read of St. Martin and Beaumont in the physiologies they studied in their school days. Beaumont was a very capable physician, and a man of the truest scientific spirit. It happened that through an accident he was given an opportunity to make the most valuable contribution to the study of the stomach of man that so far had been furnished by any investigator. The hunter, St. Martin, had suffered a gunshot wound in his stomach, and Beaumont kept him alive for years with the wound open so that he might study the movements of the man's interior organs. For the first time, here was a human body with a window in it, so to speak, and through this window the scientist patiently watched and studied for years. Of course, however, the window gave only a limited view of what was going on inside this particular house of life, and a great number of Beaumont's ideas and theories have been proven erroneous; nevertheless, he obtained much important knowledge. When Dr. Beaumont peered through that curious window which he made in the stomach of Alexis St. Martin, he noticed that when the latter drank a glassful of water at the usual temperature of freshly drawn well water, the temperature of the food undergoing digestion fell immediately to 70. The process of digestion was

checked absolutely and did not resume until the body had regained its proper temperature, which it did not do for more than half an hour.

Another way in which drinking at meals proves harmful is because of the fact that particles of food not thoroughly masticated are washed from the mouth into the stomach. If any drink at meals is taken at all, it should be a few minutes before eating. Of course, sipping of a little water will not be harmful, if care is taken not to sip at the time when food is in the mouth. It will be found, however, that unless the meal is composed of very dry foods, there will be little inclination to drink at meals. When, however, the food is rendered either fiery or irritating with spices, and other stimulating condiments, it is small wonder that there is an imperious demand for water or liquid of any kind to allay the irritation.

HOW THE BODY PRODUCES “APPETITE JUICE”

He who is really hungry, however, has no need of condiments, and usually small relish for them.

The old saying that hunger is the best sauce is one of those proverbs of the people which modern science is proving to be firmly established on truth. No sauce can equal appetite. Experiments by Professor Pawlow of St. Petersburg, Director of Department of Experimental Physiology in the Imperial Literary School of Medicine, have shown that there is a real “appetite juice” formed by the body when it is hungry.

Appetite, and hunger, are not synonymous terms with the mere habit-craving for food which most people consider to be either appetite or hunger. Real hunger, or appetite, only comes to the body when the body has earned it. There must be an expenditure of tissue, which the body requires to be repaired; or there must be a real need for energy to carry on work before the body will manifest its need for energy-supplying material. In other words, the body cares nothing about our likes or dislikes, our whims or our fancies, in the nature of food, save when it has a real need for food. Professor Chittenden demonstrated that most people simply eat the entire round of meals from mere habit. The disturbance when for any reason they miss one or two meals from the accustomed routine is simply the outcry of a habit and not the outcry of a real need. While Dr. Kellogg advises that no meal be

missed, yet he also strongly advises us not to eat unless really hungry, merely drinking a little fruit juice or something of the kind at the meal hour in order to keep up the normal action of the digestive organs.

The digestive juice which is manufactured by the body when it is really hungry and food has been given to it has been shown by Pawlow and Hanecke to be the most important element in digestion. The chemical juices produced in the stomach and intestines while food is in them is of small importance and value compared with the juices that are formed while food is being chewed when the body has a good appetite or is really hungry.

This juice begins to flow at the very sight of food, and continues to from three to five minutes after beginning mastication. The production of juice in the stomach is stimulated by the contact of food with the mouth, and only during that contact; so it is obvious that the longer the food is held in the mouth, if it is held there in enjoyment, and the more completely it is chewed, so long as chewing is accompanied by taste, the more thoroughly are the flavors set free by the act of chewing, and the higher becomes the stimulating effect of these flavors upon the psychic centers which cause the appetite juice to flow into the stomach.

These facts prove the dependence of gastric digestion, or stomach digestion, upon mastication. Pawlow was experimenting with gastric juice when he hit upon this demonstration; and he has concluded that we cannot have gastric digestion at all well without thorough mouth digestion; that the complete mastication of food, in other words, is the thing necessary to prepare the stomach to receive the food. Thus, if you chew your food well, the food will be predigested in the mouth, and when it enters the stomach it will find already there waiting for it not only enough gastric juice to digest it, but just the particular kind of gastric juice that is needed.

Pawlow turned this discovery of his to a very practical use. He has a dozen or more healthy dogs which he calls his Dog Dairy. From these dogs he collects daily a quart or more of gastric juice, or appetite juice; and the dogs produce this large quantity without taking a particle of food into their stomachs. The juice is carefully filtered, and bottled and shipped all over the world to those physicians who are in touch with Pawlow and his work, and by them are administered to human patients. It is given to those patients

who are deficient in gastric juice, and is used in very obstinate cases of indigestion.

Pawlow collects his juice by having openings made in the throat and in the stomachs of the dogs. When the dogs are hungry they are given food of kinds which they particularly like, and they are allowed to smell the odor and to become excited over the prospect of eating it before they are actually allowed to have it. With the first sight and odor of this food, the dogs begin to secrete the appetite juice, which flows from the opening made in their stomachs through tubes into receptacles. Then when they begin to eat their food, the food does not reach the stomach at all, but simply passes through the openings in the throat into a receptacle before the dog, and the dog can go on eating the same meal over and over again. They thus enjoy themselves thoroughly for a long time. When the appetite juice ceases to flow, the process of feeding them in this manner stops, and they are given a real meal.

VIII

HEALTH AND THE MIND

This account of Professor Pawlow's experiment leads directly to the all important subject of the influence of mental states upon digestion and assimilation. Dr. Saleeby has published a book called "Worry, the Disease of the Age"—the very title of which shows the attitude of physicians upon this question; and the bad opinion which mankind has always entertained of such states of mind as "the blues" has now been scientifically justified. The effects of pain and pleasure upon digestion have been demonstrated by actual experiments in the laboratory of the St. Petersburg professor.

A vivid account of these experiments has been given to the writers by Dr. J. H. Kellogg, who witnessed them about a year ago. Dr. Kellogg writes:

"Professor Pawlow took Professor Benedict and myself into a quiet corner of his laboratory, and there we found a dog that had his salivary glands or ducts arranged so that by means of little tubes passing through the skin all the saliva, instead of passing down his throat, passed out through the tubes and could be collected in small glass bottles suspended beside his neck.

"The dog had been prepared beforehand by the attendant. Little empty bottles were attached to the collecting tubes, and as soon as the dog saw Pawlow, he seemed to be very happy, and wagged his tail, and his eyes gave evidence of satisfaction; but there was no flow of saliva until Professor Pawlow brought near to his nose a bottle containing some powdered meat. He took out the cork in the presence of the dog, turned out a little of it in his hand, shook it in the bottle and brought it near to the dog's nose. The dog began to sniff it, licked his chops, snapped his jaws, reached out after it, and in less than two minutes the saliva began to flow very profusely, and it was not more than fifteen or twenty seconds before the saliva was pouring down into the bottles.

“Professor Pawlow, then, after holding the bottle out before the dog for about thirty seconds, put the stopper into it, and put it behind him out of sight, and in a very few seconds the saliva ceased to flow. Then he brought it back again, showed it to the dog, brought it near his nose, allowed him to smell it but kept it just out of his reach all the time, and the saliva poured out again freely. He continued this until the dog finally made up his mind he was not going to get any meat, and when the powder was brought near to him he paid no attention to it, but turned his head around and looked very disappointed and very ugly, and at that point, the saliva ceased to flow.

“That was a very remarkable thing to me. The meat was right there, he could smell it, but he knew he was not going to get it, so he was angry, and as his state of mind changed, the secretion of saliva was wholly arrested. I was very much surprised. Of course, I believe thoroughly in the importance of being in a happy state of mind when eating, but I really did not appreciate thoroughly the importance of those things; I did not fully appreciate how positive an inhibitor of the activity of the salivary glands an unhappy state might be.

“But a common experiment made in India shows the same idea. When an Anglo-Indian has lost anything of value, he has his whole family of servants brought to him to find out which one has stolen it. A common test is to stand them all up in a row, and then to give each one a morsel of dry rice to chew. They must chew this rice for five minutes, and then the master goes around and examines each man’s mouth. The mouth which is dry is the mouth of the culprit, and the state of that man’s mind has the effect of arresting the flow of saliva. Pawlow has shown that this is a positive physiologic law and operates upon the dogs as well as upon human beings.

“Another experiment astonished me even more than this. We followed Pawlow down through a long narrow hall and upstairs into a room which was small and secluded, in a very quiet part of the laboratory, remote from any noisy occupation, and there we found a brown dog standing on a high table. It was a delicate and very intelligent looking animal. The attendant sat near by, and the dog was prepared as the other had been. As we came in, the Professor beckoned to us to sit down on a little bench beside the wall and indicated that we should be quiet. He stepped up to the dog, looked at him,

and the dog recognized him with a smile in the dog's way of smiling!—and presently the saliva began to flow.

“Professor Pawlow was very much surprised. We had come into the room and he had offered the dog nothing, but the saliva was flowing. That was contrary to his expectation. He looked with considerable astonishment at the attendant. The attendant quietly said, ‘You have been feeding meat to the other dog, and he smells the meat on your hands.’

“The dog had such a keen sense of smell that the odor of meat on Pawlow's hands even at a distance of several feet was sufficient to cause the saliva to flow. So he went out, washed his hands and came back. At this time, not a drop of saliva was flowing. The arrangement was such that every particle secreted must come outside of the mouth into these bottles. While we were waiting in silence, watching the dog quietly, suddenly the attendant pressed his foot without making any motion of the body at all, upon a little lever beneath his toe and the result was the causing of a high musical note to be sounded, a very high pitched tone.

“Instantly, in less than three seconds, the saliva was flowing into the tube. We waited a little while until the saliva ceased to flow, then the note was sounded again. Instantly the saliva began to flow.

“Professor Pawlow has been experimenting upon this line for a long time. Other experiments were made. One interesting experiment was with a large number of dogs. He had upon one counter a long row of dogs, about a dozen, which had their stomachs fixed in such a way, and their throats fixed also in such a way, that upon the secretion of the gastric juice in the stomach the juice would flow out into a flask.

“The dogs were suspended in a sort of harness. They had had their throats fixed so that food instead of going into the stomach came out at the throat. So as the dog ate the food, the food fell back into the plate and the dog continued eating the same breakfast over and over. These dogs had been eating the same breakfast for four hours, from six to ten o'clock in the morning, and they were still eating, and just as hungry as ever because there was no food entering their stomachs at all and their appetites were growing keener every moment, and they were having a wonderfully good time. I

thought that some people I have met might enjoy such an arrangement. This really has the same effect without having your throat cut.

“I noticed that if these dogs got disgruntled, or tired, or dissatisfied, then the gastric juice would cease to flow. Sometimes the food, having been chewed a very long time, lost its flavor, and the dogs secreted no more juice; then the attendant would come along and put a little fresh food into the plate and the dogs would seize this with great avidity, and the gastric juice would begin to flow again in a perfect stream.

“These experiments have demonstrated in the most positive manner the definite connection there is between psychic conditions and the process of digestion, and have shown us that the food must be palatable, that it must address the olfactory sense agreeably, and that the mind must be in a happy state in order that the digestive process may proceed.”

And then Dr. Kellogg goes on to tell of the work of Professor Cannon, of Harvard University, who actually has made visible the digestive processes in the stomach by means of the X-ray. By feeding cats food colored with certain substances which are impervious to the X-rays, he was enabled to photograph all the actual movements of the organs concerned in the acts of digestion. It was demonstrated that certain emotions, such as anger and fear, positively stopped the whole process of digestion.

Depressing thought will affect injuriously the circulation of the blood; it will also affect the breathing. The mere attitude of the body assumed by the despondent person has its bad influence. The head droops in a melancholy fashion—and this very attitude prevents normal action of the lungs and the blood veins. Depressing thoughts destroy the appetite; and when the body does not receive its proper nourishment, the blood becomes impoverished.

“Any severe anger or grief is almost certain to be succeeded by fever in certain parts of Africa,” says Sir Samuel Baker, in the *British and Foreign Medico Chirurgical Review*. “In many cases, I have seen reasons for believing that cancer had its origin in prolonged anxiety,” says Sir George Paget, in his “Lectures.” “The vast majority of the cases of cancer, especially of breast or uterine cancer, are probably due to mental anxiety,” says Dr. Snow, in the *London Lancet*. “Diabetes from a sudden mental shock is a true, pure type of physical malady of mental origin,” says Sir B.

W. Richardson in "Discourses." "I have been surprised how often patients with primary cancer of the liver lay the cause of this ill health to protracted grief or anxiety. The cases have been far too numerous to be accounted for as mere coincidences," says Murchison.

"Eruptions on the skin will follow excessive mental strain. In all these and in cancer, epilepsy and mania from mental causes there is a predisposition. It is remarkable how little the question of physical disease from mental influence has been studied," says Sir B. W. Richardson.

"My experiments show that irascible, malevolent and depressing emotions generate in the system injurious compounds, some of which are extremely poisonous; also that agreeable, happy emotions generate chemical compounds of nutritious value, which stimulate the cells to manufacture energy," says Elmer Gates, the celebrated American scientist. Gates' experiments show with minute exactitude just how it is that one's impalpable thoughts and emotions affect the battle of the blood, and his work makes it easier for one to understand and appreciate the portion of truth underlying such manifestations as the New Thought and Christian Science movement. There can be no doubt that men and women have practically remolded their bodies and changed the whole course of their lives by using the impalpable yet potent force of their wills. Sometimes these have been men and women seemingly without a vestige of will; and yet, by comprehending the necessity for will, they took the first steps towards attaining possession of it. Many very remarkable stories could be told illustrating this point. Professor William James, of Harvard, introduced one of the writers to a man who had been afflicted with what had seemed a helpless case of mental trouble, accompanied by physical ailments which were rapidly breaking him down; and this man had effected a complete cure through his own unaided efforts. He resolved that he could be cured, and cured he was.

We remember another instance; this time of a consumptive; a man who was so far gone that all the physicians gave up his case as hopeless. To all intents and purposes he was already a dead man, when there came to him the light of a new hope. He had spent a great deal of money in taking various "treatments" for tuberculosis, without deriving permanent benefit, and then had come to believe utterly that in only one way was there hope

for the consumptive, namely, by living entirely in the open air. When seemingly at his last gasp he arrived at a branch of the Battle Creek Sanitarium at Boulder Creek, Colorado. In certain photographs of this establishment you may see on a bare hillside that stands back of the building, a narrow foot-path. This path has many turnings and windings in its lower course, but towards the top of the hill it aspires upward in a straight line. That trail was made by the consumptive who had determined that he would live, crawling on his hands and knees up the side of the hill. He positively refused to go under a roof for any consideration whatsoever. His meals were brought to him where he lay on the road side. At first he was so weak that he could only go a few feet in the course of a day, and had to drag himself along in a wavering line. But he began to improve—he went on improving—until, finally, along the track on which he had crawled he was running at top speed.

And a little while ago this man was one of the athletes who took part in Professor Irving Fisher's endurance competition between flesh-eating athletes and vegetarians; and he proved to be best of them all! He doubled the best record made by any Yale man in the deep-knee bending contest. The most enduring Yale man was able to make the deep-knee bend—which is a very severe test of physical endurance—twelve hundred times. The consumptive who had cured himself went twenty-four hundred times. He thinks nothing of a ten or fifteen mile run before breakfast in the morning.

It is important to apply these truths to the question of nutrition. It is positively harmful to eat food when one is gloomy or low spirited or worried or angry.

You may object to this that you cannot at will make an optimist of yourself at meal times, and turn on a flow of good humor as you draw water from a tap. But you can at least refrain from eating, and if you do you will discover that the real hunger which is bound to develop is a very strong emotion. It will drive away any ordinary attack of the blues very quickly; and will call up pleasant anticipations of the joy of food to assist the digestive processes.

IX

THE CASE AS TO MEAT

“I wish there was a science of nutrition worthy of the name,” writes Bernard Shaw in a private letter. “The mass of special pleading on behalf of meat eating on the one side and vegetarianism on the other, which calls itself the science of metabolism to-day, seems to me to be so corrupt as to be worthless.” The fact that Shaw himself is a fervent vegetarian lends additional significance to this statement. Until quite recently the advocacy of either dietary has been based upon considerations the opposite of physiologic. It has been the sentimental aspects of the controversy—vegetable versus animal foods—which have received most emphasis. The vegetarian supported his position on the ethical ground that the eating of animal food, involving as it does the taking of life, is wrong. On the other hand, the advocate of meat eating based his arguments on the support given to it by common custom, and a belief that a meat diet is that which supplies vigor and manly force. As Dr. Woods Hutchinson, the most prominent of the champions of meat eating, puts the case: “Vegetarianism is the diet of the enslaved, stagnant, and conquered races, and a diet rich in meat is that of the progressing, the dominant and the conquering strains. The rise of any nation in civilization is invariably accompanied by an increasing abundance in food supplies from all possible sources, both vegetable and animal.”

At the same time, even Dr. Hutchinson admits that human life can be maintained upon a vegetarian diet. “Nearly one-half of the human race,” he writes, “has been compelled from sheer necessity to prove that thesis in its actual experience; but we find absolutely no jot of evidence in support of the contention that there is any advantage or superiority in the vegetable diet as such—no more than that there is any inherent superiority in a pure animal diet as such.... There is no valid or necessary ground, so far as we have been able to discover, for the exclusion of any known article of food, whether vegetable or animal, from our diet list in health.”

Dr. Hutchinson’s views were printed in a popular magazine, and have been very widely quoted, but he seems to have written without paying

attention to a number of scientific investigations which suggest ample grounds for the radical reduction of the meat portion of the ordinary diet. Among these are the experiments of Dr. Horter of New York, Professors Mendel, Chittenden and Fisher of Yale, Dr. Fenton B. Turck, and such world-known physiologists as Combe of Lausanne, and Metchnikoff, Gautier, and Tissier of Paris. The elaborate researches of Dr. Kellogg of Battle Creek are dismissed by Woods Hutchinson, because of the fact that Dr. Kellogg not only upholds the exclusion of meat from the diet for purely scientific reasons, but also on ethical grounds. The writers of this book, however, have discarded meat from their dietary for scientific reasons, paying as little attention to the ethical side of the question as Dr. Hutchinson could desire. They will give in this place a brief summary of these scientific reasons.

THE BELGIAN EXPERIMENTS

We have already told of the experiments whereby Professor Fisher of Yale proved the superior endurance of vegetarians over meat-eaters. It happens that experiments of the same nature were carried on at about the same time by two women scientists in Belgium, Dr. J. Ioteyko, head of the laboratory at the University of Brussels, and Mlle. Varia Kipiani. They studied the question of vegetarianism by several methods, and became convinced that the vegetarian régime is a more rational one.

Their experiments were for the most part comparisons of strength and endurance between men and women subsisting on the usual high proteid, or flesh diet, and men and women who for longer or shorter periods had abstained entirely from meat. The results tally remarkably with those obtained by Professor Fisher. So far as strength was concerned, very little difference was discovered between vegetarians and “carnivores.” In endurance, on the other hand (and it is endurance that most people need) a very remarkable difference was found, the vegetarians surpassing the carnivores from 50 to 200%. The Brussels investigators found also that the vegetarians recuperated from fatigue far more quickly than the meat eaters, a discovery which was one of the most remarkable features of the Yale experiments.



**MR. SINCLAIR'S CHILDREN,
Brought up according to good health principles.**

In commenting upon the Belgian experiments, Professor Fisher writes:

DR. TURCK'S INVESTIGATIONS

It is possible that flesh-eating, as ordinarily practiced, is injurious both because of excessive proteid and because meat, as such, contains poisonous elements. It is well known that Liebig came to repudiate the idea that the extractives of meat were nutritious, and that investigation has shown them to be poisonous. Professor Fisher also points out that Dr. F. B. Turck has found that dogs, mice, and rats fed on meat extractives exhibit symptoms of poisoning and often die. The poisonous effect is aggravated by intestinal bacteria, which find in these extractives an excellent culture medium. Dr. Turck concludes:

“(1) It is clearly evident from these experiments, which correspond to the investigations of others, that the injurious effects of meat are due not so much to the muscle proteid, myosin, as to the extractives.

“(2) That the injurious effects of the extractives are increased through the action of intestinal bacteria.”

Dr. Turck does not find any evidence that the extractives in small quantities are injurious.

Dr. Turck therefore concludes that the “high liver” who uses much flesh and also an excess of starch and sugar is a “bad risk” for life insurance companies. He recommends, if meat is to be used, that the extractives first be removed by special processes, which he explains.

These investigations, with those of Combe of Lausanne, Metchnikoff and Tissier, of Paris, as well as Herter and others in the United States, seem gradually to be demonstrating that the fancied strength from meat is, like the fancied strength from alcohol, an illusion. The “beef and ale of England” are largely sources of weakness, not strength.

THE DANGER OF INFECTION FROM MEAT

It has always been conceded that by eating raw or underdone beef or pork one may acquire tape worms; and that in eating raw or underdone pork one runs the same risk of contracting that incurable malady, trichinosis. The danger from these sources, however, is comparatively slight, since most people eat their meat well cooked; but in the view of many modern scientists all meat eaters are open to a particular form of germ infection which involves all kinds of meat, fish, flesh and fowl, cooked as well as uncooked.

Everybody knows how readily meats of all kinds, and particularly seafood, such as fish, oysters and clams, undergo putrefaction. The processes of decay in fish and animals begin within an hour or two after death, under the influence of putrefactive bacteria, which are always present in the colon, or large intestine of animals, upon the skin and in the atmosphere about them. Ordinary cooking does not destroy them, for they are able to stand the ordinary cooking temperature. Salt and smoked fish, and other meats have these germs present in vast multitudes; and beef and game that is “hung” for a long time in order to become “tender,” are so far advanced in decay before they are brought to the table that every minute particle of them is alive with these germs.

These facts are granted by all; but the physiologist who favors the use of meat, says that unless excessive quantities are consumed, the healthy person undergoes little risk. The argument is, that when the germs are swallowed into the stomach they are there destroyed by the action of the gastric juice, which is germicidal; but experiments have lately proved that some of these germs escape destruction by the gastric juice, and find their way to the colon, where they continue to multiply in the mucous which covers the intestinal wall, and thus maintain constant and active putrefactive processes in that part of the body.

THE NUMBER OF GERMS WE EAT

Dr. Kellogg of Battle Creek has lately made public the results of a carefully conducted series of observations made by Dr. A. W. Nelson, bacteriologist of the clinical laboratory of the Battle Creek Sanitarium. Various specimens of meat were purchased in the ordinary way in the market, wrapped in clean paper, and immediately taken to the laboratory,

where samples were removed for observation under the microscope. The meat was then taken to the diet kitchen and well cooked, after which cultures were again made.

The germs found in meat are classed as aerobes and anaerobes. The aerobes are for the most part acid-forming germs, and comparatively harmless. But the anaerobes are poison-forming germs, and are the agents of putrefaction and of various diseases. They are to-day considered as the most potent causes of many chronic maladies, and especially of that most common of diseases, intestinal autointoxication.

Dr. Nelson found that in one specimen of raw beef, there were present per moist gram of material 105,000 aerobes and 90,000 anaerobes. On the outside of the beef after it had been fried, there were no germs present, but on the inside of the fried beef, he found 3000 aerobes and 2000 anaerobes per gram. With three other specimens of beef, that were broiled, and boiled, and roasted, respectively, the results were generally similar. Of all modes of cooking, roasting seems to have least effect upon the bacteria, for in specimen No. 3, while there were fewer bacteria than in specimen No. 1 before cooking, there were found after it had been well roasted 150,000 aerobes and 160,000 anaerobes.

In fresh fish raw there were found 870,000 anaerobes per gram; in sardines in oil, 14,000,000; while in codfish that had been soaked to remove the salt, there were found 47,600,000. In another experiment specimens of meat were secured such as were served on the dining tables of one of the prominent city hotels, and taken at once to the laboratory, where without delay bacterial cultures were made. A specimen of sirloin steak was found to contain 378,000,000 anaerobes per gram of moist material.

An interesting experiment which showed the increase of anaerobes or poison-forming germs in dead flesh, was that made with two chickens of equal size, one of which was drawn, and the other undrawn. Both were placed under the same conditions in a room the temperature of which was maintained at 70° Fahrenheit. Bacterial cultures were made at frequent intervals, with results as given in the following table, the figures showing the number of bacteria per gram of moist material.

	No. 11	Drawn	No. 12	Not Drawn
	Aerobes	Anaerobes	Aerobes	Anaerobes
3 hrs after death	4,500	5,650	5,000	6,500
2d day	8,500	9,000	10,000	12,000
3d day	17,000	16,000	60,000	20,000

It must be remembered that these chickens were freshly killed, and that the anaerobes had no such opportunity to increase as in ordinary market beefs.

Specimens of several other kinds of meat were purchased in the market, and at once taken to the laboratory for study. Cultures were made immediately on reaching the laboratory, and again after the meat had been allowed to stand (covered) at room temperature for twenty hours. The following table shows the results of the bacterial counts:

BACTERIA PER GRAM (MOIST)

Immediately after purchase

Specimen	Aerobes	Anaerobes
No. 13 Large sausage	560,000,000	420,000,000
No. 14 Small sausage	834,400,000	663,000,000
No. 15 Round steak	420,000,000	560,000,000
No. 16 Roast beef	252,000,000	560,000,000
No. 17 Smoked ham	47,320,000	43,120,000
No. 18 Hamburger steak	138,000,000	129,000,000
No. 19 Pork	635,600,000	126,040,000
No. 20 Porterhouse steak	31,920,000	30,800,000

After being kept at room temperature for twenty hours.

Specimen	Aerobes	Anaerobes
No. 13 Large sausage	770,000,000	490,000,000
No. 14 Small sausage	770,000,000	640,400,000
No. 15 Round steak	750,000,000	840,000,000
No. 16 Roast beef	728,000,000	750,000,000
No. 17 Smoked ham	616,000,000	750,000,000
No. 18 Hamburger steak	784,000,000	700,000,000
No. 19 Pork	952,000,000	1,036,000,000
No. 20 Porterhouse steak	336,000,000	700,000,000

These experiments were made in the winter time, when, because of the diminished amount of dust in the air, germs are less abundant. Even in the

winter time, however, certain meat products simply swarm with germs. A specimen of raw liver examined in January was found to contain 269,800,000 bacteria per ounce or gram. Raw sausage contained 48,280,000 bacteria per ounce or gram.

“A food which introduces these deadly organisms, the anaerobes, at the rate of ten to twenty-five billions to the ounce, as do pork, beef and sausage, must certainly be classed as unclean,” said Dr. Kellogg, in summing up the report on his experiments. “When thousands are daily indulging themselves in this dietary, what wonder that Bright’s disease, enteritis, and other maladies due to germs and germ poisons are so rife and so rapidly increasing? It is quite as important to keep the inside of the body in a sweet, clean and wholesome condition as to maintain a wholesome state of the external portion of the body.”

CANCER AND MEAT EATING

That nothing could seem more definite than the connection between cancer and the practice of eating inferior meat, is the conclusion reached by Dr. G. Cook Adams, who made a series of statistical studies under the direction of the Chicago Board of Health. “There cannot be the slightest doubt,” says this expert, “that the great increase in cancer among the foreign born of Chicago over the prevalence of that disease in their native countries, is due to the increased consumption of animal foods, particularly those derived from diseased animals.” This conclusion substantiates the original deductions made by Dr. Adams from investigations carried on over a number of years in Australia and London.

Dr. Woods Hutchinson stated that the rise of any nation in civilization is invariably accompanied by an increased abundance in food supply; and the rise of these foreign born in Chicago in civilization substantiates Dr. Woods Hutchinson’s views. Receiving more wages than in their native homes, where their diet was simple, they are enabled to indulge in a meat diet denied them in Europe. The result is an increase in the death rate from cancer between the years 1856 and 1866 of 680%, while from 1866 to 1905 the increase was 232%.

In 1905 cancer was responsible for one in every twenty-three deaths, while in 1906 one death in every 21.8 was due to this horrible disease. The Italians and the Chinese were the only two of all the races represented in Chicago that do not show a far greater death rate from cancer than in their own homes. The Italians keep up the use of macaroni and spaghetti, while the Chinese adhere to their native diet of rice. The nations showing the higher mortality consume large quantities of canned, preserved, dried and pickled meats, sausages, etc. It was also shown that the bulk of the fresh meat prepared at the plant of a slaughtering company was stock condemned by official inspectors, and this was the meat eaten by the poor.

INVESTIGATIONS IN NEW YORK

Dr. W. H. Guilfoy, of the New York Health Department, recently published the results of investigations of the death rate among foreigners in New York, and showed that cancer, heart disease and chronic Bright's disease have increased alarmingly in recent years, and his statistics show that foreigners of flesh eating nations reveal the highest rates for the three diseases mentioned, in marked contrast with nations that consume from 50 to 400% less meat per capita. The following list shows the exact comparison:

DEATHS PER 100,000 AMONG FLESH-EATING FOREIGNERS

	Cancer.	Heart Disease.	Chronic Bright's Disease.
Irish	166.6	381.2	410
German	151.9	231.5	212
English	140	207	209
Bohemian	246	237.4	255.7

DEATHS PER 100,000 AMONG NATIONALITIES NOTED FOR SMALL CONSUMPTION OF MEAT

	Cancer.	Heart Disease.	Chronic Bright's Disease.
Austro-Hungarian	151.5	190.7	131.2
Swedish	84.7	69.7	99.6
Polish	130	170	121

Another argument which the opponents of meat-eating bring forward, is based upon the fact that in eating flesh which contains blood, we consume a great deal of waste material and poisons from the body of the animal. When the blood flows from the heart outward to each organ of the body it is a life-stream containing life-giving oxygen and particles of fresh food material for the use of the tissues, but when it flows back it is freighted with the elements of disease and death, with poisonous substances which are the bi-products of organic activity, and which, if retained in the body for any length of time invariably cause disease. The rapidity with which the blood becomes impure and poisonous may be easily noted by winding a string about the finger, when the flesh will quickly turn a blue color. Animals die as men and women die, with their ailments within them, and if you eat of them you eat the products of their disease process. Tuberculosis is known to be one of the maladies sometimes transmitted by the use of flesh. Numerous epidemics of typhoid fever have been traced to the use of oysters.

THE PROTEID ARGUMENT

It had generally been assumed by physiologists that the great virtue of meat lay in the greater digestibility of its proteid matter. Recent experiment investigations, however, have shown that the vegetable proteids are as a rule not less digestible than those from animal sources. The vegetable proteids are often packed away and enveloped in cellulose or other material difficult of digestion, or are permeated with fats, as in some of the nuts; but modern methods of preparing grains for the market, and also the thorough cooking of them, remove this difficulty.

The deficiency of ordinary vegetable dietaries in proteids has been a ground for criticism by the opponents of this regimen. Since, however, the researches of Chittenden, Mendel, Metchnikoff, Dr. Folin, and others have shown us that we need much less proteid than the elder school of physiologists so long supposed, this objection loses its weight. And, furthermore, there are many nut foods which are even richer in proteids than cooked meats. Cooked meat contains 25% of proteids, while peanut butter contains 29%. The edible portion of walnuts contains 27%, and the edible portion of pine nuts 35%.

To sum up the argument in this matter it is our belief that modern science has demonstrated that excessive meat eating is dangerous, because of its high proteid content and its liability to germ infection; and, also, that we can obtain all the elements which meat contains from other kinds of food which are not open to the objections fairly to be made against the use of meat. Nevertheless, here, as elsewhere, it may be said that “Fletcherism,”—complete mastication—is again the key that unlocks the solution of this problem for many. Thorough mastication leads to the use of less meat; it also gives the germicidal saliva a chance to kill harmful germs; and it aids the digestive organs very materially. Eat meat—says the rational physiologist—if you feel you must, or if it is difficult to abandon its use, but be careful to chew it well.

It is true, to be sure, that the digestion of proteid is accomplished not by saliva, but by stomach juices, which would seem to be an argument in favor of bolting meat (as the dog does), but the mere maceration of the meat by the teeth, if nothing more, is a help to the stomach in its work of digestion.

X

THE CASE AGAINST STIMULANTS

The dominant note of the discussion that for years has been waged in scientific and medical circles as to the effect of alcohol on the human constitution has been, to the puzzled layman at any rate, the insistent, reiterated cry of the fundamental “mystery” of alcohol. Alcohol is poison! cries one school. It is not anything of the sort, being, as a matter of fact, a food! retorts the opposing school. Its use in health or its administration to patients sick of any ailment is hardly short of a crime, declares one leading physician. Tut! tut! alcohol in moderation does no harm, and it is invaluable in the treatment of many diseases! replies another. And so the arguments proceeded.

Summing up his views of the deliberations of the British Association for the Advancement of Science, recently held at Leicester, England, and which formed a storm center for the great alcohol debate, a noted chemist in London “Science,” said that we know how far the sun is, and can tell the weight of the earth, predict when the next comet may be expected, and give true answers to many other important questions, but we do not know “anything to speak of” on the subject of alcohol. As to the discussions that have waged at Leicester and elsewhere on the question of the medical use of alcohol, the general impression left on the world of laymen is that they all (the noted authorities) disagreed with one another more or less, and that nobody can declare with any scientific authority whether alcoholic liquor is good for us or bad for us.

We propose here to describe the work of one scientist who has made experiments which enable him to declare with authority that alcohol is injurious. This investigator is Charles E. Stewart, M. D., of the Battle Creek Sanitarium. He has closely studied the work of Sir Edward Wright, London, the discoverer of “Opsonins”; and his experiments were suggested by those of Wright. They led him to the discovery that alcohol has a harmful effect on the blood by lowering its supply of opsonins.

It has been demonstrated to the satisfaction of most students of Wright and Metchnikoff, and their allies, that the opsonins form one of the most valuable of the body's defences against disease. And if Dr. Stewart has demonstrated that alcohol poisons the opsonins, it must be admitted that at last a positive and tangible proof has been brought forward of alcohol's harmful qualities. What nourishes and strengthens the blood, helps the life force within us; what weakens or poisons the blood, is an attack upon the very citadel of vitality. Alcohol, says Dr. Stewart, is such an enemy.

In such diseases as pneumonia and tuberculosis, the white cells, according to Wright, cannot effectually combat the germs unless there are plenty of opsonins present to aid them. Now, in treating pneumonia and tuberculosis, many practitioners encourage the use of alcohol. Dr. Stewart believed that alcohol was injurious. Having heard Sir W. Edward Wright's lectures, he asked himself the question:

“Can the evil effects of alcohol be due to its lowering of the opsonic power of the blood?”

He instituted a series of experiments to determine, if possible, the facts in the case. He first of all administered to four persons who all their lives had been total abstainers, two ounces each of port wine. The normal opsonic power of each of these individuals had been determined as being 75 or above—that is to say, it was well above the point at which the opsonic power must be maintained in order that the white cell may do effective fighting. At the time when the subjects took the port wine, the first subject had a normal amount of opsonic power to resist the germ of tuberculosis which may be expressed by the term 1.13., and a normal power of resistance to the pus germ, which infects wounds, of 1.06. After drinking the wine, both those powers of resistance were lowered most perceptibly; the first to .85, and the second to .67. Similar results, in greater or less degree, followed in all other cases. The port wine decreased the power of the blood to make opsonic sauce for the white cells.

In a second series of experiments, two ounces of Scotch whisky were taken an hour apart; that is, the normal index was taken, and immediately afterwards an ounce of the Scotch whisky was taken, an hour later another ounce, and an hour after this the index was taken again. The results here

were similar. For the germs of tuberculosis it was discovered that the opsonic power had dropped 10% and for the streptococci (or pus-forming) germs about 8%.

In another experiment where two ounces of sherry wine were used, the opsonic power for the germs of tuberculosis dropped 11% and for the streptococci 5%.

In another experiment where four ounces of champagne were taken, the opsonic power dropped 9% for the germs of tuberculosis and 19% for the streptococci germ. Many other experiments were performed, but they gave practically the same results. The opsonic power decreased in proportion to the amount of alcohol contained in the liquor.

Dr. Stewart carried on his experiments in the laboratory of the Battle Creek Sanitarium, with the assistance of Dr. A. W. Nelson. He reported his results to the American Society for the Study of Alcohol and Drug Neuroses:

“I realize that there are a great number of factors which influence the opsonic power of the blood, and that there is considerable variation in even what may be considered normal cases, but, notwithstanding these variations, there is a sufficient uniformity to enable us to make some very valuable deductions. I feel justified in concluding that alcohol has a marked influence in reducing the vital forces of the body, thereby greatly interfering with the natural power of the body to remedy ailments. Since Wright has shown that out of all comparison the most valuable asset in medicine lies in raising the anti-bacterial power of the blood, the administration of alcohol, which according to these experiments, is pro-bacterial, and as such a strong liability instead of an asset, should be eliminated from our therapeutics, at least so far as internal administration in infectious diseases is concerned.

“While only a comparatively few experiments have been made, the results obtained have been uniform, and justify, I believe, the preliminary report of it given to the medical profession and the public with the hope that it may encourage others to pursue the work further in this direction.

“Heretofore, when any statement was made to the effect that alcohol caused this or the other disease, or ailment, or harmful effect of any sort on

the human constitution, the reply could be and was made that the case could not be proven; that there were always circumstances which might be construed as showing that other factors besides alcohol influenced the situation. Now, however, I believe that we have opened up a line of investigation which will place the proofs against alcohol on a solid scientific basis by demonstrating its injurious effect on the blood, which is the life."

TEA AND COFFEE

In the same laboratory where Dr. Stewart placed his case against alcohol, experiments are being made which show in the same direct way that such drinks as tea and coffee also lower the opsonic power of the blood. Into the United States alone are imported more than one billion pounds, or five hundred thousand tons of tea and coffee each year. It is estimated that tea and coffee contain from three to six per cent. of poison. Therefore, more than fifteen thousand tons of poison, "so deadly that twenty grains might produce fatal results if administered to a full-grown man in a single dose"—in all more than ten billion deadly doses of poison, or, "fully six times as much as would be required to kill every man, woman and child on the face of the earth," are brought into this country every year, as component parts of substances which are commonly regarded as pleasant foodstuffs.

This is the case stated against coffee and tea in its broadest and most emphatic form. The opponents of the use of tea drinking term both tea and coffee "drugs." What is commonly thought to be the pleasantest property of both tea and coffee, namely, their ability to banish one's sense of fatigue, is regarded by the critics of the tea and coffee drinking habits as perhaps the most sufficient evidence of their poisonous character.

"No one would doubt for a moment," says one such critic, "the poisonous nature of a drug capable of producing irresistible drowsiness in a person who is not weary, as morphine would, for instance. Vice versa, the power of a drug to produce wakefulness in a person strongly inclined to sleep as the result of fatigue is equal evidence of its poisonous character. The sallow complexion common among women of the higher classes who have reached middle life, the almost universal nervousness among American women, and many common digestive disorders, and the increasing prevalence of

nervous or sick headaches, afford to the experienced physician ample evidence of the toxic or poisonous character of tea and coffee.”

Tea and coffee contain (in addition to caffeine) tannic acid, and various other volatile poisons, each of which produces characteristic harmful effects. The volatile oils give rise to nervous excitability, and after a time provoke serious nervous disorders. Caffeine is a narcotic, which has been shown to diminish the activity of the peptic glands—and thus seriously to interfere with the normal operation of the organs of digestion. The eminent physiologist, Wolfe, showed by experiments that three grains of caffeine—an amount that might easily be imbibed in an ordinary cup of tea or coffee—very substantially impairs the quality of the gastric juices, lessening their total acidity. Roberts’ experiments showed that tea and coffee interfere with the action of the saliva upon the starch of the food, and at times may even wholly destroy its effect.

XI

DIET REFORM IN THE FAMILY

The reader is now familiar with the new ideas upon the subject of human nutrition. It is obvious, of course, that if these ideas should ever come into general acceptance, there would be enormous changes in the every-day habits of human beings. And we can well imagine that a person might be fully convinced of the soundness of all the arguments which have been advanced in this book, and yet shrink in dismay from the complications incidental to applying them.

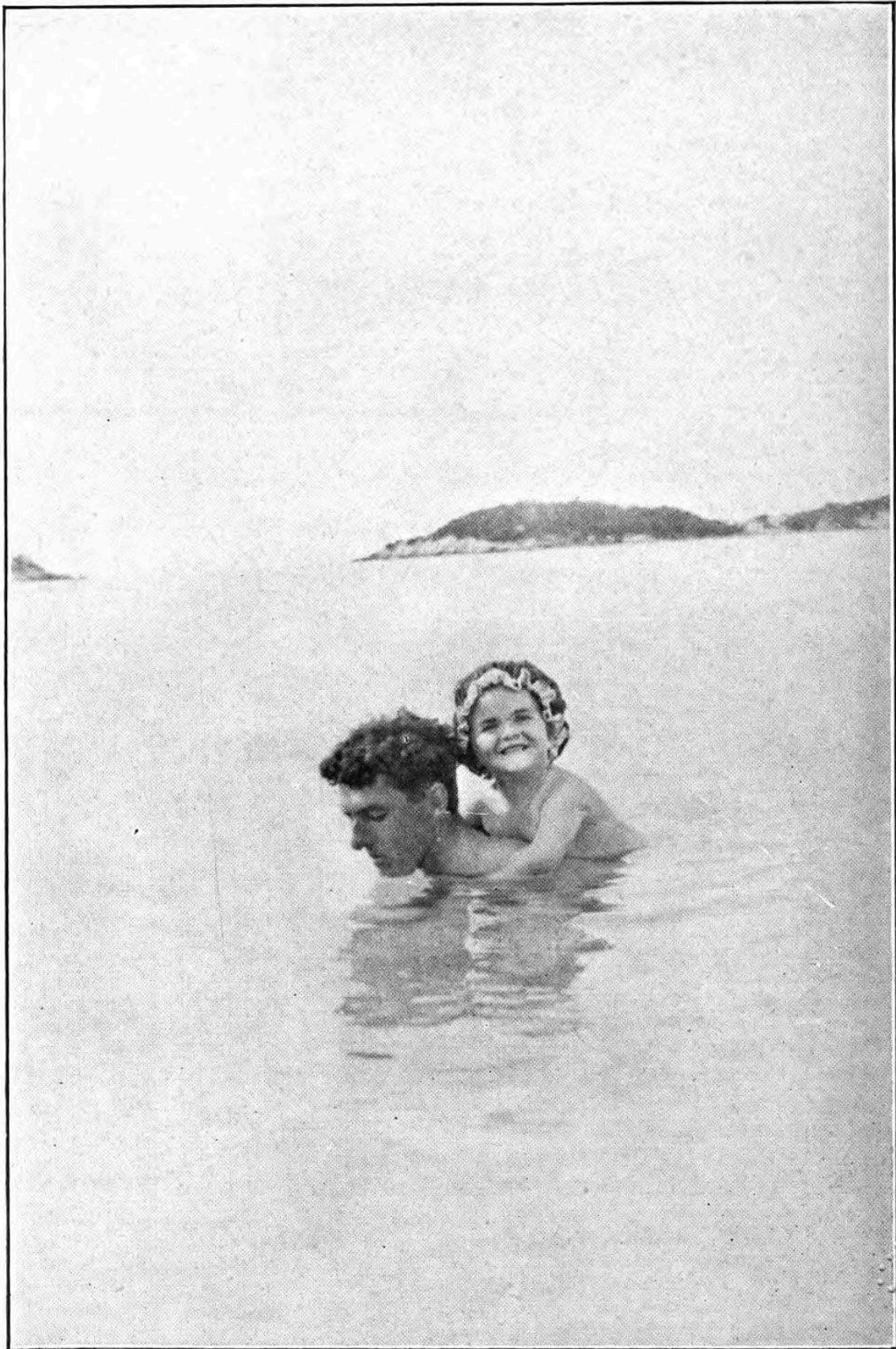
We ourselves have faced these difficulties in many forms. We have wished to have two meals, and yet felt obliged to have three, because all our friends had them, and we did not wish to be hermits. We have wished to avoid meat, and yet have eaten it, because it was on the table, and we did not like to startle our hostess—and perhaps find ourselves involved in an argument about vegetarianism, in the course of which we had either to permit a good cause to go down into defeat, or else to tell facts about meat which would take away every one's appetite for meat, and for vegetables as well. But in the end, the desire for health has conquered all other motives with us, and we have broken with every trace of the old ways. It seemed to us that we would help and interest others if we gave some account of how the new ideas have worked out in practice, and the daily regimen of a family which adopts them.

This book is written in Bermuda, where the writers have been living in co-operation, along the lines worked out at Helicon Hall, only upon a much smaller scale. Their party consists of eight adults and three children—this including two governesses, a secretary, and a servant. They live in an isolated neighborhood, upon the waterfront. Most of the party sleep out of doors on the broad verandas of the house, while the wide doors and windows of the other rooms afford ample ventilation. Daily sea-bathing is the habit of all of the group.

The married women of the party assume in turn the direction of our dietaries; that is to say, they choose the menus, and attend to the ordering of the food supplies. We eat but twice a day, and the menus are made up entirely of fruits, grains, nuts and vegetables, with the occasional use of eggs. We obtain from the Battle Creek Sanitarium a great number of the foods we use, availing ourselves of its splendidly managed food-department. The children eat three times a day, but their breakfasts are very light, consisting of orange juice and a fig or two, or perhaps a banana. The children have this light breakfast immediately after arising. At ten o'clock comes the principal meal of the day for the whole household. An effort is made to make this meal "well balanced"; that is to say, to have the proportion of proteids, carbohydrates and fats. There are usually not more than two, or at the most, three cooked dishes. Sometimes the main dish is a soup; sometimes it is baked or boiled macaroni with tomato dressing; sometimes it is bean or pea croquettes; sometimes it is scrambled eggs, or the yolks of hard boiled eggs.

We have a constant supply of fresh vegetables, the justly celebrated Bermuda onion; beets, turnips, egg plant, raw cabbage, potatoes, white and sweet, rice, hominy, green peas, tomatoes, and lettuce.

We have corn pones, corn bread, brown bread containing oatmeal, ordinary white bread, and oven toast—that is to say, slices of bread baked in the oven until it is brown all the way through. From Battle Creek we have malt honey, malted nuts, ripe olives, olive oil, fig and prune marmalades made without cane sugar, various crackers and grain preparations, and several other nut products. The Sanitarium health-chocolate, a sweet made without the use of cane sugar, and with chocolate divested of its caffeine, also appears on our table. We have eliminated dessert at dinner, having learned not only at Battle Creek, but in the sore school of experience, that the heterogeneous mixtures of cream or milk and cane sugar and various mushy stuffs, along with butter or lard, in the shape of pies and puddings and cakes, are extremely undesirable foods. We find the sweet, pure taste of malt honey an adequate and highly satisfactory substitute.



The Daily Swim

Fruits rarely appear on the table at dinner, since we do not wish to mix them with vegetables. They make their appearance in great abundance at supper, which we have at five o'clock. At this meal we have various cooked fruits, such as prunes or apricots or baked or stewed apples; and of uncooked fruits, oranges, apples, figs, bananas, grapes, and whatever else the market affords. With these we have zweibach and common bread or crackers. At both meals appears Yogurt, an acidulous and agreeable beverage which gratefully checks thirst and in itself nourishes, and is also the vehicle whereby millions of beneficial germs are introduced into the body.

The work of preparing and serving these two meals is done by one person—and that person has time left to play tennis and go in swimming with the rest of us. The total cost of the food is less than thirty dollars a week; cooked and served, its cost is about three dollars and a quarter a week per person. In this connection it should be explained that Bermuda prices, for even the commonest things, are in excess of prices in New York. We pay five cents each for eggs and twelve cents a quart for milk. We have oranges by the barrel, but they come from California, or from Jamaica by way of New York. We have olive oil at four dollars a gallon, and sterilized butter at fifty cents a pound. And in addition the figures quoted include expressage and steamer charges, and ten per cent. duty as well. We mention these things for the light they throw upon the relative costs of the vegetarian and carnivorous life.

The reader will also wish to know about the health of a family living in this manner. When we came here all our children were half-sick from too long contact with cities, and we were not used to the climate, and so one of them caught a severe cold. With this exception there has not been a day's sickness among them, nor the remotest trace of an ailment. If we were to describe their looks the reader might attribute it to parental blindness, and so the proper plan seems to us to insert a picture of them, and let the reader come to his own conclusions.

For the guidance of any housewife who may wish to try our regimen, we give a few typical menus, and also recipes for some of the favorite dishes of our family. We are all hungry when mealtime comes in our household, and

we enjoy the surprises of the menu with all the zest that we ever welcomed roast turkey and pumpkin pies in the old days. And this seems in some magical way to be true, not only of ourselves, but also of such guests as happen along. It is worth noting that three different persons, who have never before known or thought anything about vegetarianism, have stayed with us for periods of several months; and all of them have fallen into the ways of our household, have been well and strong, and untroubled by craving for meat—and in two cases have found, to their great dismay, that they were gaining in weight upon two “low proteid” meals a day!

The first of the tables which follow contains a typical menu for a week; and the second gives an extra list of dinners. The third shows what we do upon some special occasion; it was the banquet which we prepared for Mark Twain—only, alas, his physician had ordered him to be home by sundown, and he couldn’t stay to partake of it.

Inasmuch as all people cannot change their meal hours in accordance with those we have suggested, we give these menus upon the basis of three meals a day, with the various food elements properly balanced. We have also included simple desserts, for the benefit of those who do not care to dispense with this feature. The menus in our own home are similar to these, with the exclusion of the breakfasts and the dessert.^[1]

^[1] Very good vegetarian cook books are those entitled “Science in the Kitchen,” and “Healthful Cookery,” both of them by Mrs. E. E. Kellogg, the wife of the superintendent of the Battle Creek Sanitarium. Some of the books which are listed in another place as being those which a student of the new art of health may read will also furnish many good recipes. The “Art of Living in Good Health,” by Dr. Daniel S. Sager, will be found especially helpful in this regard. We give in the Appendix three simple menus of the Battle Creek Sanitarium. These menus have the food values indicated, and will be found very useful in giving a rough idea of the number of calories contained in ordinary foods.

MONDAY

Breakfast

Oranges
Poached eggs
Graham gems

Dinner

Lima beans, dried or fresh
Baked potatoes
Mixed nuts
Whole wheat bread
Lettuce salad
Tapioca pudding

Supper

Oven toast brown bread
Cottage cheese
Apple sauce
Almond cream
Figs
Bananas

TUESDAY *Breakfast*

Grape fruit
Corn meal mush with cream
Buttered toast

Dinner

Baked macaroni
Mixed nuts
Brown bread
Tomato salad with mayonnaise dressing
Indian meal pudding

Supper

Zweibach
Brown bread
Ripe olives
Stewed prunes
Dates
Bananas
Hot malted nuts

WEDNESDAY *Breakfast*

Baked apples and cream
Omelet
Pop overs

Dinner

Peas patties with tomato sauce
Baked sweet potatoes
White bread
Boiled onions
Baked custard

Supper

Oven toast
Whole wheat bread
Nut butter
Stewed fruit
Cottage cheese
Apples
Bananas

THURSDAY *Breakfast*

Oranges
Hominy with cream
Currant puffs

Dinner

Bean and nut croquettes with cream sauce
Baked egg plant
Graham bread
Boiled rice
Dates with whipped cream

Supper

Oven toast
Graham bread
Honey
Ripe olives
Apple sauce
Grapes
Bananas

FRIDAY *Breakfast*

Grapes
Scrambled eggs
Whole wheat gems

Dinner

Vegetable soup
Assorted nuts
Beet and lettuce salad with mayonnaise dressing
Corn pones
Cottage pudding

Supper

Golden maize crackers
Graham bread
Nut butter
Canned fruit
Bananas and apples

SATURDAY *Breakfast*

Grape fruit
Toasted corn flakes with cream
Buttered toast
Marmalade

Dinner

Baked beans
Cabbage slaw
Baked potatoes
Mashed turnips
Brown bread
Baked apples with cream

Supper

Oven toast
Brown bread
Cottage cheese
Sliced pineapple
Bananas
Figs

SUNDAY *Breakfast*

Grapes
Soft boiled eggs
Corn meal gems
Orange marmalade

Dinner

Pea and tomato soup
Succotash
Corn bread
Potato salad
Baked bananas
Mixed nuts and raisins

Supper

Zweibach
Oatmeal bread
Malted nuts
Ripe olives
Canned fruits
Bananas
Dates

EXTRA DINNERS

Yolks hard boiled eggs
Baked potatoes
Beets
Prune pudding
Vegetable soup
Cabbage salad
Corn bread
Baked custard

Scrambled eggs
Baked lyonnaise potatoes
Beet and lettuce salad
Dates with whipped cream

Macaroni with tomato sauce
Whole wheat gems
Egg salad
Apple tapioca pudding

Baked beans
Tomato, chili sauce
Mashed turnips
Lettuce with French dressing
Lemon jelly

Pea soup
Corn pones
Potato and onion salad
Cabinet pudding

Peas patties with tomato sauce
Mashed potatoes
Carrots with butter sauce

Baked nuttolene with cream sauce
Baked sweet potatoes
Stewed tomatoes
Baked apples and cream
Lima beans (fresh or dried)
Baked sweet potatoes
Lettuce
Corn pones
Stuffed dates

Baked beans
Lettuce
Corn (canned or sweet)
Nuts and raisins

RECIPES

Vegetable soup: Cut in dice three turnips, three carrots, three onions, three potatoes. Cover with water and simmer for thirty minutes. Cook one can of tomatoes, or one quart of fresh tomatoes, strain and thicken a little with flour. Add to vegetables and cook thirty minutes. Add butter and sprinkle with parsley.

Corn pones: Three cups corn meal, 1 heaping teaspoon salt, 1 tablespoon sugar, 1 heaping tablespoon butter. Add boiling water until meal is scalded, pat it into flat, thin cakes and bake three-quarters of an hour.

Mayonnaise dressing: Yolk of egg; add 1½ cups olive oil, drop by drop, stirring in one direction. Juice of two small lemons, 1 teaspoon salt.

Macaroni with tomatoes: Half package macaroni; drop into a kettle of boiling water. Boil vigorously for thirty minutes. To one can tomatoes add two onions chopped fine. Simmer until onions are done, then strain and thicken with flour. Put macaroni into colander and rinse with cold water. Add the tomato sauce and simmer gently for fifteen minutes. It is well to do this in double boiler to prevent burning.

Bean or pea soup without meat or pork: Soak two cups of split peas over night. In the morning slice and add two large onions and simmer for several hours. Strain.

Beans baked without pork: Use butter or nut butter instead.

Bean and nut croquettes: Cook dried beans until soft. Strain through colander to remove all skins. Add equal parts of walnut meat ground in chopper; season with salt and a little sage. Mix with beaten egg. Form into croquettes and bake until dry and nicely browned. Serve with tomato or cream sauce.

Baked egg plant: Boil egg plant until tender; pare and mash; mix with bread crumbs and eggs, and bake until nicely browned. A little finely chopped onions may be added if desired.

Peas cutlets: One cup pea pulp, one cup steamed rice, one grated onion, one-half teaspoon sage, one-half cup tomato juice, one-third cup browned flour. Mix together and mold in cakes two-thirds of an inch thick. Bake half an hour. Serve with tomato or cream sauce.

XII

BREATHING AND EXERCISE

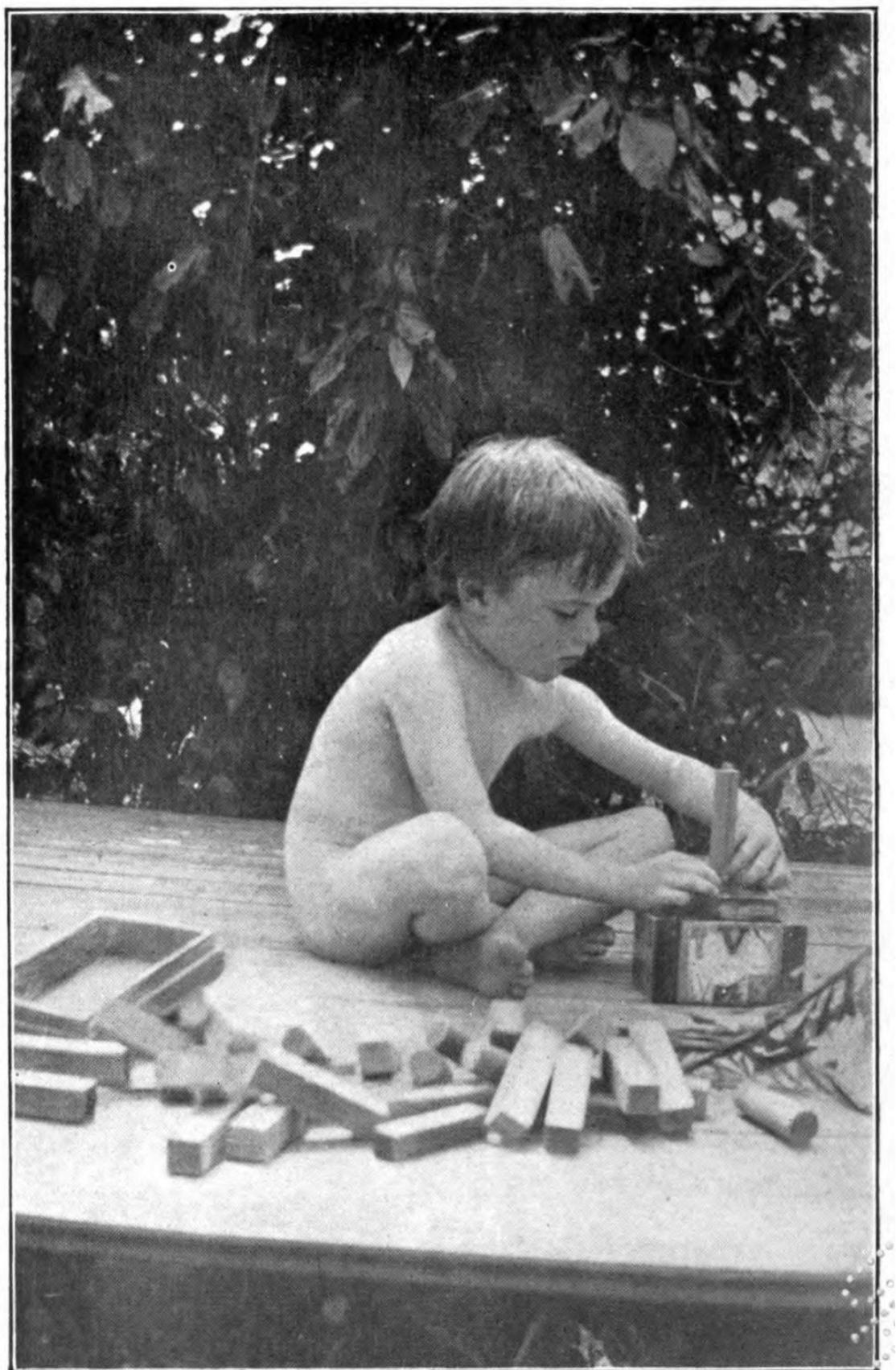
We have devoted most of our space to the problems of nutrition, since nutrition is the most important factor in the question of how to keep in health. We wish now to speak of other matters, of great importance in the art of keeping well; these are breathing, bathing, and exercise.

Many people have lived for more than a month without food. You can go for days without water. But if you are deprived of air for but a few minutes, your death is certain. Sixteen to eighteen times a minute the normal person respire, one breath being taken for every four beats of the heart, the central engine of life. Each time you breathe, the amount of air which passes into the lungs is about twenty-five cubic inches; which represent, however, but a small part of the actual capacity of the lungs. The average man can take into the lungs with an ordinary inspiration one hundred or more cubic inches, and is able to force out an equal amount with an ordinary expiration. If you have striven your utmost to expel all the air possible from your lungs, there will still remain about one hundred cubic inches of air within them. The total lung capacity of the average man is about three hundred and twenty-five cubic inches, or nearly one and a half gallons of air.

THE INDISPENSABILITY OF OXYGEN

Sunlight is the basis of all life. It is sunlight which plants absorb, and which they transform into materials which go to make up the living tissues of all things. The place of breathing in the process of life is manifold. But its primary function is to make available for the body's uses the sunlight, or energy, which is stored up in the food we eat. It does this by means of the oxygen which it contains, and the purpose of breathing is to obtain from the air an adequate supply of oxygen. Oxygen is one of the essential materials required for the support of life. Without oxygen the whole life process would come to an end. From every breath that is taken into the body, about one and a quarter cubic inches of oxygen must be obtained by the body, to

keep up the fire of life within us. You cannot burn a match, or your reading lamp in the evening, unless there is an adequate supply of oxygen; and even so does the body require this indispensable and all powerful element in order to maintain itself.



FRESH AIR IN BERMUDA

We have noted the fact that of the myriads upon myriads of swarming cells which the blood contains, a large proportion are the oxygen-conveyers. When you take air into your lungs, these cells absorb the precious element, and rush with it to all parts of the body. After distributing the oxygen wherever it is needed, they pick up for the return journey to the lungs all manner of débris and gases—the poisons which are produced by the organs of the body as they carry on their work. As Metchnikoff has shown us, it is the accumulation of poisons produced by the activity of our various organs which, unless properly disposed of, or kept below excessive quantities, bring about premature old age, the majority of all diseases, and early death. The amount of poisons which the average person throws off from the body with a single breath, as has been shown by delicate laboratory experiments, is enough to contaminate and render unfit for breathing three cubic feet or three-quarters of a barrel of air. Assuming an average of twenty breaths per minute, which is the normal rate for breathing for adults, the amount of air each person contaminates per minute will be sixty cubic feet, or one cubic foot a second.

If you hold your breath for a minute, you will be conscious of an extremely unpleasant feeling, which is the way in which the body manifests its urgent need for oxygen. The need of ventilation is not merely the need of oxygen, however. There may be plenty of oxygen in the air of a room which has been closed for some time, and which has been breathed in and out of the lungs of the people in the room; the trouble is that this oxygen is unfit for breathing, being full of impurities thrown off by the bodies of these people.

HOW TO CALCULATE ROOM VENTILATION

Dr. Kellogg has supplied some exceedingly useful calculations of the degree of ventilation needed in rooms of various sizes. “Every one,” he says, “should become intelligent in relation to the matter of ventilation, and should appreciate its importance. Vast and sometimes irreparable injury frequently results from the confinement of several scores or hundreds of people in a school room, church or lecture room, without adequate means of removing the impurities thrown off from their lungs and bodies. The same

air being breathed over and over becomes intensely charged with poisons which render the blood impure, lessen resistance and induce susceptibility to taking cold and to infection with germs of pneumonia, consumption and other infectious diseases which are always present in a very crowded audience room.

“Suppose, for example, a thousand persons are seated in a room forty feet in width, sixty in length, and fifteen in height; how long a time would elapse before the air of such a room would become unfit for further respiration? Remembering that each person spoils one foot of air every second, it is clear that one thousand cubic feet of air will be contaminated for every second that the room is occupied. To ascertain the number of seconds which would elapse before the entire air contained in the room will be contaminated, so that it is unfit for further breathing, we have only to divide the cubic contents of the room by one thousand. Multiplying, we have $60 \times 40 \times 15$ equals 36,000, the number of cubic feet. This, divided by one thousand, gives thirty-six as the number of seconds. Thus it appears that with closed doors and windows breath poisoning of the audience would begin at the end of thirty-six seconds, or less than one minute. The condition of the air in such a room at the end of an hour cannot be adequately pictured in words, and yet hundreds of audiences are daily subjected to just such inhumane treatment through the ignorance or stupidity of architects, or the carelessness of janitors, or the criminal negligence of both.”

TUBERCULOSIS POINTS THE MORAL

No circumstance has been more successful in impressing the great importance of fresh air and adequate ventilation upon the public mind than the success which has attended the open air cure for consumption. This is a mode of treatment of comparatively recent adoption in America, but it is by this time generally recognized as really the only possible cure for tuberculosis. The mortality from this disease is greater than any except pneumonia; another disease that proper breathing habits will do much to avert. In America one person in every nine dies of tuberculosis; and of the deaths which occur between the ages of fifteen and thirty-five, one-third are due to the great white plague. We give these figures on the authority of

Professor Irving Fisher of Yale, who is Secretary of the New Haven AntiTuberculosis Association. His interest in this disease is that of one who has had it, and who has cured it by the open air treatment. Of the authors of this book, one has had an experience similar to Professor Fisher. There is nothing academic about this insistence on the need of fresh air and proper breathing habits; literally, and in the fullest degree, it is a question of life and death whether you shall breathe properly, and have good air to breathe, or whether you shall not.

HOW BREATHING AIDS THE BATTLE OF THE BLOOD

To return for a moment to the processes of breathing, we find that the act of inflating the lungs is a blood-pumping process as well. This blood-pumping process has a great effect upon the struggle of the white soldiers of the blood to maintain the body against the inroads of disease. Each time that the wall of the chest is elevated after the lungs have been emptied, a suction force is exerted upon the large veins which enter the chest, especially those which come in through the abdominal cavity. "At the same moment," to quote Dr. Kellogg again, "the downward pressure of the diaphragm by which the liver, stomach, and other abdominal organs are compressed against the muscular walls of the abdomen, serves to force the blood from below upward, emptying the venous blood of the abdominal cavity into the chest, thus helping it toward the heart. The more tense and well developed the muscles of the abdominal wall and the stronger the muscles of respiration, the stronger will be this upward movement of the blood. When the abdominal muscles are weakened by improper dress, by corsets, tight lacing, or by wearing of belts or bands or by sedentary habits, especially sitting in a stooped position, the weakened muscles yield to the downward pressure of the diaphragm, thus neutralizing to a large degree the beneficial influence of this action. This condition is unquestionably a cause of chronic disease of the liver and stomach, inactive bowels, and possibly lays the foundation of cirrhosis of the liver, spleen, and other grave disorders of the abdominal region."

It is very obvious how deep breathing will thus influence the vigor of the blood's army of cells. Deep breathing forces the blood to rush into the lungs, there to be charged with oxygen. Without this oxygen the white cells

die. Vigorous breathing also directly aids digestion, and promotes the absorption of food materials. Those who have slow digestion will find that breathing exercises will be of especial benefit. In ordinary breathing of a quiet person, the movements of the chest are so slight as to be scarcely noticeable, but when vigorous breathing is indulged in, the diaphragm as it moves up and down kneads the stomach and its contents and, very materially, assists the digestive organs.

HAVE FRESH AIR AT NIGHT

During sleeping hours the breathing movements are slighter and slower than when one is awake and active. It is necessary that the activity of the body should be lessened in order that rest may be secured; and yet the work of the liver, kidneys, and other organs which are engaged in throwing off poisons goes on continually; as does also the repairing work of the living cells, which are forever building up the parts of the body broken down by work or sickness. For some six to nine hours the body is thus occupied in resting and repairing itself, in order that on the next day it may respond like a living machine to the demand of the conscious mind. We should do all in our power to help on this recuperative process; and no way will be more effective than to sleep, out of doors, or with the head at a window, or at least in a well-ventilated room.

There are a great number of breathing exercises described in various books on the subject, but the best breathing exercise is natural breathing. If the head is kept erect, and the shoulders low so that the chest is upright; if breathing is carried on through the nostrils, and the habit of deep breathing carefully cultivated—there will be no need for special exercises, save in the case of invalids. The most effective of all breathing exercises is to run or walk rapidly, or walk up a hill, or up stairs, if these be in the open air, with the head well back. This exercise heightens the action of the lungs, and all parts of the body are flooded with fresh air.

HOW EXERCISE AIDS THE BATTLE OF THE BLOOD

The question of breathing properly is intimately bound up with the question of exercise. The best of all exercise is play. All games in the open

air which a person takes part in for the love of them far surpass the cleverest and most scientific sets of rules which physiologists have ever evolved. Unconscious performance of all the functions of the body is the ideal of hygiene. Exercise aids the battle of life within us in a direct manner. Exercise breaks down worn out tissue, making room for new and healthy tissue. It increases the rate of oxidation or burning up of fuel within us, and this in its turn enables the body to get rid of waste of material. Exercise also increases the strength and endurance of the muscles and fibres.

When muscles become weak, they relax and allow various portions of the body to drop into positions which are not only ungraceful, but are decidedly injurious. When the muscles are not used and become flabby, the shoulders get rounded and drop forward through the weakness of the muscles which are intended to hold them back in position. The ribs which form the framework of the chest not being properly sustained by the muscles attached to them, gradually fall inward, thus flattening the chest, and compressing the lungs. There is a very close connection between gracefulness of carriage and sound bodily health.

The person who lounges, or slouches, be it ever so picturesquely, does so at the expense of the body. Proper exercise will prevent these physical defects, and will remedy them in most persons who have not yet attained middle age. Even in advanced years, say the physiologists, much may be done to correct these physical deformities by properly directed and systematic exercises.

EXERCISES MAKE NEW BLOOD

Exercise has another most important task in supplying an adequate amount of blood to the bones of the body, in order that these bones may carry on their work of manufacturing fresh blood for the use of the body. Unless these bones are bathed with the already existing blood of the body, which carries to them oxygen and nourishment, the process of manufacturing new blood, which goes on within the marrow of the bones, would quickly cease. It has been demonstrated by science that muscular activity increases the blood flow through the muscles as many as six times.

Here, then, lies perhaps the first hope for supplying new blood to any body which has begun to deteriorate through the accumulation of poisons emanating from the large intestine, or from the other organs. Exercise will supply the blood-producing bone marrow with six times as much raw material to make new blood as a sedentary mode of life would produce, and at the same time this six-times-strengthened flood will wash out of the crevices of the bones and muscles and fibres the stored up poisons. For these purposes, the exercises which move the large muscle masses are the most helpful. Dr. Benton A. Colver, of the Battle Creek Sanitarium, to whom we are indebted for assistance in preparing this chapter, names the following exercises as being beneficial for this purpose:

Low knee bending, stretching and heel sinking, and heel raising; lying on the floor with the weight supported by toes and hands, and lowering and raising the body; raising the body by the arms, holding to a bar above the head; walking with a vigorous stride, and running and swimming.

Of all these exercises, swimming is theoretically the best, for the reason that it exercises equally all the muscle masses in the body, and requires the best balanced of all movements. Walking and running come next in the order of excellence, simply for the reason that they can be carried on best in the open air and without the bother that may accompany the performance of more formal exercises.

EXERCISES WHICH RETARD AUTO-INTOXICATION

Another way in which exercise directly helps the battle of the blood within us, is by assisting such organs of body-poison elimination as the spleen, the liver, and the portal system of veins. It is in these organs that the exhausted blood is broken up and cast off. The blood in these organs is loaded with broken down tissue and other waste material from the body, and is contaminated with gases and poisons. In the body of the person who leads a sedentary life a great volume of blood settles in these organs and is prematurely put out of use.

Proper exercises will empty this great tank of stagnant blood as easily as a sponge is emptied by the pressure of the hand. This passive blood, having access to all the organs of digestion, is largely responsible for the supply of

inferior digestive juices, and thus is a leading factor in indigestion, loss of appetite, and such diseases as catarrh of the stomach and bowels. If, however, this blood is pumped on as it should be to the heart and lungs, there to be cleansed, the fresh blood rushes in to fill its place, armed by the activity of the lungs with its life-giving ammunition of oxygen.

By persistently keeping up this emptying and filling of the portal veins, and of the spleen and liver, the old cinders left from the oxidation of food are washed away, new digestive juices are formed, and the whole tone of the body is improved. For such purposes such exercises as the following are extremely valuable:

Stand erect and, with the hands on hips, bend the trunk forward, backward, and sideward, keeping the legs stiff. Trunk rotation, performed by bending forward and then describing as large a circle as possible with the head thrown first to the right and then to the left, and bending the trunk backward as far as possible when that segment of the arc is reached; lying on the back and raising first the head, second the feet, with bent knees, and third, the feet with straight legs. These exercises stretch the diaphragm against the liver and portal vein, and thus squeeze out the blood from these organs and send it back to the heart and lungs.

EXERCISES WHICH PROMOTE DEEP BREATHING

A third manner in which exercise directly assists the battle of the blood is by increasing heart action and deep breathing. Exercises which accomplish these functions insure an abundant supply of oxygen to the blood and the tissues. In this manner, more heat will be produced in the active tissues, and the blood current will carry this beneficial glow of heat to the most distant parts. This toning up of the "heating system" will be evidenced by the appearance of perspiration. When skin and lung activity are thus increased, the accumulated wastes of the body are quickly eliminated.



OUTDOOR EXERCISE.

The person who takes vigorous exercise in the open air such as playing games like tennis or golf, or who walks vigorously, will have no need for formal breathing exercises. For those, however, who cannot readily obtain outdoor exercises the natural way, the following chest movements and breathing exercises are recommended. They should be taken with the body free from tight clothing, and either in the open air or a well-ventilated room. First, raise the hands above the head as far as they can reach, and then bring them forward and upward several times, and then upward and downward on the side of the head, inhaling on the uplifting of the arms, exhaling on the sinking of the arms. When the arms are lifted above the head, opportunity is given for the air inhaled to reach the upper part of the lungs, parts which in the sedentary person are very rarely used, and where usually the germs of tuberculosis begin their evil work. Arm extension forward, breathing deeply with arms carried sideward and backward, at shoulder height. If those who exercise in their rooms will be careful to breathe only through the nose and will keep the head erect, they will find that the performance of almost any set of exercises will serve also as breathing exercises, since they will increase lung activity.

XIII

BATHING AND CLEANLINESS

The soldiers of the body which carry on for us the battle against disease, old age, and death, have as great and as constant a need of water as do the human soldiers, part of whose equipment is always the indispensable canteen. Water is needed by the body in many ways, but it is especially required by the blood. Water is the solvent in which float the white and red corpuscles of the blood, and the many nutritive elements which the blood carries through the body, and the particles of waste material which it bears to the lungs to be burnt up, or to the other excretory organs to be ejected. By the aid of water, the minute particles of food which are broken up and transformed by the chemical processes of the body are conveyed to the most distant fibre of the intricate human mechanism, wherever repair or new growth is required. No other element of nature could so well carry on this function as water. It is so limpid and mobile that it can move through the most delicate and intricate network of veins, and can find its way by osmosis or percolation into such parts as are inaccessible by openings.

The human body is constantly throwing off water. A large portion is lost by evaporation from the skin, upon which it is poured out by millions of what might be termed little sewer pipes or sweat ducts, for the purpose of washing away impurities from the system. The kidneys remove a considerable quantity, bearing with it poisonous elements in solution, the product of various vital activities. In other ways water is removed from the body, to the amount of about five pints in twenty-four hours. This loss must be made good in order that the requisite fluidity of the blood shall be maintained; and the need of the body is expressed by thirst. Beverages which contain other substances, as flavor, or as part of some mixed drink, are useful as thirst quenchers just in proportion to the amount of water which they contain.

Physiologists point to the evaporation of water from the surface of the human body as being one of the most perfect adaptations of means to ends exhibited in the whole circle of life. The vital activities of the body occasion

the constant production of heat. At times the heat is greater than is needed, and would destroy the vitality of certain tissues if it were not speedily conducted away, just as too much heat in a stove would melt the iron of the stove. The evaporation of water from the skin accomplishes this heat dispersal. When external heat is great, perspiration in the normal, healthy person is more active than when external heat is less than that in the body, and, by this provision of Nature, the temperature of the body is maintained at about 100° Fahrenheit under all circumstances, and thus man is enabled to exist under such great extremes of heat and cold as are found in nature.

There are numerous other ways in which water is essential to the process of life within us. The free drinking of water greatly favors the elimination from the system of the products of waste. It hastens tissue change, and encourages the assimilation of food. And apart from its use internally, it has also a very great value as a means of applying heat to or abstracting it from the body for remedial purposes, to say nothing of the functions it performs as a cleansing agent. Of late years the value of water in therapeutics has become generally recognized by the medical profession, and all over the world its use as an active agent has increased. Indeed, in the view of some physiologists, ordinary pure cold water is by far the most powerful and useful of all known healing agencies. It heals not by any strange or occult power, but by co-operating with the natural forces of the body, by aiding to the utmost those physiological processes by means of which the body sustains itself in health, and resists the encroachments of disease by the means of its bodyguard of blood cells, and by maintaining at its high pitch its innate vital resistance. When the Austrian Priessnitz first began the use of water in his mountain village a century ago, the world believed that the wonderful cures he wrought were accomplished by mystical charms or incantations by which he was supposed to communicate to the water its healing power. Modern science, however, has revealed the secret of water's potency as a curative agent, and hydrotherapy, or curing by water, is now as well recognized as almost any other branch of medical science.

THE VALUE OF BATHING

The daily cold bath is one of the best ways of keeping the doctor at a distance. Cold water has the property of increasing vital work of all kinds.

When it is applied to the skin “impulses are sent inward that awaken every organ of the body,” says Kellogg. Let us see what takes place: when a person dips his body into cold water, as in sea bathing, or when he steps into the bath at home, the first thing he does, which in fact he finds himself doing involuntarily, is to draw in a deep breath.

“Oooh-h-h!” he says, but he says it with an indrawing breath. The lungs swell out, the heart begins to pound away with unusually increased vigor and strength, and every part of the system is stimulated. Cold bathing and deep breathing are two valuable things which go inseparably together. The deep breathing increases lung activity, and the lungs bring in more oxygen; the heart circulates the blood with greater force, and hence more and better blood is carried to every tissue of the body. The result is a stirring up of the bodily forces, and a distribution throughout the system of a larger amount of highly vitalized and oxygenated blood.

It has been shown definitely that cold bathing increases enormously the number of white blood corpuscles in the blood. Whether this result is accomplished by the birth of new cells, or by the calling forward of cells from remote parts of the body into the general stream of the blood, is not generally known; but the fact remains that counts of the blood cells taken just before and just after the body has been stimulated by cold water show a decided increase in the army of the warrior cells.

The benefit of sea bathing comes not from the salt in the air or in the water, as some people suppose, but simply from the cold water. The reaction from the dip into the cold water, which is brought about by the blood rushing to the surface to supply the heat which has been taken from it by the application of the water, is one of the most valuable of all curative processes. It is this reaction that sends the blood cells scurrying actively throughout the whole fortress of the body.

HOW COLD BATHING AIDS NUTRITION

Another way in which the application of cold water promotes the functions of life is by the stimulation of the secretion of gastric juice which it accomplishes. It thus helps on actively the digestive processes by which

food is absorbed and taken into the blood. The liver and the salivary glands are stimulated in the same way.

When applied to the face, cold water stirs up the flagging energies of the brain, by invigorating the blood. A dash of cold water upon the chest produces a stimulation of all the bodily forces, which a tired person will find more valuable than any pick-me-up or tonic or cup of tea, or nip of whisky or other alleged stimulant could possibly be. Applied over the heart, this organ is made to beat with greater steadiness and vigor. Application to the stomach causes increased production of pepsin and acid or gastric juices. Over the bowels it stimulates intestinal activity; over the loins it increases the action of the kidneys. A cold compress, or a douch over the liver will cause increased liver activity. Every organ in the interior of the body may be thus aroused to increased activity by a simple application of cold water upon the skin overlying the organ, for thus a rush of blood will be caused to that particular portion. It is necessary that the application should be brief, three or four seconds to as many minutes. These short cold applications of water to the skin will increase immediately the activity of any sluggish part, or of any organ whose function we wish to increase as a means of aiding the body in its battle against the causes of disease.

The whole nervous system derives benefit from the stimulation of brief cold baths. This is one of the most valuable functions of water. Hydrotherapy has come to be a most valuable adjunct to the treatment of all nervous diseases. A slow stomach may be wakened up and set to doing effective work by a general cold bath taken daily, or by a local application of cold water. A cold water bag over the stomach for half an hour just before meal time is a wonderful appetite awakener, which may be used by persons whose circumstances preclude them from the general cold bath and the exercises which cause a natural desire for food.

The best of all prescriptions for cold feet is to stand in very cold water a half inch deep and rub one foot with the other in alternation for five minutes. Hydrotherapy is the principal curative agent employed in the great Battle Creek Sanitarium, and its branches throughout the world, and in his book "Rational Hydrotherapy," Dr. Kellogg has presented in a shape that makes the knowledge available to everybody the modes of treatment which may be employed at home. "A good way," says Dr. Kellogg, upon whom

we draw for information in the preparation of this chapter, “is to stand in the bath tub with the cold water faucet open and the plug out.” It will not be long before the feet will be red and will fairly burn with the afflux of fresh, warm blood which will rush to the feet.

Hot water can be used in conjunction with cold water, since heat tends to lessen vital work, and so heat may be employed when it is desired to diminish organic activity. Pain is one direct evidence of excessive activity. Heat is nature’s great remedy for internal pain. Heat cuts off the influence of cold and at the same time diverts the blood to the surface of the body. Cold, on the contrary, usually increases pain when the seat of it is some internal organ. Sometimes heat and cold are applied at the same time, as for a toothache, for instance, when a hot fomentation is applied to the cheek and an ice bag to the neck under the jaw. Pain in the pelvis is almost always relieved by a very hot foot bath or leg bath, which relieves the congestion by diverting the blood into the legs, and thus removes the condition which was responsible for the pain.

THE CARE OF THE TEETH

It appears to be a fact that in the United States the profession of dentistry, both mechanical and medical, has been carried to its highest point. No doubt Americans will cheerfully assure themselves that American brains and “bustle” are responsible for this condition. But the truth can not be quite so comforting; the great development of dentistry in this country must be due to the demand for it; and the demand for it evidences a state of affairs that is far from reassuring.

So rapid has been the increase of degeneration of the teeth in modern times, that many physiologists have seriously asked the question, “Will the American race become toothless?” To-day, while artificial teeth are manufactured from such a variety of substances and sold at such a variety of prices, it would seem that Americans are becoming a race of “store teeth” men and women.

As with all other branches of hygiene, dentistry is now beginning to discover the ideal of *prevention*; recognizing that the sanitary care of the mouth is a more important object than the most cunning imitation of teeth,

or the most ingenious masterpieces in bridge and crown work. Under the leadership of a man who will be recognized in the future as a pioneer in the cause of health, Dr. D. D. Smith, of Philadelphia, a large and rapidly growing body of dentists have formed what is termed the Prophylactic School, the development of which will result not merely in the prevention of a great deal of disease of the teeth and mouth, but of all the body. Physicians in general, and even most dentists, have only begun to recognize the part which the mouth plays in the causation of diseases.

At the present time, there are, roughly speaking, about 14,000 dentists in the United States, who annually extract twenty million teeth, manufacture and insert three million artificial teeth, and hammer into the cavities of diseased teeth at least three tons of pure gold, to say nothing about the many tons of mercury, tin, and other metals employed in fillings. When the principles of the Prophylactic School spread, it is safe to say that while the importance of the dentist will become even more generally recognized than it is to-day, nevertheless he will pull fewer teeth, and use less gold and other metals. The principle upon which Dr. Smith, and his rapidly growing band of followers, build their work, is an intelligent recognition of the fact that there are in the human mouth to-day, as has been the condition through all the centuries, highly malignant features of general infection and causes of numerous diseases which until now have been wholly unperceived and neglected. The ordinary physician tells his patient to poke out his tongue, when he looks for an index to that patient's general condition of health, but he does not look above or below or around or about the tongue, where, in a great number of cases he might find not merely the symptoms but the cause of his patient's ailment.

“THE VESTIBULE OF LIFE”

To show some of the common mouth conditions that make it almost an ideal medium for bacterial culture, we quote the following paragraph from Dr. Smith, adding the fact that his statement is one with which all up-to-date physicians concur:

“The mouth, with its large extent of dentate surface, becomes quickly infested and infected with all manner of bacterial formations, decomposing particles of food, stagnant, septic matter from saliva, mucous and sputum,

not infrequently with pus exudations from irritated and inflamed gum margins, gaseous emanations from decaying teeth and putrescent pulp tissue, salivary calculus (tartar), nicotine, and the chemical toxins, or poisons, of decomposition which result from a mixing of mouth secretions, excretions and food remains in a temperature constantly maintained at the high normal of ninety-eight degrees Fahrenheit. While this may seem a formidable array, it fails to prevent any of the sources of infection connected with untreated teeth; and incredible as it may appear, these conditions are found not in the lower classes alone, but in general mouth conditions in high and low born, fastidious and boor, king and peasant.”

“Try to estimate the amount of poisonous products that would be generated if such a surface were smeared over with the various foods from the dining table, and these allowed to decompose,” says Dr. Alfred C. Fones,^[2] “and a fair idea may be obtained of the amount of decomposition that is taking place in unsanitary mouths. Nor is this simile forceful enough, for the food in the mouth is in one of the most favorable environments known for the activity and virulency of germ life, so that the products generated would be far more numerous, more poisonous and irritating in every action, than such products from food decomposing in the open air.”

^[2] In his essay “Clean Methods, The First Law of Hygiene.”

HOW MOUTH INFECTION SPREADS

Mouth infection, due to the teeth, sees its most critical period during that of childhood and early youth, a period in which the mouth under present conditions is almost entirely without intelligent care. Children’s mouths, says Dr. Smith, are frequently veritable crucibles in which are generated chemical agents and compounds highly detrimental to the teeth themselves, and not less to the general health of the child. The poisons arising from decaying food particles and decaying teeth themselves, vitiated salivary and mucous secretions, germ life upon the teeth and gums, and breaths loaded with emanations from stagnant septic material, all with the high temperature of ninety-eight degrees, insinuate into the general circulation of the blood a constantly increasing infection, which will later on find expression in many diseased conditions, and often in chronic and fatal disorders. It may appear,

as it commonly does, in stomach or kidneys, in lungs or nervous system, in heart, brain, or skin, in any organ or tissue, indeed, to which mouth toxins are directly or indirectly conveyed. Experience has shown that it is not only possible, but entirely practicable to arrest and prevent teeth diseases in the mouths of children, and at the same time to keep the mouth aseptic or free from germ life.

Not only does an infected mouth work havoc to the body of which it is the vestibule, but it spreads disease about it. The original experiments of Koninger have shown that in a room where there is no current of air perceptible, a person coughing or sneezing can scatter germs to a distance of more than twenty-two feet. They are conveyed through the air by means of little droplets of saliva. These globules are microscopic balloons, having a bubble of air in the center, and remain in suspension but a short time. Ordinary breathing will scatter these droplets to a considerable distance, but, of course, their germ-carrying capabilities are most marked during coughing and sneezing. The more microbes the mouth contains the greater the danger of infection. Washing the mouth has the effect of decreasing the microbes of such diseases as diphtheria and consumption, and other bacilli susceptible of being scattered abroad in these salivary droplets. Placing the hand or a handkerchief over the mouth prevents the emanation of droplets charged with bacilli. So well is this fact of droplet germ infection recognized, that in many operating rooms no one present is allowed to speak during operations. Chronic headaches, neurasthenia, constipation, coughs and colds, and many other grave troubles, have all been helped and many times cured by “oral prophylaxis” or proper mouth treatment.

The practical application of the discoveries and recommendations of the new school of dentists can be expressed very simply and briefly, and if followed out, will undoubtedly prove of tremendous service to the white cells in the battle of the blood. It must be remembered that proper mastication of food, which we have seen to be a leading principle of the new hygiene, cannot be carried out unless you have a good and healthy mouth. Five brushings a day at home is the ideal and proper care for every mouth, for those who eat through the ordinary routine of three meals a day. The first thing in the morning the teeth should be thoroughly brushed with tepid water to remove the decomposed mucous and saliva produced in the mouth during sleep. After eating the teeth should be cleansed with the help

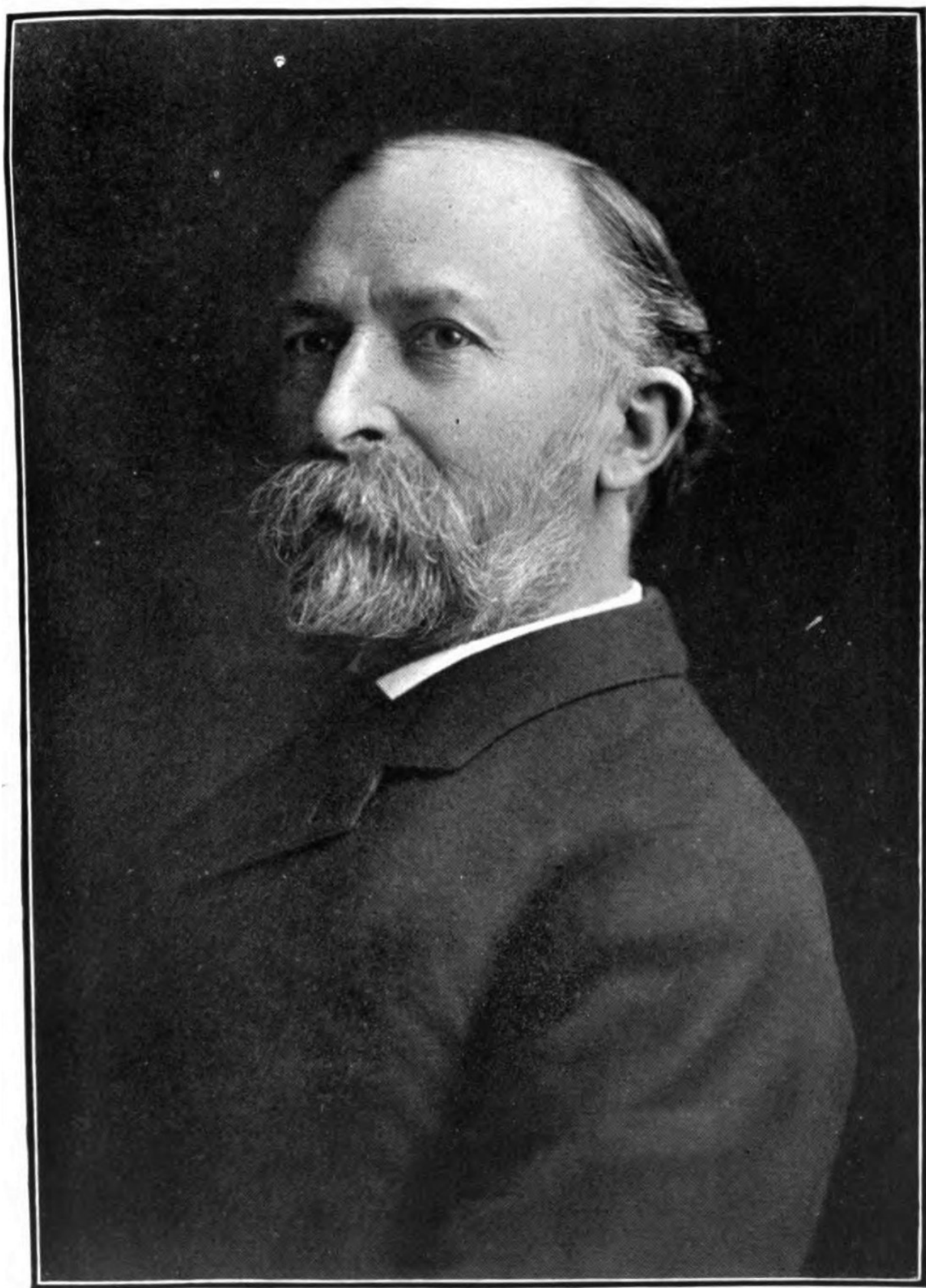
of a dentifrice. The thorough removal of grease is a chemical process, not to be accomplished by mere brushing, and therefore requires a solvent such as is contained in a good dentifrice. Such duties soon become habits; and if they are based upon common-sense, the health which they will bring will more than compensate for the trouble involved.

XIV

A UNIVERSITY OF HEALTH

There have been frequent references in this book to the Battle Creek Sanitarium, and to Dr. J. H. Kellogg, its superintendent. We have written here of the art of staying well, but many people are sick, and are in need of special advice and assistance; to such we believe that we can do no greater service than to tell them of this Sanitarium and its work.

The institution is not a commercial one; its founder is one of the great humanitarians of the time, as well as one of the great scientists. None of its thousand odd men and women workers receive more than a bare living for their services, and the institution is legally so constituted that all its profits must be turned into the work. Therefore, we hold it to be a public duty to spread as widely as possible the facts relating to it. Mr. Horace Fletcher has called Battle Creek the “Mecca of Health.” More aptly still, the Sanitarium has been named a “University of Health”; and no image could be more essentially true.



**DR. J. H. KELLOGG,
Of the Battle Creek Sanitarium.**

For, while the people at Battle Creek realize that the record of the institution for more than forty years in curing sick people is one to which they may point with pride, yet in their view this good work is but a trivial thing in comparison with their principal object, which is the conversion of those who come to them to be cured, into home teachers and missionaries of the truths of right living. It is wonderful to observe to what a great extent success has already rewarded their efforts, to see the signs which indicate the growth of public interest in their work.

Dr. Kellogg took charge of the institution which is now known as The Battle Creek Sanitarium thirty-two years ago. The institution at that time was a small two-story building, known as a water-cure or health institute, with three or four cottages and twelve patients. With the changing of the name and management, and the application of scientific methods, a new era of prosperity began, and the work has steadily progressed ever since.

The Battle Creek Sanitarium was the first attempt to assemble in one place all rational means of treating disease in combination with the regulation of diet and habits of life, and giving special emphasis to physiologic or natural methods of cure. The institution has for many years been recognized as the leading establishment of the sort in the world.

From the beginning, the Sanitarium has been non-sectarian in character. Although a deeply religious spirit pervades the place, the institution is not and never has been under the control of any denomination. For many years it was closely affiliated with the Seventh-day Adventist denomination, because of the preponderance of persons belonging to this denomination among its managers and employees. For years, however, this affiliation has ceased to exist.

The institution is non-dividend paying. That is, it is a strictly altruistic or philanthropic enterprise. The charter which it received from the State requires that its earnings shall be devoted to the development of the enterprise and the maintenance of its charities. Dr. Kellogg receives no compensation for his labors in connection with the institution, and the thirty or forty physicians and business managers who are associated with him in

his work likewise accept very meager compensation for their labors. Dr. Kellogg has for many years received a liberal income from the sale of his books, foods, and from his various inventions, but the income from these sources, as well as from the institution itself, has been devoted to the carrying forward of the humanitarian work to which he has devoted his life. The Haskell Home for Orphans, The Bethesda Rescue Home, the Life Boat Mission in Chicago, The American Medical Missionary College, and other charitable and philanthropic enterprises are allied enterprises which have grown out of the work which began at the Battle Creek Sanitarium.

The institution has never been endowed, and therefore, if the work was to grow, it was necessary to make money. The authors of this book have seen and read the legal documents by which Dr. Kellogg turned over to the American Missionary Association nearly everything of which he was possessed. The value of his work as a surgeon, estimated at prevailing rates for such work, would be at least fifty to sixty thousand dollars yearly. He touches not a cent of this money, nor does he touch his salary as superintendent—which he himself placed at the figure of twelve hundred dollars. There are many other physicians connected with the institution who, as specialists in New York or Chicago, would be in receipt of large incomes, but they are as content as is Dr. Kellogg to accept a bare pittance, finding their joy in the work they are doing.^[3]

^[3] The reader must be warned that there are many charlatans and shrewd business men who have taken advantage of the work of Dr. Kellogg and of the prestige of the name “Battle Creek.”

The energy displayed by the faculty and staff of the University of Health in carrying on their work is nothing less than astonishing. During one week when the writers were at the Sanitarium, there were more than a thousand patients all told, including the non-paying ones. There are many days when Dr. Kellogg operates from early in the morning until late at night, having very many highly difficult and dangerous operations to perform, for he is well known as a surgeon. After such a long day in the operating room, without a break for food or rest, he will give one of his lectures to the patients, or go the rounds of the wards, winding up the day by attending to a mass of business or writing or studying in his laboratories. He works continually, day in and day out, for eighteen hours a day; and this he has

done for the past thirty-five years or so. He wrote one bulky book containing much technical and scientific matter in ten days, using three or four stenographers, and working in stretches of twenty hours at a time. He has never taken a holiday. All of his many journeys abroad or in this country are on matters connected with his mission in life; and while on his journeys he is continually writing or studying, and carrying on the direction of his multitudinous affairs by letter or telegraph. Yet to-day, at the age of fifty-five, he shows no signs of diminution of energy; no signs of nervous breakdown, or of the ailments which bring thousands of business men and women to him for treatment.

He himself thinks that there is nothing very remarkable in all this. He attributes it to his abstention from meat, from tea and coffee, alcohol and tobacco. He never eats more than one “hearty” meal a day; his second meal, when he takes one, consisting of a little fruit. His sole regret is that during the first fourteen or fifteen years of his life he ate meat. He believes that any child, if it begin right, can, when it grows up, do all that he is doing.

“I was,” he said to a friend, “a puny, undersized, ailing child; born when my father was more than fifty. It was the accepted opinion that I would not live to be a man which I fully believed. I had an appetite for knowledge and resolved that since I was to die early I must study and work very hard in order to accomplish a little something before I died. So I would study until one to three o’clock in the morning; then rise at six. From the age of ten I have fully supported myself. All this deliberate stealing of time from sleep resulted in a permanent stunting of my growth. And as I went on in life, I kept up the same habits of night work. And yet, I have only once been troubled by an illness; which came upon me a few years ago as a result of overwork. But which I got rid of; and now I am in better bodily condition than I was twenty-five years ago. But I was not handicapped by a great number of things that are bars to other workers, over which they stumble. I have slept when I could in the open air; I have drawn from air, water, light, heat, and proper exercise, the benefits that inhere in them; and I have nourished my body on wholesome foods. I mention these points with insistence—these points that seem so freakish to many people—simply because to me they are fundamental points in the physiologic, or natural, way of healing and of living.”

Dr. Kellogg publishes a big magazine of large circulation named *Good Health*; and in this he teaches that health is not a mere negation of ailments—a state of being free from rheumatism, or consumption, or biliousness, or any other of the “thousand natural shocks that flesh is heir to”—but that it is being wholesome, happy, sane, complete, a unit—a man or woman eating, drinking, sleeping, working, playing, functioning in all parts as naturally, as inevitably, as easily and as unconsciously, as a flower grows.

One of the writers has told of his experience many years ago, when he went to a physician and requested to be helped in keeping well. He went to Battle Creek Sanitarium on account of the illness of his wife, and when one of the physicians proposed to him that he himself undergo the treatments, he answered (having in mind this earlier experience, and of the doubts it had bred in him), “There is nothing the matter with me at present that I know of.” The answer of the Sanitarium physician was, “The less there is the matter with you the better, from our point of view.” And so he realized that at last he had found a place where his own idea of health-preservation was understood.

He accepted joyfully the offer to assist him in getting a scientific understanding of his own bodily condition. A drop of his blood was taken and analyzed, microscopically and chemically. He went to the diet table, and for three days ate precisely measured quantities of specified foods; during the period all his excretions were weighed and analyzed and examined under the microscope. A thorough physical examination was made, and also a series of tests, upon a machine invented by Dr. Kellogg, to register the strength of each group of muscles of the body. The results of all these examinations were presented to him in an elaborate set of reports and charts, together with a prescription for treatments, diet and exercise. He had stated that there was nothing the matter with him, so far as he knew. He found that anaerobes—the dangerous bacterial inhabitants of the intestinal tract—numbered something over four billion to the gram of intestinal contents—a gram being about a thirtieth part of an ounce. During the six weeks of his stay at the Sanitarium the more important of these tests were repeated weekly; and when he left, the number of anaerobes had been reduced nearly ninety per cent.

Dr. Kellogg terms the system of treatment employed by the Sanitarium the Physiologic Method, and he writes of it as follows:

“The Physiologic Method consists in the treatment of the sick by natural, physical, or physiologic means scientifically applied.

“The haphazard or empirical use of water, electricity, Swedish movements, and allied measures is not the Physiologic Method. It is no method at all. It is empiricism, at best; at its worst, it is quackery. The application of the Physiologic Method requires much more than simply a knowledge of the technique of baths, electricity, movements, etc. It requires a thorough knowledge of physiology, and an intelligent grasp of all the resources of modern medical science. For, while the Physiologic Method depends for its curative effects upon those natural agencies which are the means of preserving health, and which may be relied upon to prevent disease as well as to cure, it recognizes and employs as supplementary remedies, all rational means which have by experience been proved to be effective.

“The Physiologic Method concerns itself first of all with causes. In the case of chronic maladies, these will generally be found in erroneous habits of life, which, through long operation, have resulted in depreciating the vital forces of the body and so deranging the bodily functions that the natural defenses have been finally broken down and morbid conditions have been established.

“Chronic disease is like a fire in the walls of a house which has slowly worked its way from the foundation upward, until the flames have burst out through the roof. The appearance of the flame is the first outward indication of the mischief which has been going on; but it is not the beginning. It is rather the end of the destructive process.

“The Physiologic Method does not undertake to cure disease, but people who are diseased. It recognizes the disease process as an effort on the part of the body to recover normal conditions,—a struggle on the part of the vital forces to maintain life under abnormal conditions and to restore vital equilibrium.

“At the outset of his course of treatment, the patient is instructed that his recovery will depend very largely upon himself; that the curative power does not reside in the doctor or in the treatment, but is a vital force operating within the patient himself. The Physiologic Method is based upon this fact.



A GROUP AT THE BATTLE CREEK SANITARIUM (DR. KELLOGG ON THE RIGHT).

“So the treatment of a patient consists, first of all, in the exact regulation of all his habits of life, and the establishment of wholesome conditions. The simple life and return to Nature are the ideals constantly held up before him. He must work out his own salvation; he must ‘cease to do evil and learn to do well’; he must cease to sow seeds of disease, and by every means in his power cultivate health.”

XV

HEALTH REFORM AND THE COMMITTEE OF ONE HUNDRED

We have set forth the underlying principles of the new art of health; and we have shown how these principles may be applied by individuals, and how they have been formulated and taught at the great University of Health at Battle Creek. It remains to give an account of a great national movement which has for its aim the spreading of a knowledge of the new hygiene in a semi-political way, a circumstance which to our minds proves that not only this nation but the whole of modern civilization is on the eve of a great revolution in its habits of living, and that this revolution will have for its rallying cry the word "Knowledge." And more especially, "Knowledge of Our Bodies, and of How to Care for Them."

The state of ignorance of the majority of people concerning the workings of their own bodies and the way to take care of them is to-day one of the greatest barriers to human progress. Few people realize that they ought to care for their bodies; or that they ought to know about their bodies until they are actually broken down. Men use their intelligence more aptly elsewhere; but all progress in other directions, in the arts and crafts and the labors of modern industry, will go for nothing if we do not learn to apply our intelligence to the matter of health.

More and more does the need for knowledge press home upon us. It is impossible for the race to survive unless that knowledge is spread. Our ancestors, it is true, knew less of their bodily make-up and bodily care than we do, but our ancestors did not need it so much. They were country dwellers, and people of the open air; they were not slaves of machinery and of office routine.

Dr. J. Pease Norton, Assistant Professor of Political Economy at Yale University, recently read before the American Association for the

Advancement of Science, a paper which vividly summed up the situation which confronts us. He said:

“There are four great wastes to-day, the more lamentable because they are unnecessary. They are preventable death, preventable sickness, preventable conditions of low physical and mental efficiency, and preventable ignorance. The magnitude of these wastes is testified to by experts competent to judge. They fall like the shades of night over the whole human race, blotting out its fairest years of happiness.

“The facts are cold and bare—one million, five hundred thousand persons must die in the United States during the next twelve months; equivalent to four million, two hundred thousand persons will be constantly sick; over five million homes, consisting of twenty-five million persons, will be made more or less wretched by mortality and morbidity.

“We look with horror on the black pages of the Middle Ages. The black waste was but a passing cloud compared with the white waste visitation. Of people living to-day, over eight million will die of tuberculosis, and the federal government does not raise a hand to help them.

“THE NEGLECT OF HEALTH A NATIONAL EVIL”

“The Department of Agriculture spends seven million dollars on plant health and animal health every year, but, with the exception of the splendid work done by Doctors Wiley, Atwater, and Benedict, Congress does not directly appropriate one cent for promoting the physical well-being of babies. Thousands have been expended in stamping out cholera among swine, but not one dollar was ever voted for eradicating pneumonia among human beings. Hundreds of thousands are consumed in saving the lives of elm trees from the attacks of beetles; in warning farmers against blights affecting potato plants; the importing Sicilian bugs to fertilize fig blossoms in California; in ostracizing various species of weeds from the ranks of the useful plants, and in exterminating parasitic growths that prey on fruit trees. In fact, the Department of Agriculture has expended during the last ten years over forty-six millions of dollars. But not a wheel of the official machinery at Washington was ever set in motion for the alleviation or cure of diseases of the heart or kidneys, which will carry off over six millions of

our entire population. Eight millions will perish of pneumonia, and the entire event is accepted by the American people with a resignation equal to that of the Hindoo, who, in the midst of indescribable filth, calmly awaits the day of cholera.

“During the next census period more than six million infants under two years of age will end their little spans of life while mothers sit by and watch in utter helplessness. And yet this number could probably be decreased by as much as half. But nothing is done.

“In the United States alone, of the eighty millions living to-day, all must die, after having lived, say a little more than three billion, two hundred million years of life, on the average slightly more than twoscore years. Of these years, one billion, six hundred million, represent the unproductive years of childhood and training.

“Consider that the burden of the unproductive years on the productive years is 20-20, or say 100 per cent. Could the average length of life be increased to sixty years, say to forty-eight billion years lived by eighty millions of people, the burden of the unproductive years would fall to 50 per cent. In the judgment of men competent to hold opinions, this is not impossible.”

It was the reading of this paper, which led to the formation of the Committee of One Hundred on National Health, of which Professor Irving Fisher of Yale is president, and which includes among its members such men and women as Ex-President Eliot of Harvard, Dr. Lyman Abbott, Miss Jane Addams, Luther Burbank, Horace Fletcher, Professor Chittenden, Dr. Kellogg, and Dr. Trudeau.

The primary and immediate purpose of the Committee's work is to promote the idea of a national Bureau of Health; but the field open to the committee includes the whole subject of public sanitation and hygiene. President Roosevelt has formally endorsed the work, in a letter from which the following is an extract: “Our national health is physically our greatest national asset. To prevent any possible deterioration of the American stock should be a national ambition. We cannot too strongly insist on the necessity of proper ideals for the family, for simple life and for those habits and tastes which produce vigor and make more capable of strenuous service

to our country. The preservation of national vigor should be a matter of patriotism.... Federal activity in these matters has already developed greatly, until it now includes quarantine, meat inspection, pure food administration, and federal investigation of the conditions of child labor. It is my hope that these important activities may be still further developed.”

And in his notable message to the country, rather than to Congress, which he issued in December, 1907, President Roosevelt wrote: “There is a constantly growing interest in this country in the question of public health. At least, the public mind is awake to the fact that many diseases, notably tuberculosis, are national scourges. The work of the State and City Boards of Health should be supplemented by the constantly increasing interest on the part of the national government. The Congress has already provided a Bureau of Public Health, and has provided for an hygienic report. There are other valuable laws relating to the public health connected with the various departments. This whole branch of the government should be strengthened and aided in every way.”

As somebody said before, these things are no more true because a President has said them; but the fact that President Roosevelt has said them, has given wide publicity to them, and impressed them upon the public consciousness.

The knowledge that economic conditions;—the way in which men and women live because they have to so live in order to earn a living, is the fundamental factor in the case of public health, is something that is bound to become recognized as the growth of knowledge goes on. It will only be a question of time before men and women will see that in order to have health, it will be necessary to organize all the affairs of life with a view to the well-being of humanity as a whole.

In order to make effective the work of the Committee of One Hundred, its President, Irving Fisher, assisted by Professor Norton, organized the American Health League, which has absorbed the Public Health Defense League, an organization formed for the purpose of fighting the patent medicine evil, and awakening public interest in matters of hygiene. The Health League already numbers nine or ten thousand citizens, who are pledged to give financial and moral support to the work of the Committee

of One Hundred in its efforts to establish a national Bureau of Health. The League is rapidly increasing in membership, for a spirit of interest in hygiene is abroad in the land. Local advisory committees have already been formed in more than two hundred cities and towns, and it is planned to prosecute the work of multiplying these branch committees until every town in the United States shall be represented in the membership. The Committee of One Hundred publishes the magazine *American Health* as its official organ, and all American men and women who are interested in the spread of the new hygiene are invited by the Committee to correspond with its Executive Secretary, Drawer 30, New Haven, Conn.

Connected with the advisory and other subcommittees, are committees of writers, editors, and newspaper men, numbering many of our most prominent penmen and pressmen, and the power of molding public opinion through this channel alone is very great. There is now being organized a Council on Co-operation, to consist of the leading officers of American religions, fraternal, learned, secret, and educational organizations; and also a Council of Research, to consist of leading investigators interested in original research along public health lines.

In other words, the Committee of One Hundred has grown to a compact, well-organized, rapidly-spreading, national Army of Health. It has grown within a wonderfully short period, simply because there was a great and pressing *need* for it.

Professor William H. Welch, a member of the Committee of One Hundred, and Professor of Pathology at Johns Hopkins University, has put himself on record as saying that if the nation were to apply in practice the existing knowledge of hygiene, the nation's death rate would be cut in two. In commenting on this statement, Irving Fisher said:

"The greatest asset of all, the physical health of our citizens, is still neglected. Professor Nicholson, an economist of Scotland, has estimated that the living capital of Great Britain is worth five times the physical capital. That is, if we capitalize each man's working capacity and add together this capitalization throughout the whole realm of Great Britain, the value of the population so obtained is five times the value of all the land and all the railroads and all the buildings, and all the iron mines and all the

other capital which is ordinarily called wealth. If we could make this human capital within the United States double its present worth (it is already five times that of the inanimate capital), it is evident what an enormous improvement would ensue as compared with the possible improvements in saving arid lands, and other physical resources. Our health has much more than a money value. But these calculations show that even on the most materialistic method of reckoning, there is truth in Emerson's statement, "the first wealth is health."

APPENDIX

DIET LIST

	Proteid	Carbo- hydrate	Fat	Water	Mineral Matter	Food Value per pound calories
Broiled tenderloin steak	23.5	0	20.4	54.8	1.2	1300
Lamb chops, broiled	21.7	0	29.9	47.6	1.3	1665
Smoked ham, fat, edible portion	14.3	0	52.3	27.9	3.7	2485
Roast turkey, edible portion	27.3	0	18.4	52.0	1.2	1295
Fricassee chicken, edible portion	17.6	2.4	11.5	67.5	1.0	855
Cooked bluefish, edible portion	26.1	0	4.5	68.2	1.2	670
Canned salmon, edible portion	21.8	0	12.1	63.5	2.6	915
Fresh oysters, solid	6.0	3.3	1.3	88.3	1.1	230
Boiled hen's eggs	13.2	0	12.0	73.2	0.8	765
Butter	1.0	0	85.0	11.0	3.0	3605
Full cream cheese	25.9	2.4	33.7	34.2	3.8	1950
Whole cow's milk	3.3	5.0	4.0	87.0	0.7	325
Wheat flour, entire wheat	13.8	71.9	1.9	11.4	1.0	1675
Boiled rice	2.8	24.4	0.1	72.5	0.2	525
Shredded wheat	10.5	77.9	1.4	8.1	2.1	1700
Macaroni	13.4	74.1	0.9	10.3	1.3	1665
Brown bread	5.4	47.1	1.8	43.6	2.1	1050
Wheat bread or rolls	8.9	56.7	4.1	29.2	1.1	1395
Whole wheat bread	9.4	49.7	0.9	38.4	1.3	1140
Soda crackers	9.8	73.1	9.1	5.9	2.1	1925
Ginger bread	5.8	63.5	9.0	18.8	2.9	1670
Sponge cake	6.3	65.9	10.7	15.3	1.8	1795

Apple pie	3.1	42.8	9.8	42.5	1.8	1270
Custard pie	4.2	26.1	6.3	62.4	1.0	830
Indian Meal pudding	5.5	27.5	4.8	60.7	1.5	815
Fresh asparagus	1.8	3.3	0.2	94.0	0.7	105
Fresh lima beans	7.1	22.0	0.7	68.5	1.7	570
Dried lima beans	18.1	65.9	1.5	10.4	4.1	1625
Cooked beets	2.3	7.4	0.1	88.6	1.6	185
Fresh cabbage, edible portion	1.6	5.6	0.3	91.5	1.0	145
Dried peas	24.6	62.0	1.0	9.5	2.9	1655
Green peas	7.7	16.9	0.5	74.6	1.0	465
Boiled potatoes	2.5	20.9	0.1	75.5	1.0	440
Fresh tomatoes	0.9	3.9	0.4	94.3	0.5	105
Baked beans, canned	6.9	19.6	2.5	68.9	2.1	600
Apples, edible portion	0.4	14.2	0.5	84.6	3.0	290
Bananas, yellow, edible portion	1.3	22.0	0.6	75.3	0.8	460
Oranges, edible portion	0.8	11.6	0.2	86.9	0.5	240
Peaches, edible portion	0.7	9.4	0.1	89.4	0.4	190
Fresh strawberries	1.0	7.4	0.6	90.4	0.6	180
Dried prunes, edible portion	2.1	73.3	0.0	22.3	2.3	1400
Almonds, edible portion	21.0	13.3	54.9	4.8	2.0	3030
Peanuts, edible portion	25.8	24.4	38.6	9.2	2.0	2560
Pine nuts, edible portion	33.9	6.9	49.4	6.4	3.4	2845
Brazil nuts, edible portion	17.0	7.0	66.8	5.3	3.9	3265
Soft-shell walnuts, edible portion	16.6	16.1	63.4	2.5	1.4	3285

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AND HOW WE WON IT, WITH AN ACCOUNT OF THE NEW
HYGIENE ***

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